Informatica® Data Ingestion and Replication
October 2026

Application Ingestion and Replication

Informatica Data Ingestion and Replication Application Ingestion and Replication
October 2026

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Preface

Read *Application Ingestion and Replication* to learn how to configure application ingestion and replication tasks in the Data Ingestion and Replication service.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Application Ingestion and Replication

Application Ingestion and Replication can transfer data from Software-as-a-Service (SaaS) and on-premise applications to on-premise databases and cloud-based data lakes, data warehouses, and event streaming platforms.

The SaaS and on-premise applications used in your business or organization store large amounts of business-critical data on a daily basis. You can use Application Ingestion and Replication to transfer the data stored by your applications to targets that can handle large volumes of data. After you transfer the data to the target, you can consolidate the data and use it for various purposes, such as advanced data analytics and data warehousing.

Application Ingestion and Replication can perform the following types of load operations:

- *Initial load*. Loads source data read at a single point in time to a target. After the data is loaded, the ingestion job ends. You can use this load type to materialize a target to which incremental changes will be sent later.
- *Incremental load*. Loads data changes continuously or until the ingestion job is stopped or ends. The ingestion job loads the changes that have occurred since the last time it ran or from a specific start point. You can use this load type to keep data in the target up to date so that you can make informed decisions for your business or organization based on the latest data.
- *Initial and Incremental load*. Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.

For more information about the sources and targets supported for each load type, see "Application Ingestion and Replication sources - preparation and usage" on page 12 and "Application Ingestion and Replication target types - preparation and usage" on page 39.

Use cases

Application Ingestion and Replication can fulfill multiple requirements for your business or organization.

You can use Application Ingestion and Replication for the following use cases:

- **Real-time data synchronization**. Democratize operational data to the rest of the organization with subsecond latency, enabling real-time analytics, monitoring, and reactive applications.

- **Stream data change events in real time to keep dashboards and reports continuously up-to-date**. Keep disparate systems continuously in sync with the single source of truth to support use cases such as predictive maintenance, preventive health care, and personalized user experience.
- **Audit and Compliance**. Create an immutable, time-ordered log of all data modifications for auditing and compliance purposes.

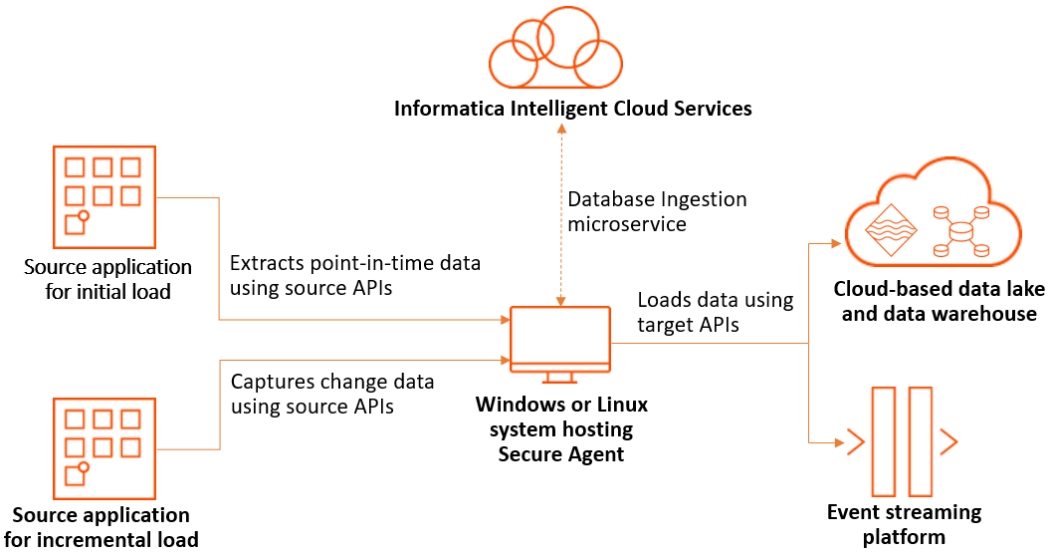
System requirements

The following table lists the Application Ingestion and Replication minimum system requirements for the Secure Agent:

Component	Minimum requirement
Cores per CPU	8 minimum, 16 recommended if you need to process a large number of source objects in an initial load
Memory	32 GB
Disk space	5 GB per job, based on a row size of 2 KB

Application Ingestion and Replication architecture

The following diagram shows the Application Ingestion and Replication architecture:



Application Ingestion and Replication requires the following components to run an application ingestion and replication job:

- **Secure Agent**: Secure Agent is the program that runs tasks and enables secure communication between your organization and IDMC. You must download the Secure Agent on all the systems from which you want to use Application Ingestion and Replication. When you run an application ingestion and replication job, the metadata of the ingestion task is pushed to the Secure Agent which enables the ingestion job to process the data.
- **Database Ingestion service**: The Database Ingestion service is a microservice that is used by Secure Agent to run both application ingestion and replication and database ingestion and replication tasks. The Database Ingestion service is automatically downloaded to the system when you download the Secure Agent.
- **IDMC interface**: IDMC provides a web-based interface to create and deploy an application ingestion and replication task. When you deploy the task, an executable ingestion job is created for the task.
- **Source and target APIs**: The application ingestion and replication job uses the source APIs to retrieve the data from the source objects and the target APIs to load the data to the target. For an incremental load operation, the ingestion job identifies the changes that are made to the source objects after a specific date and time and retrieves the changes at intervals defined in the associated ingestion task.

General limitations and guidelines

Before you configure application ingestion and replication tasks for initial, incremental, or combined initial and incremental load operations, review the following limitations and guidelines:

- You can't run application ingestion and replication tasks on a Hosted Agent. You can use either a local Secure Agent installation or serverless runtime environment hosted on Microsoft Azure. Data Ingestion and Replication does not support serverless runtime environments hosted on AWS. To find out if you can use the serverless runtime environment with your source, see *Connectors and Connections*.
- If a Secure Agent in a runtime environment that is assigned to a application ingestion and replication task terminates, you cannot undeploy the associated ingestion job, update the task to specify another runtime environment, and deploy the task again. In this situation, perform one of the following actions:
  - Assign a different Secure Agent to the runtime environment. Ensure that the new Secure Agent is running. Then restart the associated ingestion job.
  - Copy the task. In the task copy, specify another runtime environment that has an active Secure Agent. Then deploy the task and run the associated ingestion job.
- If you run multiple incremental load or combined initial and incremental load jobs against the same source, you cannot move capture processing of some source objects from one job to another job or merge the jobs into a single job, without resynchronizing the source and target objects being processed by the new job. Each job maintains a unique recovery checkpoint. If you try to move capture processing of an object from one job to another job, the correct recovery information for that table is not maintained, potentially causing data loss or duplicate events on the target.
- Application Ingestion and Replication uses the minimum source constraints needed for replication when generating target objects. If a source has a primary key, the job preferentially uses the primary key and no other constraint when creating the target object.
  Source object properties that are not required for replication, such as for storage and partitioning, are not reflected on the target

  .

- Because Data Ingestion and Replication seeks to maintain consistency between the source and target when replicating data, it does not add or recognize any constraints on the target, such as those for nullability, foreign keys, and defaults. Doing so could lead to inconsistencies between the source and target and potential replication errors. For example, errors could occur if the NOT NULL constraint is added to a target column that receives data from a nullable source column, or if the DEFAULT constraint is added to a target column that receives source data that doesn't match the default.

  Note: If source constraints exist at the time of task deployment, they're not re-created at the target because they might not have existed previously when data was added to a source column and captured.
- If you use a Secure Agent group with multiple agents and the active agent goes down unexpectedly, application ingestion and replication jobs can automatically switch over to another available agent in the Secure Agent group. The automatic switchover occurs after the 15-minute heartbeat interval elapses. Automatic switchovers are subject to following limitations for application ingestion and replication jobs:
  - The job cannot have persistent storage enabled.
  - The job cannot have an SAP ODP source or a Kafka target.
  - If the database ingestion service is in a stopped or error state, you need to manually stop and then resume the job.

Application Ingestion and Replication sources - preparation and usage

The sources that Application Ingestion and Replication support depend on whether the application ingestion and replication tasks transfer a point-in-time snapshot of data in a batch initial load operation or load incremental change data from a specific start point.

The following table lists the source types that Application Ingestion and Replication support along with the types of load operations supported for each source type:

Source type	Supported load operations
Marketo	Initial load, incremental load, and combined initial and incremental load
Microsoft Dynamics 365	Initial load, incremental load, and combined initial and incremental load
NetSuite	Initial load, incremental load, and combined initial and incremental load
Oracle Fusion Cloud Applications	REST: Initial load, incremental load, and combined initial and incremental load BICC: Initial load, incremental load, and combined initial and incremental load
Salesforce	Initial load, incremental load, and combined initial and incremental load
Salesforce Marketing Cloud	Initial load
SAP ODP	SAP ECC: Initial load, incremental load, and combined initial and incremental load SAP S4/HANA: Initial load, incremental load, and combined initial and incremental load
SAP OData V2	SAP S4/HANA: Initial load, incremental load, and combined initial and incremental load.

Source type	Supported load operations
SAP Mass Ingestion	SAP ECC: Initial load, incremental load, and combined initial and incremental load for Oracle database. SAP S4/HANA: Initial load and incremental load for HANA database.
ServiceNow	Initial load, incremental load, and combined initial and incremental load
Workday	Initial load, incremental load, and combined initial and incremental load

To determine the connectors to use for the source types, see *Connectors and Connections* > *Applications Ingestion and Replication connectors*.

Marketo sources

Consider the following guidelines when you use Marketo sources:

- The first time an incremental load job runs to capture change data for a Marketo source, the job retrieves and loads only the change records that are created after the date and time specified in the associated ingestion task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- Incremental load jobs configured for Marketo sources can replicate only insert and update operations performed on the source objects.
- Application Ingestion and Replication does not replicate the relationships defined between Marketo objects. The target does not contain the relationships that exists between the objects on the source.
- By default, Marketo instances are limited to 500 MB of downloaded data per day. Additionally, bulk API downloads are limited to a window of 31 day period.
- Some Marketo objects do not allow APIs to directly retrieve the data stored in them. To retrieve the data from such objects, APIs require certain filters as input parameters. The filters that the APIs use are fields of other Marketo objects. Application Ingestion and Replication does not replicate the data stored in such Marketo objects that are dependent on other objects.
- Application ingestion and replication jobs do not propagate the data stored in custom objects that are not linked to any lead object. Additionally, application ingestion and replication jobs do not propagate the data stored in custom object fields with the display name ID.
- By default, Application Ingestion and Replication can capture change data for the following Marketo objects:
  - Campaign
  - Email Templates
  - Folder
  - Landing Pages
  - Landing Page Templates
  - Program
  - Segmentations
  - Smart Campaign
  - Smart Lists

- Snippets
- Static Lists

Note: To capture change data for the Campaign, Email Templates, Folder, Landing Pages, Landing Page Templates, Segmentations, and Snippets objects, you must set the marketo.include.additional.cdc.objects custom property to TRUE. You can set the custom property on the **Source** tab of the application ingestion and replication task wizard.

Microsoft Dynamics 365 sources

Consider the following guidelines when you use Microsoft Dynamics 365 sources:

- Application ingestion and replication tasks for Microsoft Dynamics 365 sources support only sales tables.
- The first time an incremental load job runs to capture change data for a Microsoft Dynamics 365 source, the job retrieves and loads only the change records that are created after the date and time specified in the associated ingestion task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- Incremental load jobs for Microsoft Dynamics 365 sources don't capture the change data for tables without the `modifiedon` column.
- Incremental load jobs and combined initial and incremental load jobs for Microsoft Dynamics 365 sources don't detect and replicate the changes that are made to the source schema.
- Incremental load jobs and combined initial and incremental load jobs for Microsoft Dynamics 365 sources don't capture the DELETE data operation for a CDC cycle.
- If a source table contains locale ID and timezonecode values, Application Ingestion and Replication propagates the values to the target without converting them to the actual values that they represent.
- Application Ingestion and Replication does not propagate columns of the following data types:
  - CUSTOMERTYPE
  - PARTYLISTTYPE
  - CALENDARRULESTYPE
  - UNKNOWN
- For OWNERTYPE and Lookup source fields, Application Ingestion and Replication adds a target column named `<ColumnName>_display_name` that holds the readable value resolved from the referenced record. The original column keeps the Dynamics 365 GUID that the field stores, so the target holds both the reference and its resolved value.
  For Choice, Yes-No, State, and Status columns, Application Ingestion and Replication replicates the display-name column to the target as a string column named `<column>name`, next to the numeric code column.

NetSuite sources

Consider the following guidelines when you use NetSuite sources:

- The first time an incremental load job runs to capture change data for a NetSuite source, the job retrieves and loads only the change records that are created after the date and time specified in the associated ingestion task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.

- Incremental load jobs configured for NetSuite sources do not capture change data for the tables that do not contain a primary key.
- Incremental load jobs do not capture change data for the tables that do not contain at least one of the following timestamps:
  - The timestamp of when its records were created.
  - The timestamp of the last time its records were modified.

  Note: For the tables that contain only the timestamp of when its records were created, incremental load jobs capture only the insert operations performed on the records.
- When an application ingestion and replication job retrieves a document or image from the source, the job propagates the unique identifier of the file cabinet that contains the document or image instead of propagating the binary content of the file.
- Application Ingestion and Replication does not capture the deletion of records from the source tables.
- Incremental load jobs and combined initial and incremental load jobs configured for NetSuite sources do not detect and replicate the deletion and renaming of source fields.
- If a NetSuite Mass Ingestion connection configured for a NetSuite.com data source is already used in an application ingestion and replication task and if the task is in any status other than Undeployed, you must not modify the connection to access a NetSuite2.com data source. Instead of modifying the existing connection, create a new connection for the NetSuite2.com data source and then configure a new application ingestion and replication task using the new connection.
- Application Ingestion and Replication does not propagate data stored in the following NetSuite tables:
  - Account_activity
  - Account_period_activity
  - Case_types
  - Case_origins
  - CompanyAddressbook
  - CompanyAddressbookEntityAddress
  - Deleted_records
  - MAP_wave_custbody17
  - MAP_wave_custbody_cseg4
  - MAP_wave_custbody_csegcs_multsel

Oracle Fusion Cloud sources

Application Ingestion and Replication uses Business Intelligence Cloud Connector (BICC) or REST approach to transfer data from the source to the target.

Application Ingestion and Replication can replicate data of the following Oracle Fusion Cloud Applications Suite modules and applications:

Module	Application
Enterprise Resource Planning (ERP)	Cloud Financials
	Procurement

Module	Application
	Project Management
	Risk Management and Compliance
Human Capital Management (HCM)	Human Resources
Industry	Higher Education
	Public Sector
	Digital Experience for Communications
	CX for Consumer Goods
	CX for Financial Services
Sales	Loyalty
	Partner Relationship Management
	Customer Data Management
	Sales Automation
	Subscription Management
Service	B2B Service
Supply Chain & Manufacturing (SCM)	AI Apps
	Cloud Service Logistics
	Fusion Cloud Inventory Management
	Maintenance
	Manufacturing
	Order Management
	Product Lifecycle Management
	Supply Chain Collaboration
	Supply Chain Planning

Source preparation

- Configure the application ingestion and replication user with following roles assigned in Oracle Applications Cloud to access the BICC Console and the data files generated by BICC and stored on the UCM server:

  Access to BICC Console

  Assign the user either the **BI Administrator** role or a custom **BIACM_ADMIN** role with the following memberships:

  ESS Administrator Role: To create and manage schedules for global data extracts or jobs.

  ORA_ASM_APPLICATION_IMPLEMENTATION_ADMIN_ABSTRACT: To access global offerings, data stores, or jobs.

  For more information on how to create these roles and add memberships, see Provision a User with Access to BICC in the Oracle documentation.

  Access to BICC-generated extract files on UCM

  Assign the user the **BICC_UCM_CONTENT_ADMIN** role with this membership:

  OBIA_EXTRACTTRANSFORMLOAD_RWD: To view and download extract files stored on UCM.

  For more information on how to create this role and add the membership, see Provision a User with Access to BICC Content on UCM in the Oracle documentation.

- To configure chunking for advanced data extraction in Oracle Fusion Cloud sources in a task, complete the following prerequisites:
  - To use primary key-based chunking, ensure that the datastore contains a single numeric primary key column that does not contain null values.
  - To use date-based chunking, ensure that the datastore includes at most one TIMESTAMP or DATE column with the "isCreationDate" property set to true in the Filter and/or Chunk by Creation Date Columns section of the Advanced Extract Configuration page in the BICC Console, and then save the datastore.

  Note: Chunking funtionality is applicable to initial load tasks, and in the initial portion in combined initial and incremental load tasks configured to use the BICC replication approach.

Usage considerations for Oracle Fusion Cloud

Consider the following guidelines when you use Oracle Fusion Cloud sources:

- When you set up an incremental load and combined initial and incremental load job, consider the following guidelines to extract data based on whether you use the BICC or REST replication approach:
  BICC approach
  - Enable the **Support Incremental** option in the BICC console for the datastore to support the incremental load and combined initial and incremental load types. To enable the **Support Incremental** option, at least one of the columns in the datastore with **TIMESTAMP** or **DATE** data type must be marked as either **Incremental Filter** or **Is Effective Start Date**.

    Note: Marking multiple columns as **Incremental Filter** or **Is Effective Start Date** might lead to inaccurate data extraction due to Oracle limitations.
  - Set the Prune Time value in the Oracle Fusion BICC extract preferences to 0 minutes. By default, the prune time is set to 1440 minutes, which causes each incremental extract job to read data starting 1440 minutes before the current job time. When you set the Prune Time to 0, the column designated as **Incremental Filter** or **Is Effective Start Date** in the datastore identifies change records and extracts only those records added or updated since the last extraction, which prevents duplicate data processing. For more information, see Pruning Time in the Oracle Fusion BICC documentation.

- In Oracle Fusion BICC source jobs, a dot (.) in datastore names is automatically replaced with double underscores (__). Jobs with an Oracle Fusion BICC source don't create a subtask if you use a dot (.) as a datastore separator. Use an asterisk (*) instead of dot (.).
- Jobs do not capture or replicate delete operations performed on the source. If records are deleted from the Oracle Fusion Cloud source, the corresponding records are not deleted from the target, which might result in a data count mismatch between the source and target.

REST Approach

- By default, incremental load jobs automatically capture change data for objects that contain both the LASTUPDATEDATE and CREATIONDATE fields and ignores objects that do not contain both fields.
- If only inserts are required, and any objects contain only the CREATIONDATE field, you can capture just the inserts for those objects by setting the **oraclefusion.cdc.allow.creationdate.objects** custom property to **true** on the **Source** tab of the application task wizard. Consider the following guidelines:

  Objects with only the CREATIONDATE field allow recording of only insertions. Updates and deletions are ignored without the LASTUPDATEDATE field.

  Do not set the **oraclefusion.cdc.allow.creationdate.objects** property for a deployed task. For an existing task, ensure you undeploy, set the property, and then deploy the task again to apply this change.

- Incremental load jobs and combined initial load and incremental load jobs configured for Oracle Fusion Cloud sources do not capture and replicate the changes made to the source schema.

Salesforce sources

Consider the following guidelines when you use Salesforce sources:

- The first time an incremental load job runs to capture change data for a Salesforce source, the job retrieves and loads only the change records that are created after the date and time specified in the associated ingestion task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- By default, incremental load jobs capture change data only for the objects that contain the `lastmodifieddate` or `systemModstamp` field. However, some objects in Salesforce contain only `CreatedDate` field instead of `lastmodifieddate` and `systemModstamp` fields. To capture the change data for such objects that contain only the `CreatedDate` field, set the salesforce.cdc.allow.createddate.objects custom property to true on the **Source** tab of the application ingestion and replication task wizard. If you set the custom property to true for a deployed task, make sure that you undeploy the job and then deploy the task again for the change to take effect.
- If you want to replicate Data.com objects, your Salesforce account must be assigned with the appropriate Data.com license.
- Application Ingestion and Replication does not capture the relationships defined between Salesforce objects. The target does not contain the relationships between the source objects.
- If an object contains a compound field, Application Ingestion and Replication propagates the component fields that make up the compound field instead of propagating the compound field. For example, if the compound field `geolocation` is a combination of fields `geolocation_longitude` and `geolocation_lattitude`, Application Ingestion and Replication separately loads the data stored in the `geolocation_longitude` and `geolocation_lattitude` fields to the target instead of propagating the compound field `geolocation`.
- The masking configuration of source fields is retained on the target.
- All incremental load activities use only the standard REST API.

- The lookup fields are represented on the target by their unique alphanumeric identifiers.
- When a new field with a default value of the Function data type is added to an existing source object, Application Ingestion and Replication does not propagate the default value to any existing row on the target. However, when new rows are added to the object, the incremental jobs propagate the values stored in the field for the newly added rows.
- Application ingestion and replication jobs do not replicate the data stored in big objects that are not indexed on the Salesforce platform. Before you deploy an application ingestion and replication task for a Salesforce source, ensure that you index all the big objects which are selected by the object selection rules in the task.
- If a source field of Decimal data type contains a value whose scale exceeds the maximum limit defined for the field, application ingestion and replication jobs might fail to propagate such fields. However, you can trim the scale of the value to the defined limit while propagating the field. You can trim the scale of the value by setting the custom property salesforce.decimal.trim.scale to true.

Salesforce Marketing Cloud sources

Consider the following guidelines when you use Salesforce Marketing Cloud sources:

- Application Ingestion and Replication can ingest the data stored in the data extension objects of Salesforce Marketing Cloud.
- You can configure only initial load jobs for Salesforce Marketing Cloud sources.
- When an application ingestion and replication job replicates a data extension object on the target, the job prepends `DATA_EXTENSION__` to the name of the object. For example, if an application ingestion and replication job replicates the data extension object `Address`, the name of the object on the target is `DATA_EXTENSION__Address`.
- An application ingestion and replication initial load job with Salesforce Marketing Cloud source and Microsoft Azure Synapse Analytics target might fail to deploy for some objects.

SAP ECC and SAP S4/HANA sources using the SAP ODP Extractor connector

Consider the following guidelines when you use SAP ECC or SAP S4/HANA sources with SAP ODP Extractor connector:

- When you use an SAP source for the first time, perform the following steps before you configure an application ingestion and replication task for the source:
  1. Verify that the appropriate SAP Notes are available in the SAP server. For more information, see the *SAP Connector Guide*.
  2. Download the SAP Java Connector (SAP JCo) library from the SAP website.
  3. Copy the native and JAR files from the downloaded SAP JCo library to the following directory: `<Secure_Agent>\ext\connectors\thirdparty\infa.odp\`
  4. Copy the sap-adapter-common.jar file from `<Secure_Agent>\downloads\package-ICSAgent_<version>\package\ICS\main\bin\rdtm\javalib\sap` to the following directory: `<Secure_Agent>\ext\connectors\thirdparty\infa.odp\`
  5. Restart the Secure Agent.

- Before you configure an application ingestion and replication task for an SAP source, ensure that you release the data sources from which you want to transfer data. You can use the following functions to release the required data sources:
  - BS_ANLY_DS_RELEASE_ODP
  - RODPS_OS_EXPOSE
- The first time an incremental load job runs to capture change data for an SAP source, the job retrieves and loads the change records from the latest position in the data stream that contains source data. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- Incremental load jobs for SAP sources do not capture the change data for data sources that do not have a primary key. Additionally, the incremental load jobs do not capture change data for the following SAP data sources:
  - The data sources for which SAP does not support delta.
  - The data sources that are configured to provide change data in the form of additive images.
- When an application ingestion and replication task encounters NULL values in the SAP table source column, it does not replicate NULL with blank data to the target table.
- Application Ingestion and Replication does not detect and propagate schema changes in SAP sources.
- Application Ingestion and Replication does not propagate fields of the following obsolete SAP data types:
  - D16S
  - D34S
  - PREC
  - VARC

  Additionally, application ingestion and replication jobs do not propagate fields of the RSTR data type.

Configuring SAP User Authorization

Configure the SAP user account to process SAP table data.

The following table describes the required authorization to read from SAP tables:

Read Object Name	Required Authorization
S_BTCH_JOB	DELE, LIST, PLAN, SHOW. Set Job Operation to RELE.
S_PROGRAM	BTCSUBMIT, SUBMIT
S_RFC	SYST, SDTX, SDIFRUNTIME, /INFADI/TBLRDR, RFC1
S_TABU_DIS / S_TABU_NUM	Provide SAP table name from which you want to read data.

Installing SAP ODP Connector Transport Files

Install the SAP ODP Extractor transport files on the SAP machines that you want to access. Before you install the transports on your production system, install and test the transports in a development system.

Verify that you install the latest transport files to extract data from the SAP ODP object.

Install the following data file and cofile to read data from the SAP ODP object:

Data and Cofile Names	Transport Request	Functionality
- K900861.N75 - R900861.N75	N75K900861	Install the transports only when you want to read from an SAP ODP object that supports hierarchy. You can use the SAP ODP Extractor Connector without installing the SAP ODP Extractor transport files for objects that do not support hierarchy.

Installing Transport Files

Install the latest transport files from a Secure Agent directory to read from a Unicode SAP system. The transport files are for SAP version ERP 6.0 EHP7 system or later.

1. Find the transport files in the following directory on the Secure Agent machine:

   `<Informatica Secure Agent installation directory>\downloads\package-SAPODP.<Latest version>\package\sapodp\sap-transport`
2. Copy the cofile transport file to the `Cofile` directory in the SAP transport management directory on each SAP machine that you want to access.

   The cofile transport file uses the following naming convention: `<number>.<sap system>`.
3. Copy the data transport file to the `Data` directory in the SAP transport management directory on each SAP machine that you want to access.

   The data transport file uses the following naming convention: `<number>.<sap-system>`.
4. To import the transports to SAP, in the STMS, click **Extras** > **Other Requests** > **Add** and add the transport request to the system queue.
5. In the **Add Transport Request to Import Queue** dialog box, enter the request number for the cofile transport.

   The request number inverts the order of the renamed cofile as follows: `<sap-system><number>`.
6. In the Request area of the import queue, select the transport request number that you added, and click **Import**.
7. If you are upgrading from a previous version of the Informatica Transports, select the **Overwrite Originals** option.

SAP ECC and SAP S4/HANA sources using the SAP Mass Ingestion connector

The SAP Mass Ingestion connector supports the following SAP sources:

- **SAP ECC on Oracle database**. Supports database views (initial load), transparent tables, pool tables, and cluster tables.
- **SAP S/4 HANA using a trigger-based approach**. Supports database views (initial load) and transparent tables.

SAP ECC sources

To use SAP ECC source with the underlying Oracle database in application ingestion and replication tasks, first prepare the source database and review the usage considerations.

Source preparation

- When you use an SAP ECC source with an SAP Mass Ingestion connector for the first time, perform the following steps before you configure an application ingestion and replication task for the source:

  1. You must install the latest version of the SAP transport file on the SAP machine you want to access. For more information about accessing the SAP transport file, contact Informatica Global Customer Support.

     The following table describes the transport on the SAP development system:

Data and Cofile Names	Data and Cofile Names	Transport Request	Functionality
R<number>.EP6 K<number>.EP6	Required	EP6K<number>	Install the transports to read from SAP transparent tables, cluster tables, pool tables, and views.

     For more information about installing the transport files, see "Installing transport files" on page 26.

  2. Verify that the appropriate SAP Notes are available in the SAP server. For more information, see the *SAP Connector Guide*.

  3. Download the SAP Java Connector (SAP JCo) library from the SAP website, and perform the followintg tasks:

     a. Copy the native and JAR files from the downloaded SAP JCo library to the following directory:
     `<Secure_Agent>\ext\connectors\thirdparty\informatica.sapami\`

     b. Restart the Secure Agent.

- To use the SAP ECC with the underlying Oracle source database in application ingestion and replication tasks, define the following system environment variables on the Linux or Windows system where the Secure Agent runs:

  - ORACLE_HOME environment variable. Points to the Oracle client installation directory, one level up from the bin directory on Windows or lib directory on Linux. This environment variable is not required. However, if you do not define it, you'll need to specify the full Oracle client installation path when you define other environment variables or Secure Agent properties that specify a path to a subdirectory.
  - TNS_ADMIN environment variable. If you specify a TNS name in the **Database Connect String** property in the Oracle database properties in the SAP Mass Ingestion connection, use this environment variable to point to the directory location of the tsnnames.ora file when the file is not in the default $ORACLE_HOME/network/admin directory. The tnsnames.ora file, along with the Oracle Call Interface (OCI), is used to communicate with the Oracle source database.

  Note: In Administrator, you can set the ociPath property for the Database Ingestion agent service (DBMI agent) to point to the OCI library that contains the oci.dll or libclntsh.so file. The OCI library is used by application ingestion and replication CDC tasks to connect to Oracle. By default, Oracle uses the ociPath value of $ORACLE_HOME/lib on Linux or %ORACLE_HOME%\bin on Windows.

- Make sure the Application Ingestion and Replication user has the Oracle privileges that are required for the application ingestion and replication load type to be performed. For more information, see "Oracle privileges" on page 27.

- If you use an Amazon RDS for Oracle, make sure the Application Ingestion and Replication user has the required privileges. For more information, see "Oracle privileges for Amazon RDS for Oracle" on page 29.

- If you use Oracle 11.2.04, set the Oracle COMPATIBLE initialization parameter to 11.2.04 to ensure that Oracle has all of the most current redo logs fixes for that release.
- Ensure that the Oracle Database Client or Instant Client is installed and configured on the Secure Agent server for the Secure Agent to communicate with Oracle. If you do not already have an Oracle client installed, you can download a client and access installation information from the Oracle web site, or ask your Oracle DBA to download and configure an Oracle client.
- For incremental load or combined initial and incremental load operations that use log-based CDC, perform the following prerequisite tasks in Oracle:
  - Enable ARCHIVELOG mode for the Oracle database. If the database is not in an Amazon RDS environment, issue the following SQL statements:

    ```
    SHUTDOWN IMMEDIATE;
    STARTUP MOUNT;
    ALTER DATABASE ARCHIVELOG;
    ALTER DATABASE OPEN;
    SHUTDOWN IMMEDIATE;
    STARTUP;
    ```

  - Enable Oracle minimal global supplemental logging on the source database.
  - For incremental load operations with the source, Application Ingestion and Replication provides alternative capture methods for capturing change data from the source and applying the data to a target. Application Ingestion and Replication reads data changes from Oracle redo logs. This method requires users to have extended privileges.
  - Because Application Ingestion and Replication expects each source table row to be unique, Informatica recommends that each source table have a primary key. Application Ingestion and Replication does not honor unique indexes in place of a primary key. If no primary key is specified, Application Ingestion and Replication treats all columns as if they are part of the primary key.
- Your jobs must be able to access transaction data in Oracle online and archive redo logs. For more information, see "Oracle archive log retention considerations" on page 34.
- Application ingestion and replication jobs that use log-based CDC require read access to Oracle online and archive redo logs to read incremental change data. If the redo logs are remote from the on-premises system where the Secure Agent runs, make sure that read access to the logs is provided, for example, by using Oracle Automatic Storage Management (ASM), mounting the logs to a network file system (NFS).
- To access the redo logs by using Oracle directory objects with BFILEs, see "Configuring BFILE access to Oracle redo logs in the Oracle file system" on page 33.

Usage considerations

- When you enable HTTPS for the SAP Mass Ingestion connection, the task processes only initial loads and the initial phase of combined loads. You cannot run CDC with HTTPS enabled.
- When the Oracle database for SAP ECC source contains CHAR or VARCHAR columns contain nulls, the application ingestion and replication job does not delimit the null values with double-quotation (") marks or any other delimiter when writing data to a Amazon S3, Flat File, Microsoft Azure Data Lake, or Microsoft Azure Synapse Analytics target.
- Application Ingestion and Replication can process data across a RESETLOGS boundary. To avoid the source and targets becoming out of sync, Informatica recommends that you stop capture processing before performing a RESETLOGS and then restart capture processing after the RESETLOGS event. Otherwise, the capture process might send data to the target that is subsequently reverted by the RESETLOGS event, causing the source and target to become out of sync.
- Alternative strategies for accessing the Oracle redo logs are available.
- If an application ingestion and replication incremental load or combined initial and incremental load task contains a source table name or one or more column names that are longer than 30 characters, Oracle suppresses supplemental logging for the entire table, including primary keys and foreign keys. As a result,

most operations on the table fail. This problem is caused by an Oracle restriction. In this situation, exclude the table from capture processing or rename the long table and column names to names of 30 characters or less.

- Do not use custom data type mapping rules to map a RAW column from the source to a target CHAR or VARCHAR column. Otherwise, the deployment of the application ingestion and replication task might fail.
- Application Ingestion and Replication does not support invisible columns in the source columns, regardless of the target type. For these columns, application ingestion and replication incremental load jobs and combined initial and incremental load jobs propagate nulls to the corresponding target columns.
- Application ingestion and replication combined initial and incremental load jobs that have SAP ECC sources with Oracle database can read changes from copies of the archive redo logs. You must set the **Reader Mode** property to ARCHIVECOPY in the SAP Mass Ingestion connection properties for the Oracle database and also set the source custom property pwx.cdcreader.oracle.reader.additional with the dir and file parameters. The dir parameter points to the name of the base directory that the CDC log reader scans for the archive log copies, and the file parameter specifies a mask that is used to filter the log copies.
- Application Ingestion and Replication does not propagate fields of the following obsolete SAP data types:

  - D16S

  - D34S

  - PREC

  - VARC

  Additionally, application ingestion and replication jobs do not propagate fields of the RSTR data type.
- An SAP ECC source with an SAP Mass Ingestion connector doesn't support schema drift.

SAP S/4 HANA sources

To use SAP S/4 HANA (trigger based) in application ingestion and replication tasks, first prepare the source database and review the usage considerations.

Source preparation

- When you use an SAP S/4 HANA source with an SAP Mass Ingestion connector for the first time, perform the following steps before you configure an application ingestion and replication task for the source:

  1. You must install the latest version of the SAP transport file on the SAP machine that you want to access.
     For more information about accessing the SAP transport file, contact Informatica Global Customer Support.

     The following table describes the transport on the SAP development system:

Data and Cofile Names	Data and Cofile Names	Transport Request	Functionality
R<number>.EP6 K<number>.EP6	Required	EP6K<number>	Install the transports to read from SAP transparent tables, cluster tables, pool tables, and views.

     For more information about installing the transport files, see "Installing transport files" on page 26.
  2. Verify that the appropriate SAP Notes are available in the SAP server. For more information, see the *SAP Connector Guide*.

3. Download the SAP Java Connector (SAP JCo) library from the SAP website, and the copy the native and JAR files from the downloaded SAP JCo library to the following directory:
   `<Secure_Agent>\ext\connectors\thirdparty\informatica.sapami\`
4. The SAP Mass Ingestion connector uses JDBC to connect to SAP S/4 HANA to read data and metadata and to test connection properties. You must download the SAP HANA JDBC driver file, ngdbc.jar, and copy it to a specific subdirectory of the Secure Agent installation directory on the machine where the Secure Agent runs.
   a. Download the SAP HANA JDBC driver jar file, ngdbc.jar, to the Linux or Windows machine where the Secure Agent runs.
      Verify that you download the most recent version of the file. If you encounter issues with downloading the file, contact SAP Customer Support.
   b. Copy the ngdbc.jar file to the following directory:
      ```
      <Secure Agent installation directory>/ext/connectors/thirdparty/
      informatica.hanami
      ```
   c. Restart the Secure Agent.
5. Create a Mass Ingestion Database user. Connect to the source database as a user with admin privilege and execute the following statement:
   ```
   CREATE USER dbmi_user password "<password>" NO FORCE_FIRST_PASSWORD_CHANGE;
   ```
   This statement creates a user in the database with the default permissions, which permit basic data dictionary views to be read and the required CDC objects to be created in the user's own schema.
6. To deploy and run a database ingestion task that includes an SAP S/4 HANA source, the source connection must specify a Mass Ingestion Databases user (dbmi_user) who has the privileges to read metadata and other information from the following system views:
   - SYS.M_DATABASE - Used to fetch the database version.
   - SYS.M_CS_PARTITIONS - Used to identify if a table is partitioned. (Not applicable for SAP HANA Cloud)
   - SYS.SCHEMAS - Used to fetch the list of schemas for a database.
   - SYS.TABLES - Used to fetch a list of table names for a schema.
   - SYS.TABLE_COLUMNS - Used to fetch column metadata for a table.
   - SYS.INDEXES - Used to fetch index information for a table.
   - SYS.INDEX_COLUMNS - Used to fetch index information for a table.
7. For incremental load jobs, grant the following privileges:
   - For triggers to write rows to the PKLOG and shadow _CDC tables of thedbmi_user, grant INSERT access on thedbmi_user's schema to the user that owns the schema of the source tables (schema_user):
     ```
     GRANT INSERT ON SCHEMA dbmi_user TO schema_user;
     ```
   - To capture change data from source tables in the schema_user's schema by using triggers, execute one of the following statements:
     ```
     GRANT TRIGGER ON SCHEMA schema_user TO dbmi_user;
     ```
     This statement grants trigger access to all tables in the schema.
     - or -
     ```
     GRANT TRIGGER ON database.table_name TO dbmi_user;
     ```

This statement grants trigger access to a specific source table. Use this statement when you want to capture data from just a few selected tables. Repeat the grant for each source table of CDC interest.

8. For initial load jobs, execute one of the following grant statements to read data from source tables:

```
GRANT SELECT ON SCHEMA schema_user TO dbmi_user;
```

This statement grants SELECT access to all tables in the schema.

- or -

```
GRANT SELECT ON database.table_name TO dbmi_user;
```

This statement grants SELECT access on a specific source table. Repeat this grant for each source table from which you want to read data.

Usage considerations

Consider the following guidelines when you use SAP S/4 HANA sources with the SAP Mass Ingestion connector:

- For SAP S/4 HANA sources, the connection requires an SAP HANA JDBC custom connection property setting for encryption. Enter the following properties in the **Advanced Connection Properties** field in the SAP Mass Ingestion connection properties:

```
encrypt=true&validateCertificate=false
```

- When you enable HTTPS for the SAP Mass Ingestion connection, the task processes only initial loads and the initial phase of combined loads. You cannot run CDC with HTTPS enabled.
- Application Ingestion and Replication does not propagate fields of the following obsolete SAP data types:
  - D16S
  - D34S
  - PREC
  - VARC
  - DATN
  - TIMN
  - DECFLOAT_16
  - DECFLOAT_34
  - UTCLONG
  - STRG datatypes (For incremental load)

  Additionally, application ingestion and replication jobs do not propagate fields of the RSTR data type.
- An SAP ECC source with an SAP Mass Ingestion connector doesn't support schema drift.

For more information about configuring SAP user authorization, see "Configuring SAP User Authorization" on page 20.

Installing transport files

Install the latest transport files from a Secure Agent directory to read from a Unicode SAP system. The transport files are for SAP version ECC 5.0 or later.

1. Find the transport files in the following directory on the Secure Agent machine:

```
<Secure Agent installation directory>\downloads\package-SAPConnector.<latest version>
\package\rdtm\sap-transport\SAPTableReader
```

2. Copy the cofile transport file to the `Cofile` directory in the SAP transport management directory on each SAP machine that you want to access.

   The cofile transport file uses the following naming convention: `K<number>.EP6`.

3. Copy the data transport file to the `Data` directory in the SAP transport management directory on each SAP machine that you want to access.

   The data transport file uses the following naming convention: `R<number>.EP6`.

4. To import the transports to SAP, in the STMS, click **Extras** > **Other Requests** > **Add** and add the transport request to the system queue.
5. In the **Add Transport Request to Import Queue** dialog box, enter the request number for the cofile transport.

   The request number inverts the order of the renamed cofile as follows: EP6K<number>.
   For example, for a cofile transport file renamed as `K900215.EP6`, enter the request number as EP6K900215.

6. In the Request area of the import queue, select the transport request number that you added, and click **Import**.
7. If you are upgrading from a previous version of the Informatica Transports, select the **Overwrite Originals** option.

Oracle privileges

To deploy and run an application ingestion and replication task that has an SAP ECC source with the underlying Oracle database, the source connection must specify a Application Ingestion and Replication user who has the privileges required for the ingestion load type.

Privileges for incremental load processing with log-based CDC

Note: If the Oracle logs are managed by ASM, the user must have SYSASM or SYSDBA authority.

For an application ingestion and replication task that performs an incremental load or combined initial and incremental load using the log-based CDC method, ensure that the Application Ingestion and Replication user (*cmid_user*) has been granted the following privileges:

```
GRANT CREATE SESSION TO <cmid_user>;

GRANT SELECT ON table TO <cmid_user>;     -- For each source table created by user
GRANT EXECUTE ON DBMS_FLASHBACK TO <cmid_user>;

-- The following grant is required for combined initial and incremental loads only. Do not
-- use ANY TABLE unless your security policy allows it.
GRANT FLASHBACK ON table|ANY TABLE TO <cmid_user>;

-- Include the following grant only if you want to Execute the CDC script for enabling
-- supplemental logging from the user interface. If you manually enable supplemental
-- logging, this grant is not needed.
GRANT ALTER table|ANY TABLE TO <cmid_user>;

GRANT SELECT ON DBA_CONSTRAINTS TO <cmid_user>;
GRANT SELECT ON DBA_CONS_COLUMNS TO <cmid_user>;
GRANT SELECT ON DBA_INDEXES TO <cmid_user>;
GRANT SELECT ON DBA_LOG_GROUPS TO <cmid_user>;
GRANT SELECT ON DBA_LOG_GROUP_COLUMNS TO <cmid_user>;
GRANT SELECT ON DBA_OBJECTS TO <cmid_user>;
GRANT SELECT ON DBA_OBJECT_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_TABLESPACES TO <cmid_user>;
GRANT SELECT ON DBA_USERS TO <cmid_user>;

GRANT SELECT ON "PUBLIC".V$ARCHIVED_LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$CONTAINERS TO <cmid_user>;  -- For Oracle multitenant environments
```

```
GRANT SELECT ON "PUBLIC".V$DATABASE TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$DATABASE_INCARNATION TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$ENCRYPTION_WALLET TO <cmid_user>;     -- For Oracle TDE access
GRANT SELECT ON "PUBLIC".V$LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$LOGFILE   TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$PARAMETER TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$PDBS TO <cmid_user>;     -- For Oracle multitenant environments
GRANT SELECT ON "PUBLIC".V$SPPARAMETER TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$STANDBY_LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$THREAD TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$TRANSACTION TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$TRANSPORTABLE_PLATFORM TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$VERSION TO <cmid_user>;

GRANT SELECT ON SYS.ATTRCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO <cmid_user>;
GRANT SELECT ON SYS.COL$ TO <cmid_user>;
GRANT SELECT ON SYS.COLTYPE$ TO <cmid_user>;
GRANT SELECT ON SYS.IDNSEQ$ TO <cmid_user>;
GRANT SELECT ON SYS.IND$ TO <cmid_user>;
GRANT SELECT ON SYS.INDPART$  TO <cmid_user>;
GRANT SELECT ON SYS.OBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.PARTOBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.RECYCLEBIN$ TO <cmid_user>;
GRANT SELECT ON SYS.TAB$ TO <cmid_user>;
GRANT SELECT ON SYS.TABCOMPART$  TO <cmid_user>;
GRANT SELECT ON SYS.TABPART$ TO <cmid_user>;
GRANT SELECT ON SYS.TABSUBPART$ TO <cmid_user>;

-- Also ensure that you have access to the following ALL_* views:
ALL_CONSTRAINTS
ALL_CONS_COLUMNS
ALL_ENCRYPTED_COLUMNS
ALL_INDEXES
ALL_IND_COLUMNS
ALL_OBJECTS
ALL_TABLES
ALL_TAB_COLS
ALL_TAB_PARTITIONS
ALL_USERS
```

Privileges for incremental load processing with query-based CDC

For an application ingestion and replication task that performs an incremental load or combined initial and incremental load using the query-based CDC method, ensure that the user has the following privileges at minimum:

```
GRANT CREATE SESSION TO <cmid_user>;

GRANT SELECT ON DBA_INDEXES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECT_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECTS TO cmid_user;
GRANT SELECT ON DBA_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_USERS TO <cmid_user>;
GRANT SELECT ON DBA_VIEWS TO <cmid_user>;  -- Only if you unload data from views

GRANT SELECT ANY TABLE TO <cmid_user>;
-or-
GRANT SELECT ON table TO <cmid_user>;         -- For each source table created by user

GRANT SELECT ON ALL_CONSTRAINTS TO <cmid_user>;
GRANT SELECT ON ALL_CONS_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_ENCRYPTED_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_IND_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_INDEXES TO <cmid_user>;
GRANT SELECT ON ALL_OBJECTS TO <cmid_user>;
GRANT SELECT ON ALL_TAB_COLS TO <cmid_user>;
GRANT SELECT ON ALL_USERS TO <cmid_user>;

GRANT SELECT ON "PUBLIC"."V$DATABASE"  TO cmid_user;
```

```
GRANT SELECT ON "PUBLIC"."V$CONTAINERS" TO cmid_user;
GRANT SELECT ON SYS.ATTRCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO <cmid_user>;
GRANT SELECT ON SYS.COL$ TO <cmid_user>;
GRANT SELECT ON SYS.COLTYPE$ TO <cmid_user>;
GRANT SELECT ON SYS.IND$ TO <cmid_user>;
GRANT SELECT ON SYS.IDNSEQ$ TO cmid_user;
GRANT SELECT ON SYS.OBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.RECYCLEBIN$ TO <cmid_user>;
GRANT SELECT ON SYS.TAB$ TO <cmid_user>;
```

Privileges for initial load processing

For an application ingestion and replication task that performs an initial load, ensure that the user has the following privileges at minimum:

```
GRANT CREATE SESSION TO <cmid_user>;

GRANT SELECT ON DBA_INDEXES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECT_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECTS TO cmid_user;
GRANT SELECT ON DBA_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_USERS TO <cmid_user>;
GRANT SELECT ON DBA_VIEWS TO <cmid_user>;  -- Only if you unload data from views

GRANT SELECT ANY TABLE TO <cmid_user>;
-or-
GRANT SELECT ON table TO <cmid_user>;         -- For each source table created by user

GRANT SELECT ON ALL_CONSTRAINTS TO <cmid_user>;
GRANT SELECT ON ALL_CONS_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_ENCRYPTED_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_IND_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_INDEXES TO <cmid_user>;
GRANT SELECT ON ALL_OBJECTS TO <cmid_user>;
GRANT SELECT ON ALL_TAB_COLS TO <cmid_user>;
GRANT SELECT ON ALL_USERS TO <cmid_user>;

GRANT SELECT ON "PUBLIC"."V$DATABASE"  TO cmid_user;
GRANT SELECT ON "PUBLIC"."V$CONTAINERS" TO cmid_user;
GRANT SELECT ON SYS.ATTRCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO <cmid_user>;
GRANT SELECT ON SYS.COL$ TO <cmid_user>;
GRANT SELECT ON SYS.COLTYPE$ TO <cmid_user>;
GRANT SELECT ON SYS.IND$ TO <cmid_user>;
GRANT SELECT ON SYS.IDNSEQ$ TO cmid_user;
GRANT SELECT ON SYS.OBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.RECYCLEBIN$ TO <cmid_user>;
GRANT SELECT ON SYS.TAB$ TO <cmid_user>;n
```

Oracle privileges for Amazon RDS for Oracle

If you are using an application ingestion and replication task that has an SAP ECC source with Amazon RDS for Oracle as the database, you must grant certain privileges to the Application Ingestion and Replication user.

Important: You must log in to Amazon RDS under the master username to run GRANT statements and procedures.

Grant the SELECT privilege, at minimum, on objects and system tables that are required for CDC processing to the Application Ingestion and Replication user (*cmid_user*). Some additional grants are required in certain situations.

Use the following GRANT statements:

```
GRANT SELECT ON "PUBLIC"."V$ARCHIVED_LOG" TO "cmid_user";

GRANT SELECT ON "PUBLIC"."V$DATABASE" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$LOG" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$LOGFILE" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$TRANSPORTABLE_PLATFORM" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$THREAD" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$DATABASE_INCARNATION" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$TRANSACTION" TO "cmid_user";

GRANT SELECT ON "SYS"."DBA_CONS_COLUMNS" TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_CONSTRAINTS" TO "cmid_user";
GRANT SELECT ON DBA_INDEXES TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_LOG_GROUP_COLUMNS" TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_TABLESPACES" TO "cmid_user";

GRANT SELECT ON "SYS"."OBJ$" TO "cmid_user";
GRANT SELECT ON "SYS"."TAB$" TO "cmid_user";
GRANT SELECT ON "SYS"."IND$" TO "cmid_user";
GRANT SELECT ON "SYS"."COL$" TO "cmid_user";

GRANT SELECT ON "SYS"."PARTOBJ$" TO "cmid_user";
GRANT SELECT ON "SYS"."TABPART$" TO "cmid_user";
GRANT SELECT ON "SYS"."TABCOMPART$" TO "cmid_user";
GRANT SELECT ON "SYS"."TABSUBPART$" TO "cmid_user";
COMMIT;

/* For combined load jobs:*/
GRANT EXECUTE ON DBMS_FLASHBACK TO "cmid_user";

/*To provide read access to the Amazon RDS online and archived redo logs:*/
GRANT READ ON DIRECTORY ONLINELOG_DIR TO "cmid_user";
GRANT READ ON DIRECTORY ARCHIVELOG_DIR TO "cmid_user";
```

Additionally, log in as the master user and run the following Amazon RDS procedures to grant the SELECT privilege on some more objects:

```
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECT_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_VIEWS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
/begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_USERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
```

```
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$CONTAINERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$PARAMETER',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$SPPARAMETER',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$STANDBY_LOG',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$VERSION',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_CONS_COLUMNS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_CONSTRAINTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_OBJECTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TAB_PARTITIONS',
p_grantee => 'cmid_user',
```

```
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_USERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rd 'ALL_TABLES',
p_grantee => 'sadmin_util.grant_sys_object(
p_obj_name => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TAB_PARTITIONS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ATTRCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'COLTYPE$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'INDPART$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'IDNSEQ$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CDEF$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
```

```
p_obj_name => 'RECYCLEBIN$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
/* Only required for RDS21 which supports PDB*/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$PDBS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
```

Configuring BFILE access to Oracle redo logs in the Oracle file system

If you store redo logs in the local Oracle server file system and want to access the logs by using Oracle directory objects with BFILEs, perform the following configuration tasks:

Complete the following usual Oracle source preparation tasks, which are not specific to BFILE access:

- Define the ORACLE_HOME environment variable on the Linux or Windows system where the Secure Agent runs for Application Ingestion and Replication to use the Oracle Call Interface (OCI) to communicate with the Oracle source database.
- Make sure the Application Ingestion and Replication user has the Oracle privileges that are required for the application ingestion and replication incremental load processing. For more information, see "Oracle privileges" on page 27.
- Enable ARCHIVELOG mode for the Oracle database.
- Define the archive log destination.

  Note: For BFILE access, use a specific archive log destination directory. Do not use the Oracle USE_DB_RECOVERY_FILE_DEST keyword to store archive logs by date in a Fast Recovery Area (FRA).
- Enable Oracle minimal global supplemental logging on the source database.
- If your Oracle source tables have primary keys, ensure that supplemental logging is enabled for all primary key columns. For source tables that do not have primary keys, ensure that supplemental logging is enabled for all columns from which change data will be captured.

  Note: When you create an application ingestion and replication task, you have the option of generating a script that implements supplemental logging for all columns or only primary key columns for the selected source tables.
- Ensure that the Oracle MAX_STRING_SIZE initialization parameter is *not* set to EXTENDED. If it is set to EXTENDED, Application Ingestion and Replication will not be able to replicate inserts and updates for tables containing columns defined with large (extended size) VARCHAR2, NVARCHAR2, or RAW columns.

Additionally, for BFILE access, perform the following steps:

1. Query the Oracle database for the online and archived redo log locations in the Oracle server file system. You can use the following example queries:
   To get location of the online redo logs:

   ```
   select * from v$logfile;
   ```

   To get the archive log destination:

   ```
   select dest_id, dest_name, destination, status from V$ARCHIVE_DEST;
   ```

2. Create the ONLINELOG_DIR and ARCHIVELOG_DIR directory objects that point to the locations of log files from step 1. An Oracle directory object specifies a logical alias name for a physical directory on the Oracle server file system under which the log files to be accessed are located. For example:

```
CREATE DIRECTORY ONLINELOG_DIR AS '/u01/oracle/data';
CREATE DIRECTORY ARCHIVELOG_DIR AS '/u01/oracle/archivedata';
```

   Note: If you plan to set the reader mode to ARCHIVEONLY in the Oracle Database Ingestion connection to read changes only from archive logs, you do not need to create an ONLINELOG_DIR directory or directory object.

   The Oracle database does not verify that the directories you specify exist. Make sure you specify valid directories that exist in the Oracle file system.

3. To verify that the directory objects were created with the correct file system paths for the redo logs, issue a select statement such as:

```
select * from all_directories;
OWNER    DIRECTORY_NAME      DIRECTORY_PATH
-------- ------------------- ---------------------------------
SYS      ARCHIVELOG_DIR      /u01/oracle/data/JO112DTL
SYS      ONLINELOG_DIR       /u01/oracle/data/JO112DTL
```

4. Grant read and write access on the ONLINELOG_DIR and ARCHIVELOG_DIR directory objects to the Application Ingestion and Replication user who is specified in the Oracle Database Ingestion connection properties. For example:

```
grant read on directory "ARCHIVELOG_DIR" to "cmid_user";
grant read on directory "ONLINELOG_DIR" to "cmid_user";
```

5. In the Oracle Database Ingestion connection properties, select the **BFILE Access** check box.

Oracle archive log retention considerations

Application ingestion and replication incremental and combined initial and incremental load jobs must be able to access transaction data in Oracle online and archive redo logs. If the logs are not available, application ingestion and replication jobs end with an error.

Typically, the Oracle DBA sets the archive log retention period based on your oganization's particular business needs and Oracle environment. Make sure that the source archive logs are retained for the longest period for which you expect change capture to be stopped or latent, plus about 1 hour, so that the logs will be available for restart processing.

To determine if the current log retention policy in your environment is sufficient to accommodate application ingestion and replication change capture processing, consider the following factors:

- How long are Oracle transactions typically open on a source?
- What is the longest period of time that change capture is allowed to be down or latent, accounting for weekends and holidays?
- What is the replication latency from source to target?
- Do you run application ingestion and replication jobs based on a schedule? If yes, what type of schedule?
- Is the pwx.cdcreader.oracle.option.additional ageOutPeriod=*minutes* custom property set on the **Source** page of task wizard?

  Note: This property specifies the age at which outstanding UOWs without change records of CDC interest are removed from the calculation of the next restart point. You can use the property to prevent CDC failures that might occur if you shut down and then restart capture processing while the transaction is outstanding and the redo log in which the UOW started is not available.
- What is the redo generation rate?
- Do you ship copies of archive logs to a secondary system?

If the archive logs are not available when you need to restart capture processing in the logs, you can ask your DBA to restore them and to modify the retention period if necessary. Otherwise, perform another initial load to re-materialize the target and then start incremental change data processing again. However, in this case, you might lose some changes.

Schema change handling

You can choose how the SAP Mass Ingestion connector handles schema changes that you make to some data object types.

Schema change handling for an SAP S4 HANA source

By default, if you make changes to the schema, the SAP Mass Ingestion connector detects the schema changes and generates an alert message.

Note: You can disable the schema detection using a custom property. For more information, contact Informatica Global Customer Support.

A schema change includes one or more of the following changes to the data object:

- Fields are added
- Fields are deleted
- Fields are renamed
- Field data type, precision, or scale is updated

You can handle the schema changes based on the following conditions:

- If you make a schema change to the data object while an incremental load job is in the **Up and Running** status, the SAP Mass Ingestion connector generates an alert message. When the alert appears, the processing of the data object stops, and you must create a new initial job with the changed data object. Run the new incremental job after the initial job completes successfully.
- If you make a schema change to the data object while an incremental load job is in the **Stopped** status, you must drop the trigger and the shadow table on the data object where the changes were made. Activate the data object through the SAP user interface and create a new initial job with the updated data object. After the initial job completes successfully, run the new incremental job.

Note: Run the incremental job immediately after the initial job completes successfully to prevent data loss.

Schema change handling for an SAP ECC source

If you make schema changes to the data object, the processing of the data object stops, and the SAP Mass Ingestion connector generates an alert message. Further changes to the data object are not captured.

A schema change includes one or more of the following changes to the data object:

- Fields are added
- Fields are deleted
- Fields are renamed
- Field data type, precision, or scale is updated

You can handle the schema changes based on the following conditions:

- If you make a schema change to the data object for a combined load job, the SAP Mass Ingestion connector generates an alert message. You can then resynchronize the data object.

- If you make a schema change to the data object for an incremental load job, you must create and run an initial job with the changed data object. After the initial job completes successfully, run the new incremental job with the changed data object.

Note: Run the incremental job immediately after the initial job completes successfully to prevent data loss.

SAP OData V2 sources

Consider the following guidelines when you use Microsoft Dynamics 365 sources:

- For incremental and combined load tasks, only those SAP OData entities with delta extraction enabled inside the OData service appear in the source table selection in your task. Their associated delta link tables are not displayed. Your SAP administrator must ensure that the OData services are correctly configured in SAP Gateway Builder (SEGW) with delta extraction enabled for these tables.
  For initial load tasks, there are no such restrictions, and all relevant SAP OData tables appear for selection.

  For more information about creating and configuring OData services with ODP, see ODP-based data extraction through OData in the SAP documentation.
- When you run an incremental load job for a task that has an SAP OData V2 source and contains a DF16_DEC column in the source table, the SAP OData service maps the DF16_DEC type to a decimal with lower precision than the actual SAP column precision. As a result, data values that are valid in SAP exceed the OData-reported precision and cause the job to fail with a precision error.

ServiceNow sources

Consider the following guidelines when you use ServiceNow sources:

- The first time an incremental load job runs to capture change data for a ServiceNow source, the job retrieves and loads only the change records that are created after the date and time specified in the associated ingestion task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- The data stored in some ServiceNow objects can be retrieved only if the API client is configured with the maint and nobody roles. Application Ingestion and Replication does not retrieve data from such source objects that require the ServiceNow Mass Ingestion connection to be configured with the maint and nobody roles.
- You must have Admin privileges to access table APIs or aggregate APIs.
- Incremental load jobs configured for ServiceNow sources do not capture change data for the objects that do not contain the sys_updated_on field.
- Incremental load jobs and combined initial and incremental load jobs configured for ServiceNow sources do not detect and replicate the renaming of source fields.
- Application Ingestion and Replication retains the masking configuration of source fields on the target. For example, if a masked field on the source contains the value 1234****, the corresponding field on the target also contains the value 1234****.
- In the Secure Agent configuration properties, set the metadata_manager.jvm.ai.servicenow.enable.cdc.using.inherited.columns and task_container.jvm.ai.servicenow.enable.cdc.using.inherited.columns custom properties to **true** to include the child tables inheriting CDC columns from the parent table in the task.

- Set the ai.servicenow.enable.additional.datatypes custom property to true, to enable the following ServiceNow data types:
  - Audio
  - Basic Image
  - Encrypted Text
  - FX Currency
  - Image
  - Journal List
  - List
  - Name-Value Pairs
  - Reference
  - Video
  - Wiki

  Note: For the **Reference** data type, when the `ai.servicenow.enable.additional.datatypes` custom property is set to `true`, the `columnName_value`, `columnName_display_value`, and `columnName_link` attributes are available in the target. If the `ai.servicenow.enable.additional.datatypes` custom property is set to `false`, only the `columnName` attribute is available in the target.

Workday sources

Consider the following guidelines when you use Workday sources:

Application Ingestion and Replication provides you the option to extract Workday data through the following web services:

- **Workday Web Services**: Provides access to Workday data through SOAP APIs.
- **Workday Report-as-a-Service (RaaS)**: Provides access to data in the custom objects and fields through custom reports.

The option to select the required web service appears on the **Source** tab of the application ingestion and replication task wizard.

Guidelines for Workday Web Services

- You can use Workday Web Services to ingest the data of Workday Human Capital Management (HCM). On the **Source** tab of the application ingestion and replication task wizard, you can select the specific HCM services that you want to replicate on your target.
- Application Ingestion and Replication replicates the data of only the source operations whose name start with `Get_`.
- Application ingestion and replication jobs retrieve the source data in an XML structure and then writes the data to the target as a single object in JSON or XML format. When you configure an application ingestion and replication job, on the **Source** tab of the task wizard, you can specify the format of the target data. On the target, each record contains the following fields:
  - **WID**: Stores the unique identifier or primary key of the record.
  - **Data**: Stores the content of the record in JSON or XML format.
- Application Ingestion and Replication retains the hierarchical structure of source data on the target.
- Application Ingestion and Replication does not propagate the data stored in custom objects.

- When an application ingestion and replication job ingests an operation to the target, the operation is renamed in the following format: `<Service_Name>__<Operation_Name>`
- Workday might not allow initial load jobs to replicate the data of the Get_Budgest_Pools operation.
- Initial load jobs configured for a Workday source and a Microsoft Azure Data Lake Storage Gen2 target do not generate schema files for the following operations if they do not contain any record:
  - Benefits_Administration__Get_ACA_1095-C_Forms_Data
  - Compensation_Review__Get_Budget_Pools
  - Compensation_Review__Get_Employee_Awards
  - Recruiting__Get_Assess_Candidate
  - Recruiting__Get_Interview_Feedbacks
- The first time an incremental load job runs to capture change data for a Workday source, the job retrieves and loads only the change records that are created after the date and time specified in the application ingestion and replication task. You can specify the date and time when you configure the source in the application ingestion and replication task wizard. However, when you resume a stopped or aborted job, the job begins propagating change data from where it last left off.
- Incremental load jobs can automatically capture the insert and update operations performed on the source. The jobs perform upsert operations on the target to replicate the insert and update operations performed on the source data. The application ingestion and replication task wizard does not provide the schema drift options for Workday sources because the upsert operations automatically replicate all changes made to the source schema.
- Application Ingestion and Replication can capture change data for the following Workday services and operations:

Service	Operations
Human_Resources	Get_Job_Profiles
	Get_Organizations
	Get_Workers
Recruiting	Get_Evergreen_Requisitions
	Get_Job_Requisitions
	Get_Organizations
	Get_Positions
Staffing	Get_Organizations
	Get_Positions
	Get_Workers

Guidelines for Workday RaaS

- You can use Workday RaaS only in initial load tasks.

- On the **Source** tab of the application ingestion and replication task wizard, you can choose to read data from one or more custom reports. If you choose to extract multiple reports, create a CSV file on the Secure Agent host and list the name of the reports in the first column of the file.
- Application ingestion and replication jobs configured to use Workday RaaS cannot detect the changes made to the source schema.

Application Ingestion and Replication target types - preparation and usage

The targets that Application Ingestion and Replication support depend on the sources specified for the application ingestion and replication tasks.

The following table shows the target types that are supported (S) for each load type:

Target Type	Initial Load	Incremental Load	Initial and Incremental Loads
Amazon Redshift	S	S	S
Amazon S3	S	S	S
Apache Kafka	-	S	-
Databricks	S	S	S
Google BigQuery	S	S	S
Google Cloud Storage	S	S	S
Microsoft Azure Data Lake Storage Gen2	S	S	S
Microsoft Azure SQL Database	S	S	S
Microsoft Azure Synapse Analytics	S	S	S
Microsoft Fabric Data Warehouse	S	S	S
Microsoft Fabric Lakehouse	S	S	S
Microsoft Fabric OneLake	S	S	S
Microsoft SQL Server	S	S	S
Oracle	S	S	S
Oracle Cloud Infrastructure (OCI) Object Storage	S	S	S
PostgreSQL	S	S	S

Target Type	Initial Load	Incremental Load	Initial and Incremental Loads
Salesforce Data 360	S	S	S
Snowflake	S	S	S

The following table lists the targets that Application Ingestion and Replication support for each source type:

Source type	Supported target type
Marketo	Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake
Microsoft Dynamics 365	Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Azure SQL Database, Microsoft SQL Server, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake
NetSuite	Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake
Oracle Fusion Cloud	- Using REST API - Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Microsoft Azure SQL Database, Microsoft SQL Server, Oracle, Oracle Cloud Infrastructure Object Storage, Salesforce Data 360, and Snowflake - Using BICC - Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Microsoft Azure SQL Database, Microsoft SQL Server, Oracle, Oracle Cloud Infrastructure Object Storage, Salesforce Data 360, and Snowflake
Salesforce	Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Oracle, Microsoft Azure SQL Database, Microsoft SQL Server, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, PostgreSQL, and Snowflake
Salesforce Marketing Cloud	Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake

Source type	Supported target type
SAP	- Using SAP ODP Extractor connector - Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake - Using SAP Mass Ingestion connector - Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Azure SQL Database, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Microsoft SQL Server, Oracle, Oracle Cloud Infrastructure Object Storage, PostgreSQL, Salesforce Data 360, and Snowflake - Using SAP OData V2 connector - Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, PostgreSQL, Salesforce Data 360, and Snowflake
ServiceNow	Amazon Redshift, Apache Kafka, Databricks, Amazon S3, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake
Workday	- SOAP. Amazon Redshift, Amazon S3, Apache Kafka, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake - RaaS. Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric Lakehouse, Microsoft Fabric OneLake, Oracle, Oracle Cloud Infrastructure Object Storage, and Snowflake (for initial load)

To determine the connectors to use for the target types, see *Connectors and Connections* > *Application Ingestion and Replication connectors*.

Amazon Redshift targets

Consider the following guidelines when you use Amazon Redshift targets:

Target preparation

Ensure that the Redshift database user has the following minimum permissions required for Application Ingestion and Replication:

```
grant create on database <database_name> to <username>;
grant create on schema <schema_name> to <username>;
```

If you use Redshift role-based access control (RBAC), grant these permissions to the appropriate roles. With these permissions, users can perform all Application Ingestion and Replication operations, including deploying tasks, having target tables created at runtime, and redeploying, resynchronizing, and resuming jobs.

To check if a user already has the permissions required to access the database and schema, use the following system functions:

```
select has_database_privilege('<username>', '<databaseName>', 'create');
select has_schema_privilege('<username>', '<schemaName>', 'create');
```

Note: If you create a new database user, you must explicitly grant the permissions needed to access the existing database, schema, and table objects to that user.

Usage considerations

- Before writing data to Amazon Redshift target tables, application ingestion and replication jobs stage the data in an Amazon S3 bucket. You must specify the name of the bucket when you configure the application ingestion and replication task. The ingestion jobs use the COPY command to load the data from the Amazon S3 bucket to the Amazon Redshift target tables. For more information about the COPY command, see the Amazon Web Services documentation.
- When you define a connection for an Amazon Redshift target, provide the access key and secret access key for the Amazon S3 bucket in which you want the application ingestion and replication jobs to stage the data before loading it to the Amazon Redshift target tables.
- When you ingest data from a source to an Amazon Redshift target, the application ingestion and replication job fails if the data source contains more than 32 data fields or columns that are defined as primary keys.
- Incremental load jobs and combined initial and incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information. The data in the recovery table prevents the jobs that are restarted after a failure from propagating previously processed data again. The recovery table is generated in the schema of the target tables.

Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, and Oracle Cloud Object Storage targets

Consider the following guidelines when you use Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, and Oracle Cloud Infrastructure (OCI) Object Storage targets:

- When you configure an application ingestion and replicationtask for an Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, or Oracle Cloud Object Storage target, you can select CSV, Avro, or Parquet as the format for the output files that contain the source data to be applied to the target.
- When you write to an Amazon S3 target, you have the option to select the **OpenTable** format, with **Parquet** set as the default output type. When the task runs, the data is organized within the S3 storage according to the namespace you specify in the target properties. The **Warehouse Base Directory** defines the folder path on S3 where the target files are created. The **Connection Directory as Parent** option determines whether the folder path from the connection is appended to the Warehouse Base Directory when forming the complete file path on S3:
  - If enabled, target files are saved to the following directory: **<Directory specified in S3 connection>/<Warehouse_Base_Directory>/<files>**
  - If not enabled, target files are saved directly to the following directory: **<Warehouse_Base_Directory>/<files>**
- If you select **CSV** as the output file format, Application Ingestion and Replication creates the following files on the target for each source field:
  - schema.ini file that describes the schema of the field. The file also includes some settings for the output file on the target.
  - Output files that contain the data stored in the source field. Application Ingestion and Replication names the output files based on the name of the source field with an appended date and time.

The schema.ini file lists the sequence of columns for the rows in the corresponding output file. The following table describes the columns in the schema.ini file:

Column	Description
ColNameHeader	Indicates whether the source data files include column headers.
Format	Format of the output files. Application Ingestion and Replication uses a comma (,) to delimit column values.
CharacterSet	Character set that is used for the corresponding output file. By default, Application Ingestion and Replication generates the files in the UTF-8 character set.
COL<*sequence_number*>	Name and data type of the source field. **Note:** If you selected any of the **Add Operation...** properties under **Advanced** on the **Target** page of the task wizard, the list of columns includes metadata columns for the operation type, time, owner, or transaction ID.

Note: You must not edit the schema.ini file.

- If you select the Avro output format, you can select an Avro format type, a file compression type, an Avro data compression type, and the directory that stores the Avro schema definitions generated for each source table. The schema definition files have the following naming pattern: *schemaname_tablename*.txt.
- If you select the Parquet output format, you can optionally select a compression type that Parquet supports.
- For application ingestion and replication tasks configured for Microsoft Azure Data Lake Storage Gen2, Application Ingestion and Replication creates an empty directory on the target for each empty source field.
- For Amazon S3 targets, if you do not specify an access key and secret key in the connection properties, Application Ingestion and Replication tries to find the AWS credentials by using the default credential provider chain that is implemented by the DefaultAWSCredentialsProviderChain class. For more information, see the Amazon Web Services documentation.
- When an incremental load job or combined initial and incremental load job configured for a target that uses the CSV output format propagates an Update operation that changed primary key values on the source, the job performs a Delete operation on the associated target row and then performs an Insert operation on the same row to replicate the change made to the source object. The Delete operation writes the before image to the target and the subsequent Insert image writes the after image to the target.

  For Update operations that do not change primary key values, application ingestion and replication jobs process each Update operation as a single operation and writes only the after image to the target. Application Ingestion and Replication processes each Update operation performed on the source as a Delete operation followed by an Insert operation on the target.
- When writing the NUMBER(p,s) data type to Amazon S3 with the Apache Iceberg format, the NUMBER(p,s) type maps to Decimal in Apache Iceberg, which allows a maximum precision of 38. If the precision of NUMBER(p,s) from the source exceeds 38, the task fails.

Apache Kafka targets

Consider the following guidelines when you use Apache Kafka targets:

- Application Ingestion and Replication supports Apache Kafka, Confluent Kafka, Amazon Managed Streaming for Apache Kafka (MSK), and Kafka-enabled Azure Event Hubs as targets for incremental load jobs. All of these Kafka target types use the Kafka connection type.

  To indicate the Kafka target type, you must specify Kafka producer properties in the task definition or Kafka connection properties. To specify these properties for a task, enter a comma-separated list of *key*:*value* pairs in the **Producer Configuration Properties** field on the **Target** page of the task wizard. To specify the producer properties for all tasks that use a Kafka connection, enter the list of properties in the **Additional Connection Properties** field in the connection properties. You can override the connection-level properties for specific tasks by also defining producer properties at the task level. For more information about producer properties, see the Apache Kafka, Confluent Kafka, Amazon MSK, or Azure Event Hubs for Kafka documentation.

- If you select **AVRO** as the output format for a Kafka target, Application Ingestion and Replication generates a schema definition file for each table with a name in the following format:

  ```
  schemaname_tablename.txt
  ```

- If a source schema change is expected to alter the target in an incremental load job, Application Ingestion and Replication regenerates the Avro schema definition file with a unique name that includes a timestamp:

  ```
  schemaname_tablename_YYYYMMDDhhmmss.txt
  ```

  This unique naming pattern preserves older schema definition files for audit purposes.

- If you have a Confluent Kafka target that uses Confluent Schema Registry to store schemas, you must configure the following settings on the Target page of the task wizard:

  - In the **Output Format** field, select **AVRO**.

  - In the **Avro Serialization Format** field, select **None**.

- You can specify Kafka producer properties in either the **Producer Configuration Properties** field on the **Target** page of the task wizard or in the **Additional Connection Properties** field in the Kafka connection properties. Enter property=value pairs that meet your business needs and that are supported by your Kafka vendor.
  For example, if you use Confluent Kafka, you can use the following entry in either the Producer Configuration Properties field or Additional Connection Properties field to specify the Schema Registry URL and enable basic authentication:

  ```
  schema.registry.url=http://schema-registry:8081,
  key.serializer=org.apache.kafka.common.serialization.StringSerializer,
  value.serializer=io.confluent.kafka.serializers.KafkaAvroSerializer,
  basic.auth.credentials.source=USER_INFO,
  basic.auth.user.info=myname:mypassword
  ```

  If you use Amazon MSK, you can use the following **Additional Connection Properties** entry to enable IAM role authentication for access to Amazon MSK targets:

  ```
  security.protocol=SASL_SSL,sasl.mechanism=AWS_MSK_IAM,sasl.jaas.config=software.amazon
  .msk.auth.iam.IAMLoginModule
  required;,sasl.client.callback.handler.class=software.amazon.msk.auth.iam.IAMClientCal
  lbackHandler
  ```

  Ensure that you enable IAM role authentication on the Amazon EC2 instance where the Secure Agent is installed.

  For more information about Kafka properties, see the documentation of your Kafka vendor.

- Application Ingestion and Replication incremental load jobs can replicate change data to Kafka targets that support SASL_SSL secured access, including Confluent Kafka, Amazon MSK, and Azure Event Hubs targets. In Administrator, you must configure a Kafka connection that includes the appropriate properties

in the **Additional Connection Properties** field. For example, for Azure Event Hubs, you could use the following **Additional Connection Properties** entry to enable SASL_SSL:

```
bootstrap.servers=NAMESPACENAME.servicebus.windows.net:9093
security.protocol=SASL_SSL
sasl.mechanism=PLAIN
sasl.jaas.config=org.apache.kafka.common.security.plain.PlainLoginModule required
username="$ConnectionString" password="{YOUR.EVENTHUBS.CONNECTION.STRING}";
```

- Beginning with the July 2025 release, newly deployed jobs send checkpoint information in the Kafka header of each message. When the job is restarted, checkpoint information is retrieved from the Kafka header. This behavior allows newly deployed jobs that have a Kafka target to run on another Secure Agent in the Secure Agent group, which provides for high availability. Prior to the July 2025 release, checkpoint information was persisted only in the checkpoint file in the Secure Agent.
  If you use Kafka Access Control List (ACLs) or other controls to restrict access to the Kafka target by group name, be aware that Application Ingestion and Replication uses the default consumer group name of infaGroup to support high availability for the Kafka target. If you use another existing consumer group name, you must specify it in the Kafka group.id configuration property to enable application ingestion and replication jobs to read topics. If you do not use the default infaGroup name or specify another existing group name, application ingestion and replication tasks will fail with the following error:

  ```
  org.apache.kafka.common.errors.GroupAuthorizationException: Not authorized to access
  group: <group_name>
  ```

Databricks targets

Consider the following guidelines when you use Databricks targets:

- When you use a Databricks target for the first time, perform the following steps before you configure an application ingestion and replication task for the target:
  1. In the Databricks connection properties, set the **JDBC Driver Class Name** property to `com.databricks.client.jdbc.Driver`.
  2. On Windows, install Visual C++ Redistributable Packages for Visual Studio 2013 on the computer where the Secure Agent runs.
- For incremental load jobs, you must enable Change Data Capture (CDC) for all source fields.
- You can access Databricks tables created on top of the following storage types:

  - Microsoft Azure Data Lake Storage (ADLS) Gen2

  - Amazon Web Services (AWS) S3

  The Databricks connection uses a JDBC URL to connect to the Databricks cluster. When you configure the target, specify the JDBC URL and credentials to use for connecting to the cluster. Also define the connection information that the target uses to connect to the staging location in Amazon S3 or ADLS Gen2.
- Before writing data to Databricks target tables, application ingestion and replication jobs stage the data in an Amazon S3 bucket or ADLS directory. You must specify the directory for the data when you configure the application ingestion and replication task.

  Note: Application Ingestion and Replication does not use the **ADLS Staging Filesystem Name** and **S3 Staging Bucket** properties in the Databricks connection properties to determine the directory.
- Application Ingestion and Replication uses jobs that run once to load data from staging files on AWS S3 or ADLS Gen2 to external tables.
  By default, Application Ingestion and Replication runs jobs on the cluster that is specified in the Databricks connection properties. If you want to run the jobs on another cluster, set the dbDeltaUseExistingCluster custom property to false on the **Target** page in the application ingestion and replication task wizard.

- If the cluster specified in the Databricks connection properties is not up and running, the application ingestion and replication job waits for the cluster to start. By default, the job waits for 10 minutes. If the cluster does not start within 10 minutes, the connection times out and deployment of the job fails.
  If you want to increase the timeout value for the connection, set the dbClusterStartWaitingTime custom property to the maximum time in seconds for which the ingestion and replication job must wait for the cluster to be up and running. You can set the custom property on the **Target** page in the application ingestion and replication task wizard.
- By default, Application Ingestion and Replication uses the Databricks COPY INTO feature to load data from the staging file to Databricks target tables. You can disable it for all load types by setting the writerDatabricksUseSqlLoad custom property to false on the **Target** page in the application ingestion and replication task wizard.
- If you use an AWS cluster, you must specify the **S3 Service Regional Endpoint** value in the Databricks connection properties. For example:

  ```
  s3.us-east-2.amazonaws.com
  ```

  Before you can test a Databricks connection. you must specify the JDBC URL in the **SQL Endpoint JDBC URL** field in the Databricks connection properties. After you test the connection, remove the **SQL Endpoint JDBC URL** value. Otherwise, when you define an application ingestion and replication task that uses the connection, a design-time error occurs because Application Ingestion and Replication tries to use the JDBC URL as well as the required **Databricks Host**, **Cluster ID**, **Organization ID**, and **Databricks Token** values to connect to target, resulting in login failures.
- Processing of Rename Field operations on Databricks target tables, without the need to rewrite the underlying Parquet files, requires the Databricks Column Mapping feature with Databricks Runtime 10.2 or later. If you set the Rename Field option to **Replicate** on the **Schedule and Runtime Options** page in the task wizard, you must alter the generated target table to set the following Databricks table properties after task deployment and before you run the job:

  ```
  ALTER TABLE <target_table> SET TBLPROPERTIES (
  'delta.columnMapping.mode' = 'name',
  'delta.minReaderVersion' = '2',
  'delta.minWriterVersion' = '5')
  ```

  These properties enable the Databricks Column Mapping feature with the required reader and writer versions. If you do not set these properties, the application ingestion and replication job will fail.
- Application ingestion and replication that have Databricks targets can get schema information for generating the target tables from Databricks Unity Catalog. To enable access to information in a Unity Catalog, specify the catalog name in the **Catalog Name** field in the Databricks connection properties. The catalog name is appended to the **SQL Warehouse JDBC URL** value for a data warehouse.

  Note: Catalog use is optional for SQL warehouses and does not apply to job clusters.

  If you use Unity Catalog, a personal storage location is automatically provisioned. To use the personal staging location, in the **Staging Environment** field of the connection properties, select **Personal Staging Location**. The Parquet data files for ingestion jobs can then be staged to the local personal storage location, which has a data retention period of 7 days. By default, the staging location is `stage://tmp/<user_name>` in the root AWS or Azure location. The <user_name> is taken from the **Database Token** connection property. This user must have read and write access to the personal staging location.
- You can stage data in a Volume in Databricks. Volumes are Unity Catalog objects used to manage and secure non-tabular datasets such as files and directories. Volumes can be of the following types:

  - **Managed volumes**
    - The data is stored in a location fully controlled and managed by Databricks. If you use volumes as the staging environment, you can specify the volume path where staging files are generated in the **Volume Path** field of the connection properties. Files will be written only up to the `dataDirectory` level.

- If you not provide a path to the files within a volume, a new volume is generated automatically in the format:

  `<Volumes/<catalogName>/<databaseName>/<volume_identifier>`

- The generated volume is deleted when an initial load job is successfully completed. Only a volume that is empty is automatically deleted. The volume is not deleted when the initial load phase of a combined load job is completed.

- **Unmanaged / external volumes**

- The data is stored in user-managed storage locations in Amazon S3 or Microsoft Azure Data Lake Storage, which you mount or access from Databricks.

- You must provide the volume path where staging files are generated in the **Volume Path** field of the connection properties. Files will be written only up to the `dataDirectory` level.

- The volume is not deleted when an initial load job or the initial load phase of a combined load job is completed.

- If you generated Databricks unmanaged tables as the target tables and no longer need a target unmanaged table, use a SQL DROP TABLE statement to delete the table from the target database. You must not manually delete the external directory in Amazon S3 or Azure Data Lake Storage for the unmanaged table. If you do so and try to deploy another job that uses that table, the deployment fails with a Metadata Handler error.

Google BigQuery targets

Consider the following guidelines when you use Google BigQuery targets:

- When you use a Google BigQuery target for the first time, perform the following steps before you configure an application ingestion and replication task for the target:

  1. Download the Google BigQuery JDBC driver SimbaJDBCDriverforGoogleBigQuery42_1.6.1.1002 from the Google Cloud website.
  2. Copy only the GoogleBigQueryJDBC42.jar file from the installation zip to the following directory:

     `<Secure_Agent_installation_directory>/apps/Database_Ingestion/ext/`

     Note: If other Google BigQuery driver .jar files exist in that directory, remove them. They're now bundled with Application Ingestion and Replication.
  3. Restart the Secure Agent.

- You must have a service account in your Google account to access Google BigQuery and Google Cloud Storage.

- Ensure that you have the client_email, project_id, private_key, and region ID values for the service account. Enter the values in the corresponding **Service Account ID**, **Project ID**, **Service Account Key**, and **Region ID** connection properties when you create a Google BigQuery connection.

- If you want to configure a timeout interval for a Google BigQuery connection, specify the timeout interval property in the **Provide Optional Properties** field of the connection properties. Use the following format:

  `"timeout": "<timeout_interval_in_seconds>"`

- You must have read and write access to the following entities:

  - Google BigQuery datasets that contain the target tables.

  - Google Cloud Storage path where Application Ingestion and Replication creates the staging file.

- Application ingestion and replication jobs configured for Google BigQuery targets do not replicate the modification and renaming of source fields on the target.

- For incremental load tasks, you must enable source database Change Data Capture (CDC) on all source fields.

- You must have the following permissions to write data to a Google BigQuery table:
  - bigquery.datasets.get
  - bigquery.datasets.getIamPolicy
  - bigquery.models.*
  - bigquery.routines.*
  - bigquery.tables.create
  - bigquery.tables.delete
  - bigquery.tables.export
  - bigquery.tables.get
  - bigquery.tables.getData
  - bigquery.tables.list
  - bigquery.tables.update
  - bigquery.tables.updateData
  - bigquery.tables.updateTag
  - resourcemanager.projects.get
  - resourcemanager.projects.list
  - bigquery.jobs.create
- When you deploy application ingestion and replication jobs, Application Ingestion and Replication generates Google BigQuery target tables clustered by primary key or unique key columns, by default. Each key column must have one of the following data types that Google BigQuery supports for clustering:
  - STRING
  - INT64
  - NUMERIC
  - BIGNUMERIC
  - DATE
  - DATETIME
  - TIMESTAMP
  - BOOL
  - GEOGRAPHY

  If any column in the primary key or unique key has an unsupported data type, that column is skipped during clustering. For example, if the primary key contains the C1, C2, C3, C4, C5 columns and C2 has an unsupported data type, the target table is created with the C1, C3, C4, and C5 columns in the CLUSTER BY clause.
- By default, Data Ingestion and Replication uses the project ID defined in the Google BigQuery V2 connection properties to bill for query execution. You can override the billing project at the task level by using the writerGbqBillingProjectId custom property under **Advanced Target Properties** on the **Target Details** page.
  When you use the property, the project ID defined in the connection properties continues to serve as the data project where the target datasets and tables reside, but query costs are billed to the specified project instead.

Microsoft Azure Synapse Analytics targets

Consider the following guidelines when you use Microsoft Azure Synapse Analytics targets:

- To deploy and run an application ingestion and replication task with a Microsoft Azure Synapse Analytics target, the target connection must specify a database user who has the CONTROL permission on the target database. To grant the CONTROL permission to the user, use the following SQL statements:

  ```
  USE database_name;
  GRANT CONTROL TO user_name;
  ```

  The CONTROL permission is required for initial load, incremental load, and combined initial and incremental load jobs. The permission allows Application Ingestion and Replication to create target tables and database objects such as external data source, external file format, and database scoped credential objects if they do not exist in the database. The CONTROL permission is specifically required for creating external data source and database scoped credential objects.

  Note: You must manually create the master key. To create the master key, you must have the CONTROL permission on the database.

- Application ingestion and replication jobs first send data to a Microsoft Azure Data Lake Storage Gen2 staging file before writing the data to Microsoft Azure Synapse Analytics target tables. The staging file uses the hexadecimal x1d separator as the field delimiter. After the data is written to the target, the data stored in the table-specific directory that includes the staging files are deleted.
- If you use Microsoft Azure Data Lake Storage Gen2 with a Microsoft Azure Synapse Analytics connection, you must enable the **Hierarchical namespace** option in Microsoft Azure Data Lake Storage. With this setting, blob storage is not recommended.
- When you configure an application ingestion and replication task for a Microsoft Azure Synapse Analytics target, ensure that each source object that you select for replication meets the following criteria:
  - The object must not contain more than 1024 fields and the size of each field must be less than 500 KB.
  - The object must not contain any record that is greater than 1 MB in size.
  - The object must not contain more than 32 primary keys.
  - The cluster index key must not contain data that is greater than 900 bytes in size.
  - The primary keys of the object must be of a data type that Microsoft Azure Synapse Analytics supports for primary keys.
- Incremental load jobs and combined initial and incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information. The data in the recovery table prevents the jobs that are restarted after a failure from propagating previously processed data again. The recovery table is generated in the schema of the target tables.
- After an application ingestion and replication job loads data to a Microsoft Azure Synapse Analytics target by using external tables, the job does not drop the log tables and external tables created on the target, even though these tables might be re-created when the job starts again.
- Application ingestion and replication jobs configured for Microsoft Azure Synapse Analytics targets do not replicate the renaming of source fields on the target.

Microsoft Azure SQL Database and SQL Server targets

Consider the following guidelines when using Microsoft Azure SQL Database and Microsoft SQL Server targets with SAP (using SAP Mass Ingestion connector), Microsoft Dynamics 365, Oracle Fusion Cloud, and Salesforce sources for all load types. Microsoft SQL Database targets include on -premises, Microsoft Amazon RDS (Relational Database Service) for SQL Server, and Azure SQL Managed Instance targets.

The following list identifies considerations for preparing and using Microsoft SQL Database and Microsoft SQL Serve targets:

- The SQL Server JDBC driver is delivered with Application Ingestion and Replication. You do not need to install it separately.
- The Application Ingestion and Replication user requires following database roles, at minimum, to create target tables and write data to the tables:
  - db_datareader
  - db_datawriter
  - db_ddladmin
- In Administrator, when you define a SQL Server connection for connecting to a SQL Server target, set the following required connection properties:
  - SQL Server Version. This property is no longer used. If you select a version, it is ignored.
  - Authentication Mode. Select **SQL Server Authentication** or **Windows Authentication v2**.
  - User Name
  - Password
  - Host
  - Port
  - Database Name
  - Schema
  - Code Page

  Optionally, to use SSL encryption for the target connection, you can set the Encryption Method property to one of the following options: SSL, Request SSL, or Login SSL. If you enable encryption, also set the following properties:
  - Crypto Protocol Version
  - Validate Server Certificate. Select **True**.
  - Trust Store
  - Trust Store Password
  - Host Name in Certificate

  Other connection properties are not supported.
- Application ingestion and replication incremental load and initial and incremental load jobs with a SQL Server target generate a LOG table based on the target table schema, together with a number of additional metadata columns. The LOG table is created right before the change data is flushed to the target. The incoming DML data is inserted to the LOG table by supplying a local CSV file to Bulk Copy API of the SQL Server driver. A merge apply statement is generated based on the information in the LOG table and the DML operations are applied to the actual target table. When this is complete, the LOG table is dropped. The LOG table might cause a temporary spike in additional space or size requirements in the customer database instance when running multiple jobs or a job with multiple tables. The space and size required by the LOG table depends on the number of rows received as part of a flush cycle.
- The number of fields in a source object that an application ingestion and replication incremental load or an initial and incremental load job can propagate to a SQL Server target must not exceed 508 fields. If a source object contains more than 508 fields, the job fails while creating the LOG table.
- Application ingestion and replication incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information that prevents jobs restarted after a failure from propagating previously processed data again. This recovery table is generated in the same schema as the target tables.

Microsoft Fabric Data Warehouse targets

The following list identifies considerations for using Microsoft Fabric Data Warehouse targets:

- Application ingestion and replication tasks that have Microsoft Fabric Data Warehouse targets do not allow schema drift with **Drop Field** and **Rename Field** options.

Microsoft Fabric Lakehouse targets

Note the following guidelines and considerations for using Microsoft Fabric Lakehouse targets:

- Application ingestion and replication jobs that have Microsoft Fabric Lakehouse targets do not allow schema drift with the Modify Column option.
- The Microsoft Fabric Lakehouse target does not allow capital letters in table names. All tables are created with lowercase letters by default.

Microsoft Fabric OneLake targets

Consider the following guidelines when you use Microsoft Fabric OneLake targets:

- When you configure an application ingestion and replication task for a Microsoft Fabric OneLake target, you can select CSV, Avro, or Parquet as the format for the output files that contain the source data to be applied to the target.
- If you select **CSV** as the output file format, Application Ingestion and Replication creates the following files on the target for each source field:
  - schema.ini file that describes the schema of the field. The file also includes some settings for the output file on the target.
  - Output files that contain the data stored in the source field. Application Ingestion and Replication names the output files based on the name of the source field with an appended date and time.

  The schema.ini file lists the sequence of columns for the rows in the corresponding output file. The following table describes the columns in the schema.ini file:

Column	Description
ColNameHeader	Indicates whether the source data files include column headers.
Format	Format of the output files. Application Ingestion and Replication uses a comma (,) to delimit column values.
CharacterSet	Character set that is used for the corresponding output file. By default, Application Ingestion and Replication generates the files in the UTF-8 character set.
COL*<sequence_number>*	Name and data type of the source field. **Note:** If you selected any of the **Add Operation...** properties under **Advanced** on the **Target** page of the task wizard, the list of columns includes metadata columns for the operation type, time, owner, or transaction ID.

  Note: You must not edit the schema.ini file.
- If you select the Avro output format, you can select an Avro format type, a file compression type, an Avro data compression type, and the directory that stores the Avro schema definitions generated for each source table. The schema definition files have the following naming pattern: *schemaname_tablename*.txt.

- If you select the Parquet output format, you can optionally select a compression type that Parquet supports.
- Application Ingestion and Replication creates an empty directory on the target for each empty source field.
- When an incremental load job or combined initial and incremental load job configured for a target that uses the CSV output format propagates an Update operation that changed primary key values on the source, the job performs a Delete operation on the associated target row and then performs an Insert operation on the same row to replicate the change made to the source object. The Delete operation writes the before image to the target and the subsequent Insert image writes the after image to the target.

  For Update operations that do not change primary key values, application ingestion and replication jobs process each Update operation as a single operation and writes only the after image to the target. Application Ingestion and Replication processes each Update operation performed on the source as a Delete operation followed by an Insert operation on the target.

Using open mirroring to replicate data to Microsoft Fabric OneLake

Open mirroring enables near real-time replication to a mirrored database in Microsoft Fabric OneLake.

The data is first replicated to a landing zone associated with the mirrored database in Microsoft Fabric OneLake. This replicated data is stored in Parquet file format, which Microsoft Fabric reads from the OneLake landing zone to create or update tables in the mirrored database. Mirroring ensures that changes in the source system are quickly reflected in Microsoft Fabric to keep the mirrored data continuously up-to-date and ready for timely analysis and reporting.

Target preparation

Before you enable mirroring in a task that loads data to Microsoft Fabric OneLake, perform the following tasks:

1. Create a mirrored database in your Microsoft Fabric workspace.
   OneLake generates a unique landing zone URL based on the mirrored database name and workspace. Note this landing zone URL for use in your task configuration.
2. Replication starts automatically after you create the mirrored database. Check the mirrored database settings and ensure that the replication status is running.

For detailed information about creating a mirrored database, see the Microsoft Fabric OneLake documentation.

Task configuration

To configure an ingestion and replication task that loads data to Microsoft Fabric OneLake, perform the following tasks:

- Ensure that each source table that you select has primary or unique key columns defined to avoid replication failures. These keys are required to handle updates and deletes.
- In the task target properties, enable **Open Mirroring** and enter the **Landing Zone URL** value so that mirrored data files are written to the correct Microsoft Fabric OneLake path.

Landing zone file structure

When the task runs, the replicated data is stored in Parquet files organized by the following schema and table folders in the OneLake landing zone: `<landing-zone>/<schema>/<table>/`

Each selected source table has its own folder that contains the following files:

- A _metadata.json file specifying key columns and metadata.

- Parquet data files named sequentially with a fixed-width numeric format of 19 digits to preserve ordering. For example, `00000000000000000001.parquet`.

For incremental changes, each record in the Parquet files includes an operation row marker that indicates the type of DML operation, which can be insert, update, or delete. This marker is derived from the infa_operation_type metadata column. It enables Microsoft Fabric OneLake to apply changes correctly.

Microsoft Fabric OneLake reads these Parquet files from the landing zone to create or update tables in the mirrored database.

Limitations

The following limitations apply when you enable open mirroring:

- The _metadata.json file for each table is generated by the task. You cannot modify it while the task is deployed. To make changes, you need to update the task and then redeploy it.
- Do not modify or delete mirrored data files in the landing zone, as this might cause replication or downstream ingestion failures.
- Schema drift automatically adds new columns when they appear in the source data. However, if a column is deleted in the source, this change is not reflected in the target.
- You can mirror a maximum of 500 tables within a single mirrored database.

Oracle targets

Target preparation

- Application Ingestion and Replication requires users to have certain privileges to load data to Oracle target databases. For on-premises Oracle targets, grant the following user privileges to the Application Ingestion and Replication user (*cmid_user*) who connects to the Oracle target:

```
GRANT CREATE SESSION TO cmid_user;

GRANT SELECT ON "PUBLIC".V$DATABASE TO cmid_user;
GRANT SELECT ON "PUBLIC".V$CONTAINERS TO cmid_user;

GRANT SELECT ON DBA_USERS TO cmid_user;
GRANT SELECT ON DBA_TABLES TO cmid_user;
GRANT SELECT ON DBA_OBJECT_TABLES TO cmid_user;
GRANT SELECT ON DBA_INDEXES TO cmid_user;
GRANT SELECT ON DBA_OBJECTS TO cmid_user;
GRANT SELECT ON DBA_VIEWS TO cmid_user;

GRANT CREATE TABLE <schema.table> TO cmid_user; <--Unless you grant on ANY TABLE
GRANT SELECT ON ALL_CONSTRAINTS TO cmid_user;
GRANT SELECT ON ALL_OBJECTS TO cmid_user;

GRANT SELECT ON SYS.TAB$ TO cmid_user;
GRANT SELECT ON SYS.RECYCLEBIN$ TO cmid_user;
GRANT SELECT ON SYS.COL$ TO cmid_user; <-- If cmid_user is the owner of the target
schema
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO cmid_user;
GRANT SELECT ON SYS.OBJ$ TO cmid_user;
GRANT SELECT ON SYS.COLTYPE$ TO cmid_user;
GRANT SELECT ON SYS.ATTRCOL$ TO cmid_user;
GRANT SELECT ON SYS.IDNSEQ$ TO cmid_user;
GRANT SELECT ON SYS.ATTRCOL$ TO cmid_user;
GRANT SELECT ON SYS.IDNSEQ$ TO cmid_user;
GRANT SELECT ON SYS.IND$ TO cmid_user;

-- Grant the following if cmid_user is NOT the owner of the target schema. If you
prefer, you
```

```
-- can grant to individual target tables and indexes instead of to ANY TABLE or ANY
INDEX.
GRANT ALTER SESSION TO cmid_user;
GRANT RESOURCE TO cmid_user;
GRANT SELECT ANY TABLE TO cmid_user;
GRANT SELECT ANY DICTIONARY TO <cmid_user>;
GRANT ALTER ANY TABLE TO cmid_user;
GRANT CREATE ANY TABLE TO cmid_user;
GRANT DROP ANY TABLE TO cmid_user;
GRANT INSERT ANY TABLE TO cmid_user;
GRANT UPDATE ANY TABLE TO cmid_user;
GRANT DELETE ANY TABLE TO cmid_user;
GRANT CREATE ANY INDEX TO cmid_user;
GRANT ALTER ANY INDEX TO cmid_user;
GRANT DROP ANY INDEX TO cmid_user;
```

- For Amazon RDS for Oracle targets, log in to RDS as the master user and grant the following user privileges to the Application Ingestion and Replication user (*cmid_user*) who connects to the Oracle target:

```
GRANT CREATE SESSION TO cmid_user;

GRANT SELECT on "PUBLIC".V$DATABASE TO cmid_user;

GRANT SELECT on DBA_USERS TO cmid_user;
GRANT SELECT on DBA_TABLES TO cmid_user;
GRANT SELECT on DBA_INDEXES TO cmid_user;
GRANT SELECT ON DBA_VIEWS TO cmid_user;

GRANT CREATE TABLE <schema.table> TO cmid_user;
GRANT SELECT on SYS.TAB$ TO cmid_user;
GRANT SELECT on SYS.COL$ TO cmid_user;
GRANT SELECT on SYS.OBJ$ TO cmid_user;
GRANT SELECT on SYS.IND$ TO cmid_user;

-- Grant the following if cmid_user is NOT the owner of the target schema. If you
prefer, you
-- can grant to individual target tables and indexes instead of to ANY TABLE or INDEX.
GRANT ALTER SESSION TO cmid_user;
GRANT RESOURCE TO cmid_user;
GRANT SELECT ANY TABLE TO cmid_user;
GRANT SELECT ANY DICTIONARY TO <cmid_user>;
GRANT ALTER ANY TABLE TO cmid_user;
GRANT CREATE ANY TABLE TO cmid_user;
GRANT DROP ANY TABLE TO cmid_user;
GRANT INSERT ANY TABLE TO cmid_user;
GRANT UPDATE ANY TABLE TO cmid_user;
GRANT DELETE ANY TABLE TO cmid_user;
GRANT CREATE ANY INDEX TO cmid_user;
GRANT ALTER ANY INDEX TO cmid_user;
GRANT DROP ANY INDEX TO cmid_user;
```

Also, run the following Amazon RDS procedures to grant additional SELECT privileges to *cmid_user*:

```
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$CONTAINERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECT_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECTS',
p_grantee => 'cmid_user',
```

```
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_CONSTRAINTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_OBJECTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'RECYCLEBIN$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CDEF$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'COLTYPE$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ATTRCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'IDNSEQ$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
```

Usage considerations

- By default, Application Ingestion and Replication disables logging for the Oracle target tables to optimize performance. You can enable logging by setting the writerOracleNoLogging custom property to false on the **Target** page in the application ingestion and replication task wizard.

- Application Ingestion and Replication uses the minimum source constraints needed for replication when generating Oracle target objects. If a source has a primary key, the job preferentially uses the primary key and no other constraint when creating the target object. In application ingestion and replication tasks that have an SAP ECC source (Oracle based), if the source does not have a primary key but does have unique indexes, the task chooses the best unique index for replication in the following order of priority: 1) the unique index that has only NOT NULL columns, 2) if multiple unique indexes have only NOT NULL columns, the unique index with the least number of columns, or 3) if no source primary key or unique NOT NULL index exists, the unique index with the greatest number of columns. Only a single primary key or unique index is used.

  Source object properties that are not required for replication, such as for storage and partitioning, are not reflected on the target.
- When loading data to an Oracle target, ensure that the total number of columns in the Oracle table or view does not exceed 1000.

PostgreSQL targets

Consider the following guidelines when you use PostgreSQL targets:

- To deploy and run an application ingestion task that includes a PostgreSQL target, the target connection must specify a database user who has the required privileges.
  The user must have the CONNECT and TEMPORARY privileges for the database specified in the connection and the USAGE and CREATE privileges for the target schema specified in the target properties.

  Use the following SQL statements to grant these privileges to a user role and then assign the role to a user:

  ```
  CREATE ROLE dbmi_role;
  GRANT CONNECT ON DATABASE database TO dbmi_role;
  GRANT TEMPORARY ON DATABASE database TO dbmi_role;
  GRANT CREATE ON SCHEMA schema TO dbmi_role;
  GRANT USAGE ON SCHEMA schema TO dbmi_role;
  CREATE USER dbmi_user with PASSWORD 'password';
  GRANT dbmi_role to dbmi_user;
  ```

  Note: Privileges on the target tables that are generated when the job runs are granted to the user who runs the job.
- Application ingestion and replication incremental load jobs with a PostgreSQL target generate a LOG table based on the target table schema, with some additional metadata columns. The LOG table is created right before change data is flushed to the target. The incoming DML data is inserted to the LOG table by supplying a local CSV file to the Bulk Copy API of the PostgreSQL driver. A set of merge apply statements is generated based on the information in the LOG table, and then the DML operations are applied to the actual target table. After the DML changes are applied, the LOG table is dropped.
  The LOG table might cause a temporary spike in additional space or size requirements in the customer database instance if you run multiple jobs or a job with multiple tables. The space and size required by the LOG table depends on the number of rows received as part of a flush cycle.
- The number of fields in a source object that an application ingestion and replication incremental load job can propagate to a PostgreSQL target must not exceed 796 fields. If a source object contains more than 796 fields, the job fails while creating the LOG table.
- The data inserted in a PostgreSQL table row depends on the page size configured for the instance. The default is 8 KB. If a row exceeds the page size, the job fails and Application Ingestion and Replication displays the following error:

  ```
  SQL Error [54000]: ERROR: row is too big: size 9312, maximum size 8160
  ```

  This error occurs when the data exceeds the configured page size and PostgreSQL can't persist the data.

For example:

- If the source table has 500 INT columns, the row size is 2000 bytes (500 * 4 bytes). The row fits within the page size (8192 bytes) and is successfully inserted.
- If the source table has 500 BIGINT columns, the row size is 4000 bytes (500 * 8 bytes). The row fits within the page size and is successfully inserted.
- If the source table has 500 VARCHAR (255) columns, the row size is 9000 bytes (500 * 18 bytes). As the row does not fit the page size, an error occurs.

- PostgreSQL supports a maximum length of 63 characters for source object identifiers. The deployment of application ingestion and replication tasks that have a PostgreSQL target will fail during validation if the length of any source table or column name exceeds 63 characters.
- Application ingestion and replication incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information that prevents jobs restarted after a failure from propagating previously processed data again. This recovery table is generated in the same schema as the target tables.

Salesforce Data 360 targets

The following list identifies the guidelines and considerations for using Salesforce Data 360 targets:

- Due to a third-party limitation, Salesforce Data 360 Connector cannot perform read and write operations on the same Data Lake Object (DLO). To work around this, you must manually create a recovery table in the target and specify the recovery table name in the writerRecoveryTableNameAlias custom property before you run the task.
  Perform the following tasks to create and configure the recovery table:
  1. Create a schema file named `INFORMATICA_CDC_RECOVERY.yml` with the following content:

```
openapi: 3.0.3
components:
  schemas:
    INFORMATICA_CDC_RECOVERY:
      type: object
      properties:
        TASK_ID:
          type: string
        TYPE:
          type: string
        SCHEMA_NAME:
          type: string
        TABLE_NAME:
          type: string
        CYCLE_NUMBER:
          type: number
        SEQUENCE:
          type: string
        LAST_CYCLE_UPDATE_TIME:
          type: string
```

  2. Log in to Salesforce and open the Salesforce Data Cloud app.
  3. Navigate to **DataCloud Setup** and select the Ingestion API associated with your task.
  4. Upload the `INFORMATICA_CDC_RECOVERY.yml` schema to create the recovery table.
  5. To create a data stream for the recovery table, perform the following tasks:
     - Open the Salesforce Data Cloud app launcher and go to the **Data Stream** window.
     - Create a new data stream.

- Select **Ingestion API** as the data stream source, and proceed.
- Select the INFORMATICA_CDC_RECOVERY table for the Ingestion API.

6. To configure the primary key for the data stream, perform the following tasks:
   - In the **New Data Stream** window, create a new formula field with the following transformation formula:

     ```
     CONCAT(sourceField['TASK_ID'], "_", sourceField['TYPE'], "_",
     sourceField['SCHEMA_NAME'], "_", sourceField['TABLE_NAME'])
     ```

   - Set the formula return type as text.
   - Select this newly created formula field as the primary key.
7. Select the data space configured for your task and deploy the data stream.
8. After deployment, open the data stream to copy the **Object API Name** of the recovery table and keep it handy.
9. When you configure a database ingestion and replication task, enter the copied object API name in the writerRecoveryTableNameAlias custom property on the Destination page before you run the task.

- Salesforce Data 360 reserves certain column names such as *cdp_sys_SourceVersion*, *DataSource*, *DataSourceObject*, and *KQ_Id* for internal use. If your source object contains any of these column names, rename them before you deploy the task to avoid failure.
- In incremental load and combined load jobs, delete operations are ignored and skipped.
- For a task that loads data to a Salesforce Data 360 target, the source object must have one primary key. If the source object has no primary key or more than one primary key, the deployment fails.
- **Audit Mode** and **Soft Deletes** apply modes are not available for Salesforce Data 360 targets. These apply modes require creating objects without a primary key. Salesforce Data 360 does not permit this, because all data lake objects require a primary key.
- Application ingestion and replication tasks that have Salesforce Data 360 targets do not allow schema drift with **Drop Field** and **Rename Field** options.

Snowflake targets

Target preparation

Target preparation varies depending on whether you use the Superpipe feature for high performance streaming of data to Snowflake target tables or write data to intermediate stage files.

With Superpipe

If you use the Superpipe feature, complete the following steps:

1. Create a Data Ingestion and Replication user. Use the following SQL statement:

   ```
   create user INFACMI_User password = 'Xxxx@xxx';
   ```

2. Create a new user role and grant it to the Data Ingestion and Replication user. Use the following SQL statements:

   ```
   create role INFACMI_superpipe;
   grant role INFACMI_superpipe to user INFACMI_User;
   ```

3. Grant usage on the Snowflake virtual warehouse to the new role. Use the following SQL statement:

   ```
   grant usage on warehouse warehouse_name to role INFACMI_superpipe;
   ```

4. Grant usage on the Snowflake database to the new role. Use the following SQL statement:

   ```
   grant usage on database INFACMI_DB1 to role INFACMI_superpipe;
   ```

5. Create a new schema. Use the following SQL statements:

```
use database INFACMI_DB1;
create schema sh_superpipe;
```

6. Grant create stream, create view, and create table privileges on the new Snowflake schema to the new role. Use the following SQL statement:

```
grant create stream, create view, create table, usage on schema
INFACMI_DB1.sh_superpipe to role INFACMI_superpipe;
```

7. Set the default role for the newly created user. Use the following SQL statement:

```
alter user INFACMI_User set default_role=INFACMI_superpipe;
```

8. Define a Snowflake Data Cloud connection to the target. You must use the **KeyPair** option as the authentication method. See Connectors and Connections > Snowflake Data Cloud connection properties.
9. Generate a private key with OpenSSL version 3.x.x and one of the following ciphers:
   - AES128
   - AES128-CBC
   - AES256
   - AES-256-CBC

   Use the following openssl commands to generate and format the private key:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 AES-256-CBC -out
rsa_key.p8
```

10. Generate the public key. Use the following openssl command, in which the -in option references the file (rsa_key.p8) that contains the encrypted private key:

```
openssl rsa -in rsa_key.p8 -pubout -out rsa_key.pub
```

11. In Snowflake, assign the public key to the Snowflake user. Use the following SQL command:

```
alter user INFACMI_User set rsa_public_key='key_value’;
```

Next step: When you create an ingestion task, select the **Superpipe** option on the **Target** page of the task wizard. You can also optionally specify a **Merge Frequency** value, which controls the frequency at which change data rows are merged and applied to the Snowflake target table.

Without Superpipe

If you do NOT use the Superpipe feature for Snowflake targets, complete the following steps as the ACCOUNTADMIN user:

1. Create a Data Ingestion and Replication user. Use one of the following SQL statements:

```
create user INFACMI_User password = 'Xxxx@xxx';
```

   or

```
replace user INFACMI_User password = 'Xxxx@xxx';
```

2. Create a new role and grant the role to the Data Ingestion and Replication user. Use the following SQL statements:

```
create role INFA_CMI_Role;
grant role INFA_CMI_Role to user INFACMI_User;
```

3. Grant usage on the Snowflake virtual warehouse to the new role. Use the following SQL statement:

```
grant usage on warehouse CMIWH to role INFA_CMI_Role;
```

4. Grant usage on the Snowflake database to the new role. Use the following SQL statement:

```
grant usage, CREATE SCHEMA on database CMIDB to role INFA_CMI_Role;
```

5. Set the default role for the newly created user. Use the following SQL statement:

```
alter user INFACMI_User set default_role=INFA_CMI_Role;
```

Also, as the INFACMI_User, create a new schema:

```
create schema CMISchema;
```

Note: If the user's default role is used for ingestion tasks and does not have the required privileges, the following error will be issued at runtime:

```
SQL compilation error: Object does not exist, or operation cannot be performed.
```

For Snowflake-managed Iceberg tables

Regardless of whether you use the Superpipe option, you can choose to store data and metadata in Snowflake-managed Iceberg tables in external cloud storage instead of in Snowflake tables. To prepare for the use of Iceberg tables, perform the following tasks:

- Create and configure an external volume in Snowflake to store an identity and access management (IAM) entity that Snowflake can use to connect to your external cloud storage. The steps depend on the type of cloud storage, for example, Amazon S3, Google Cloud Storage, or Azure storage. For instructions, see the Snowflake documentation here.
- Create a role with the minimum access control privileges required to create the external volume and Iceberg tables and to use the external volume and table schema. For instructions, see the Snowflake documentation here.

Usage guidelines

- Data Ingestion and Replication provides alternative methods of moving data to a Snowflake Data Cloud target:
  - If you select the **Superpipe** option when defining an ingestion task, the ingestion job uses the Snowpipe Streaming API to stream rows of data directly to the target tables with low latency. This method is available for all load types. You must use **KeyPair** authentication.
  - If you do not use Superpipe, ingestion jobs first write the data to data files in an internal stage, which has the name you specify in the task definition.
- If you do not use Superpipe and the internal stage specified in the target properties for an ingestion job does not exist, Data Ingestion and Replication automatically creates the stage by running the following SQL command:

  ```
  Create stage if not exists "Schema"."Stage_Bucket"”
  ```

  For the command to run successfully, the following privilege must be granted to your user role:

  ```
  GRANT CREATE STAGE ON SCHEMA "Schema" TO ROLE <your_role>;
  ```
- You can enable application ingestion and replication jobs to store data in an open format in Snowflake-managed Iceberg tables in an external cloud Amazon S3, Google Cloud Storage, or Azure Storage location instead of in Snowflake. For more information, see “Using Snowflake-managed Iceberg tables to store data in external cloud storage” on page 61.
- When you define a connection for a Snowflake target, you must set the **Additional JDBC URL Parameters** field to `database=target_database_name`. Otherwise, when you try to define the target in the application ingestion and replication task wizard, an error message indicating that the list of schemas cannot be retrieved appears.
- When you define a connection for a Snowflake target using the **KeyPair** option as the authentication method and you generate the private key with OpenSSL 3.x.x version, use `AES128`, `AES128-CBC`, `AES256`, or `AES-256-CBC` cipher while generating the private key. Run the following command:

  ```
  openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 AES-256-CBC -out rsa_key.p8
  ```

  OpenSSL recommends using version 3.x.x.

  Data Ingestion and Replication supports the following ciphers:

  •AES128

- AES128-CBC
- AES256
- AES-256-CBC
- CAMELLIA-256-CBC
- CAMELLIA-256-CTR
- ARIA-256-CTR

Note: The `CAMELLIA-256-CBC`, `CAMELLIA-256-CTR`, and `ARIA-256-CTR` ciphers are not valid with password encrypted private keys. To generate an encrypted private key with these ciphers, use the following command:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -v2 CAMELLIA-256-CTR -inform PEM -out
rsa_key.p8 -nocrypt
```

- Incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information. The data in the recovery table prevents the jobs that are restarted after a failure from propagating previously processed data again. The recovery table is generated in the schema that contains the target tables.
- Application ingestion and replication incremental load and combined load jobs that have a Snowflake target create transient _LOG tables to stage data on the target instead of permanent tables, by default. To prevent permanent tables from being created, the Snowflake Time Travel and Fail-safe features are disabled by default. The use of transient tables helps reduce _LOG table storage consumption and associated costs. With transient tables, however, you can't use the Superpipe option. If you need to create permanent tables, you can set the target snowflakeStagingTransientTable custom property to false.
- In an application ingestion and replication task configured for change data capture that runs at scheduled intervals and detects changes based on record timestamps, only the most recent change for each record is captured during each interval. If a record undergoes multiple updates within the same interval, the CDC process captures only the latest change, while earlier changes made within that interval are not recorded.
- Before stopping an application ingestion and replication incremental load or combined load job that has a Snowflake target and uses the **Superpipe** option, ensure that all ingested data in the change stream has been merged into the target. Otherwise, if the job is not restarted for an extended period or before the stream expires, any data that remains in the stream is lost and can’t be recovered.
- The object must not contain any record that is greater than 16 MB.
- For Snowflake targets, you cannot alter the scale of NUMBER fields or change the data type of an existing field to a different data type because Snowflake does not support these actions.

Using Snowflake-managed Iceberg tables to store data in external cloud storage

You can enable application ingestion and replication jobs to store data and metadata in Snowflake-managed Iceberg tables in an external cloud location instead of in Snowflake tables. The external location can be in Amazon S3, Google Cloud Storage, or Azure Storage.

The external storage of data in an open format has the following benefits:

- Ability to use query processing engines other than Snowflake to process the data
- Interoperability across different data processing and analytics tools
- Better performance
- Robust security and governance

Application Ingestion and Replication automatically creates the Snowflake-managed Iceberg tables when you deploy ingestion and replication tasks enabled for Iceberg table use. You do not need to create the Iceberg tables manually.

Snowflake manages the metadata and catalog for the Iceberg tables. Snowflake uses an external volume that stores identity and access management (IAM) information to connect to the external storage location. In Snowflake, you must create and configure an external volume beforehand. For more information, see "Target preparation" in "Snowflake targets" on page 58.

When you configure ingestion and replication tasks, specify the following target custom properties to enable the use of Snowflake-managed Iceberg tables:

- **writerTargetTableFormat**. Required. Set this custom property to **iceberg** to enable the use of Snowflake-managed Iceberg tables.
- **writerTargetTableExternalVolumeName**. Required. Enter the name of the external volume that you created in the Snowflake account.
- **writerTargetTableExternalStorageBaseLocation**. Optional. The base URL for the external storage location. If you do not specify this value, the base URL from the external volume definition is used.

The ingestion and replication tasks can use any load type, any apply mode, and schema drift. The Superpipe option is allowed but not required.

The following limitations apply to using Iceberg tables:

- You must use permanent Snowflake-managed Iceberg tables, not transient or temporary tables.
- The external volume must be in the same cloud and region as the Snowflake account.
- Char and varchar data types with a limited size specification are not supported.
- Time and timestamp data types have a microsecond precision.

Configure private connectivity to Snowflake

You can access Snowflake using AWS or Azure Private Link endpoints.

The AWS or Azure Private Link setup ensures that the connection to Snowflake uses the AWS or Azure internal network and does not take place over the public Internet.

To connect to the Snowflake account over the private AWS network, see AWS Private Link and Snowflake.

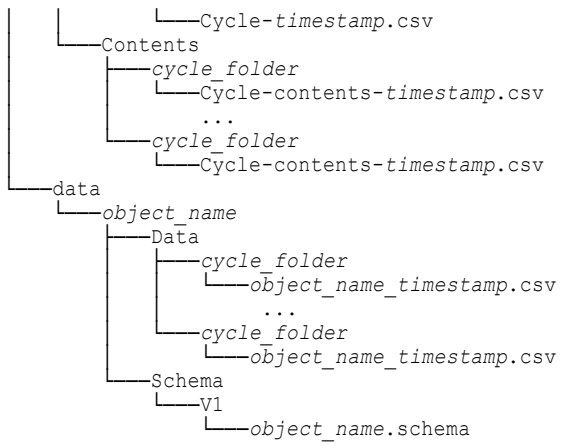
To connect to the Snowflake account over the private Azure network, see Azure Private Link and Snowflake.

Default directory structure of CDC files on Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, and Microsoft Fabric OneLake targets

Application ingestion and replication jobs create directories on Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, and Microsoft Fabric OneLake targets to store information about change data processing.

The following directory structure is created by default on the targets:

```
Bucket
└──connection_folder
    └──job_folder
        ├──cycle
        │   ├──Completed
        │   │   ├──completed_cycle_folder
        │   │   │   └──Cycle-timestamp.csv
        │   │   │       ...
        │   │   └──completed_cycle_folder
```



The following table describes the directories in the default structure:

Folder	Description
connection_folder	Contains the Application Ingestion and Replication objects. This folder is specified in the **Folder Path** field of the Amazon S3 connection properties, in the **Directory Path** field of the Microsoft Azure Data Lake Storage Gen2 connection properties, or in the **Lakehouse Path** field of the Microsoft Fabric OneLake connection properties. **Note:** This folder is not created for Google Cloud Storage targets.
job_folder	Contains job output files. This folder is specified in the **Directory** field on the **Target** page of the application ingestion and replication task wizard.
cycle/Completed	Contains a subfolder for each completed CDC cycle. Each cycle subfolder contains a completed cycle file.
cycle/Contents	Contains a subfolder for each CDC cycle. Each cycle subfolder contains a cycle contents file.
data	Contains output data files and schema files for each object.
data/*object_name*/Schema/V1	Contains a schema file. **Note:** Application Ingestion and Replication does not save a schema file in this folder if the output files use the Parquet format.
data/*object_name*/Data	Contains a subfolder for each CDC cycle that produces output data files.

Cycle directories

Application Ingestion and Replication uses the following pattern to name cycle directories:

```
[dt=]yyyy-mm-dd-hh-mm-ss
```

The "dt=" prefix is added to cycle folder names if you select the **Add Directory Tags** check box on the **Target** page of the application ingestion and replication task wizard.

Cycle contents files

Cycle contents files are located in cycle/Contents/*cycle_folder* subdirectories. Cycle contents files contain a record for each object that has had a DML event during the cycle. If no DML operations occurred on an object in the cycle, the object does not appear in the cycle contents file.

Application Ingestion and Replication uses the following pattern to name cycle content files:

`Cycle-contents-`*`timestamp`*`.csv`

A cycle contents csv file contains the following information:

- Object name
- Cycle name
- Path to the cycle folder for the object
- Start sequence for the object
- End sequence for the object
- Number of Insert operations
- Number of Update operations
- Number of Delete operations
- **Combined load jobs only**. Number of Insert operations encountered during the initial load phase
- **Combined load jobs only**. Number of Delete operations encountered during the initial load phase
- Schema version
- Path to the schema file for the schema version

  Note: If the output data files use the Parquet format, Application Ingestion and Replication does not save a schema file at the path that is specified in the cycle contents file. Instead, use the schema file in the folder that is specified in the **Avro Schema Directory** field on the **Target** page of the application ingestion and replication task wizard.

Completed cycle files

Completed cycle files are located in cycle/Completed/*completed_cycle_folder* subdirectories. An application ingestion and replication job creates a cycle file in this subdirectory after a cycle completes. If this file is not present, the cycle has not completed yet.

Application Ingestion and Replication uses the following pattern to name completed cycle files:

`Cycle-`*`timestamp`*`.csv`

A completed cycle csv file contains the following information:

- Cycle name
- Cycle start time
- Cycle end time
- Current sequence number at the time the cycle ended
- Path to the cycle contents file
- Reason for the end of cycle
  Valid reason values are:
  - **NORMAL_COMMIT**. A commit operation was encountered after the cycle had reached the DML limit or the end of the cycle interval. A cycle can end only on a commit boundary.
  - **NORMAL_EXPIRY**. The cycle ended because the cycle interval expired. The last operation was a commit.
  - **Combined load jobs only:** `BACKLOG_COMPLETED`. The cycle ended because CDC backlog processing completed. The CDC backlog consists of events captured during the initial load phase of the combined job. The backlog includes potential DML changes captured at the beginning or end of the initial load phase and during the transition from the initial load phase to the main CDC incremental processing.

- **Combined load jobs only:** `INITIAL_LOAD_COMPLETED`. The cycle ended because the initial load completed.
- **Combined load jobs only:** `RESYNC_STARTED`. The cycle ended because the object resync initiated.

Output data files

The data files contain records that include the following information:

- Operation type. Valid values are:
  - `I` for Insert operations
  - `E` for Update operations
  - `D` for Delete operations
  - **Combined load jobs only:** `X` for Delete operations encountered during the initial load phase of a combined load job
  - **Combined load jobs only:** `Y` for Insert operations encountered during the initial load phase of a combined load job
- Sortable sequence number. In combined initial and incremental load jobs, the sortable sequence number contains a 20-digit prefix that can be used to align rows with the resync version and the load job. The prefix is a combination of the following attributes:
  1. Incarnation. This nine-digit number is incremented each time the object is resynced. The initial value is 1.
  2. Schema version. This nine-digit number is incremented each time a schema drift change is propagated for the object. The initial value is 1.
  3. Phase. This two-digit number changes when transition from unload, to backlog, to CDC is performed. Valid values are:
     - `00` for Truncation, which is the first data record written during initial load or resync
     - `01` for a normal insert during initial load or resync
     - `02` for a change detected during the initial load
     - `03` for a change detected after the initial load or resync is completed but before the transition back to the main CDC phase
     - `04` for a change detected during the normal CDC phase
- Data fields

  Note: Insert and Delete records contain only after images. Update records contain both before and after images.

Custom directory structure for output files on Amazon S3, Google Cloud Storage, Microsoft Fabric OneLake, and ADLS Gen2 targets

You can configure a custom directory structure for the output files that initial load, incremental load, and combined initial and incremental load jobs write to Amazon S3, Google Cloud Storage, Microsoft Azure Data

Lake Storage (ADLS) Gen2, and Microsoft Fabric OneLake targets if you do not want to use the default structure.

Initial loads

By default, initial load jobs write output files to *tablename_timestamp* subdirectories under the parent directory. For Amazon S3 and ADLS Gen2 targets, the parent directory is specified in the target connection properties if the **Connection Directory as Parent** check box is selected on the **Target** page of the task wizard.

- In an Amazon S3 connection, this parent directory is specified in the **Folder Path** field.
- In an ADLS Gen2 connection, the parent directory is specified in the **Directory Path** field.

For Google Cloud Storage targets, the parent directory is the bucket container specified in the **Bucket** field on the **Target** page of the task wizard.

For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.

You can customize the directory structure to suit your needs. For example, for initial loads, you can write the output files under a root directory or directory path that is different from the parent directory specified in the connection properties to better organize the files for your environment or to find them more easily. Or you can consolidate all output files for a table directly in a directory with the table name rather than write the files to separate timestamped subdirectories, for example, to facilitate automated processing of all of the files.

To configure a directory structure, you must use the **Data Directory** field on the **Target** page of the ingestion task wizard. The default value is `{TableName}_{Timestamp}`, which causes output files to be written to *tablename_timestamp* subdirectories under the parent directory. You can configure a custom directory path by creating a directory pattern that consists of any combination of case-insensitive placeholders and directory names.

By default, the target schema is also written to the data directory. If you want to use a different directory for the schema, you can also define a directory pattern in the **Schema Directory** field.

You can manually enter the directory pattern for storing data or schema, or use the Edit icon located next to the **Data Directory** and **Schema Directory** fields to specify directory patterns using predefined placeholders.

The following table lists the available placeholders that you can use to build your data and schema directory path:

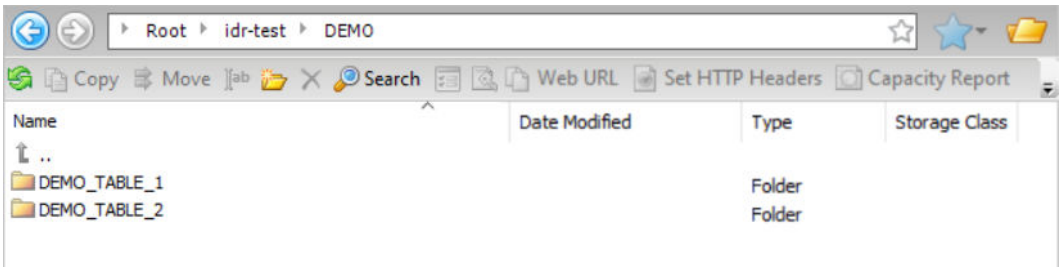
Path Type	Value
Folder Path	Enter a folder name or use variables to create a folder name. For example, to organize your data by date, schema, table, and load time, enter {yyyy}/{mm}/{dd}/{SchemaName}/{TableName}/{Timestamp}
Timestamp	You can select from the following values: - {Timestamp} for the date and time, in the format yyyymmdd_hhmissms, at which the initial load job started to transfer data to the target. - {Schema} for the target schema name. - {yy} for a two-digit year. - {yyyy} for a four-digit year. - {mm} for a two-digit month value. - {dd} for a two-digit day in the month.

Path Type	Value
Schema Name	You can select from the following values: - {SchemaName} for the target schema name. - toUpper(SchemaName) to use uppercase for the values represented by the placeholder in parentheses. - toLower(SchemaName) to use lowercase for the values represented by the placeholder in parentheses.
Table Name	You can select from the following values : - {TableName} for the target table name. - toUpper(TableName) to use uppercase for the values represented by the placeholder in parentheses. - toLower(TableName) to use lowercase for the values represented by the placeholder in parentheses.

Example 1

You are using an Amazon S3 target and want to write output files and the target schema to the same directory, which is under the parent directory specified in the **Folder Path** field of the connection properties. In this case, the parent directory is `idr-test/DEMO/`. You want to write all of the output files for a table to a directory that has a name matching the table name, without a timestamp. You must complete the **Data Directory** field and select the **Connection Directory as Parent** check box.

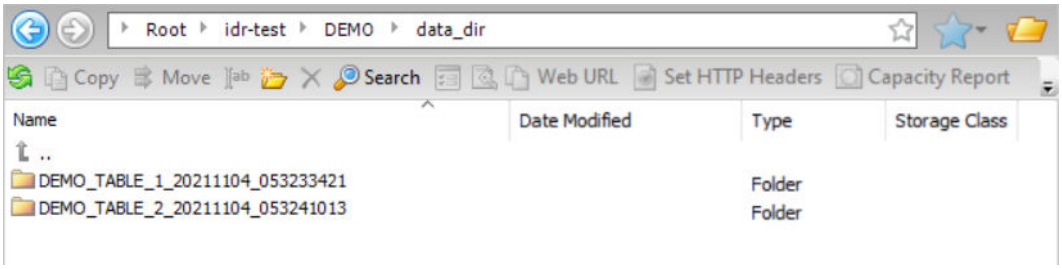
Based on this configuration, the resulting directory structure is:



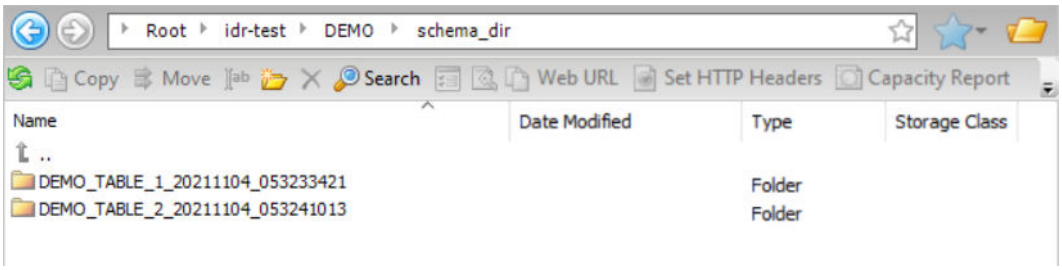
Example 2

You are using an Amazon S3 target and want to write output data files to a custom directory path and write the target schema to a separate directory path. To use the directory specified in the **Folder Path** field in the Amazon S3 connection properties as the parent directory for the data directory and schema directory, select **Connection Directory as Parent**. In this case, the parent directory is `idr-test/DEMO/`. In the **Data Directory** and **Schema Directory** fields, define directory patterns by using a specific directory name, such as data_dir and schema_dir, followed by the default {TableName}_{Timestamp} placeholder value. The placeholder creates *tablename_timestamp* destination directories.

Based on this configuration, the resulting data directory structure is:



And the resulting schema directory structure is:



Incremental loads and combined initial and incremental loads

By default, incremental load and combined initial and incremental load jobs write cycle files and data files to subdirectories under the parent directory. However, you can create a custom directory structure to organize the files to best suit your organization's requirements.

This feature applies to application ingestion and replication incremental load jobs that have a Salesforce source and Amazon S3, Google Cloud Storage, Microsoft Fabric OneLake, or Microsoft Azure Data Lake Storage (ADLS) Gen2 targets.

For Amazon S3 and ADLS Gen2 targets, the parent directory is set in the target connection properties if the **Connection Directory as Parent** check box is selected on the **Target** page of the task wizard.

- In an Amazon S3 connection, the parent directory is specified in the **Folder Path** field.
- In an ADLS Gen2 connection, the parent directory is specified in the **Directory Path** field.

For Google Cloud Storage targets, the parent directory is the bucket container specified in the **Bucket** field on the **Target** page of the task wizard.

For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.

You can customize the directory structure to suit your needs. For example, you can write the data and cycle files under a target directory for the task instead of under the parent directory specified in the connection properties. Alternatively, you can 1) consolidate table-specific data and schema files under a subdirectory that includes the table name, 2) partition the data files and summary contents and completed files by CDC cycle, or 3) create a completely customized directory structure by defining a pattern that includes literal values and placeholders. For example, if you want to run SQL-type expressions to process the data based on time, you can write all data files directly to timestamp subdirectories without partitioning them by CDC cycle.

To configure a custom directory structure for an incremental load task, define a pattern for any of the following optional fields on the **Target** page of the ingestion task wizard:

Field	Description	Default
Task Target Directory	Name of a root directory to use for storing output files for an incremental load task. If you select the **Connection Directory as Parent** option, you can still optionally specify a task target directory. It will be appended to the parent directory to form the root for the data, schema, cycle completion, and cycle contents directories. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.	None
Connection Directory as Parent	Select this check box to use the parent directory specified in the connection properties. This field is not available for the Microsoft Fabric OneLake target.	Selected
Data Directory	Path to the subdirectory that contains the data files. In the directory path, the {TableName} placeholder is required if data and schema files are *not* partitioned by CDC cycle.	{TaskTargetDirectory}/data/{TableName}/data
Schema Directory	Path to the subdirectory in which to store the schema file if you do not want to store it in the data directory. In the directory path, the {TableName} placeholder is required if data and schema files are not partitioned by CDC cycle.	{TaskTargetDirectory}/data/{TableName}/schema
Cycle Completion Directory	Path to the directory that contains the cycle completed file.	{TaskTargetDirectory}/cycle/completed
Cycle Contents Directory	Path to the directory that contains the cycle contents files.	{TaskTargetDirectory}/cycle/contents
Use Cycle Partitioning for Data Directory	Causes a timestamp subdirectory to be created for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.	Selected
Use Cycle Partitioning for Summary Directories	Causes a timestamp subdirectory to be created for each CDC cycle, under the summary contents and completed subdirectories.	Selected
List Individual Files in Contents	Lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.	Not selected if **Use Cycle Partitioning for Summary Directories** is selected. Selected if you cleared **Use Cycle Partitioning for Summary Directories**.

The directory pattern can consist of any combination of case-insensitive placeholders, shown in curly brackets { }, and specific directory names. To specify a custom data directory or schema directory

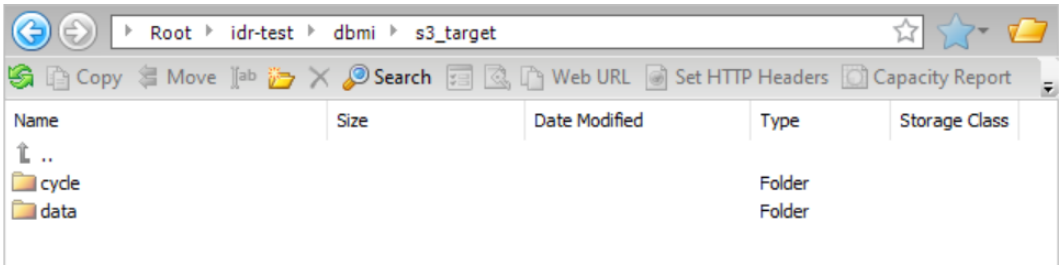
expression, use the Edit icon available at the fields and select the directory pattern from the listed placeholders:

Path Type	Value
Folder Path	Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
Timestamp	You can select from the following values: - {Timestamp} for the date and time, in the format yyyymmdd_hhmissms - {Schema} for the target schema name. - {yy} for a two-digit year. - {yyyy} for a four-digit year. - {mm} for a two-digit month value. - {dd} for a two-digit day in the month. Use {Timestamp}, {yy}, {yyyy}, {mm}, and {dd} in directory patterns to insert specific date and time information into directory names for organizing data. When you specify these placeholders in directory patterns for data, contents, and completed directories, these placeholders represent the time when the CDC cycle began. For the schema directory, these placeholders represent the time when the entire CDC job started, not just the cycle.
Schema Name	You can select from the following values: - {SchemaName} for the target schema name. - toUpper(SchemaName) to use uppercase for the values represented by the placeholder in parentheses. - toLower(SchemaName) to use lowercase for the values represented by the placeholder in parentheses.
Table Name	You can select from the following values: - {TableName} for the target table name. - toUpper(TableName) to use uppercase for the values represented by the placeholder in parentheses. - toLower(TableName) to use lowercase for the values represented by the placeholder in parentheses.

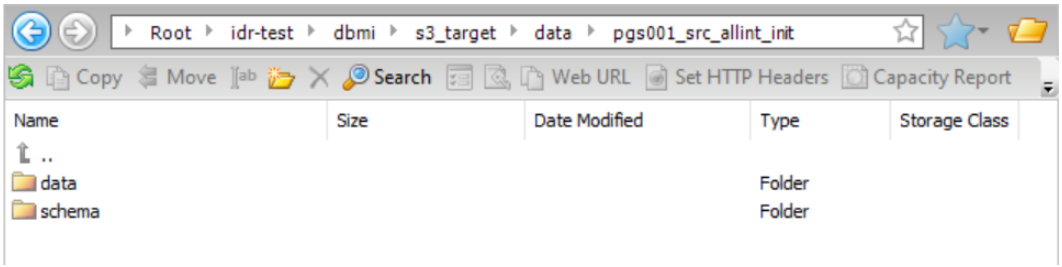
Example 1

You want to accept the default directory settings for incremental load jobs as displayed in the task wizard. The target type is Amazon S3. Because the **Connection Directory as Parent** check box is selected by default, the parent directory path that is specified in the **Folder Path** field of the Amazon S3 connection properties is used. This parent directory is `idr-test/dbmi/`. You also must specify a task target directory name, in this case, s3_target, because the {TaskTargetDirectory} placeholder is used in the default patterns in the subsequent directory fields. The files in the data directory and schema directory will be grouped by table name because the {TableName} placeholder is included in their default patterns. Also, because cycle partitioning is enabled, the files in the data directory, schema directory, and cycle summary directories will be subdivided by CDC cycle.

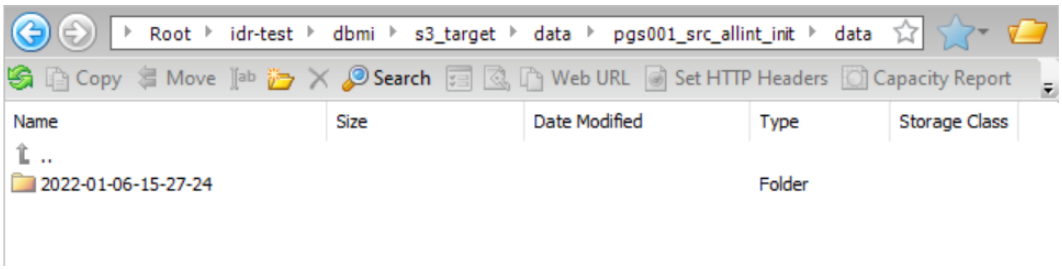
Based on this configuration, the resulting data directory structure is:



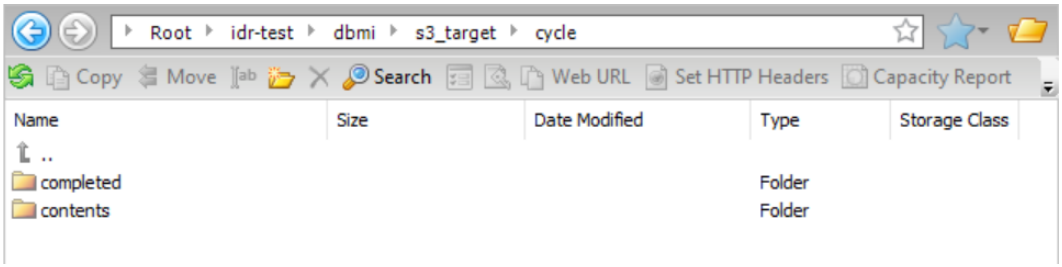
If you drill down on the data folder and then on a table in that folder (pgs001_src_allint_init), you can access the data and schema subdirectories:



If you drill down on the data folder, you can access the timestamp directories for the data files:



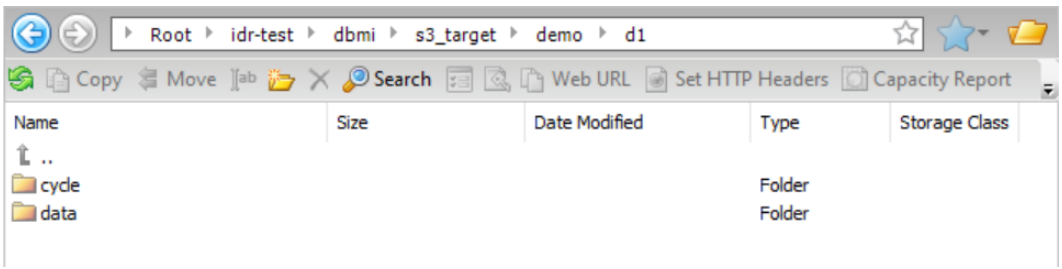
If you drill down on cycle, you can access the summary contents and completed subdirectories:



Example 2

You want to create a custom directory structure for incremental load jobs that adds the subdirectories "demo" and "d1" in all of the directory paths except in the schema directory so that you can easily find the files for your demos. Because the **Connection Directory as Parent** check box is selected, the parent directory path (`idr-test/dbmi/`) that is specified in the **Folder Path** field of the Amazon S3 connection properties is used. You also must specify the task target directory because the {TaskTargetDirectory} placeholder is used in the patterns in the subsequent directory fields. The files in the data directory and schema directory will be grouped by table name. Also, because cycle partitioning is enabled, the files in the data, schema, and cycle summary directories will be subdivided by CDC cycle.

Based on this configuration, the resulting data directory structure is:



Avro data types

Application Ingestion and Replication supports only some of the primitive and logical data types that Avro schemas provide.

A primitive data type is a type that allows you to represent a single data value. A logical type is an Avro primitive or complex type with extra attributes to represent a derived type. This topic applies to all target types that support Avro or Parquet output format.

The following table lists the primitive Avro data types that Application Ingestion and Replication supports:

Primitive data type	Description
INT	32-bit signed integer
LONG	64-bit signed integer
FLOAT	Single precision (32-bit) IEEE 754 floating-point number
DOUBLE	Double precision (64-bit) IEEE 754 floating-point number
BYTES	Sequence of 8-bit unsigned bytes
STRING	Unicode character sequence

The following table lists the logical Avro data types that Application Ingestion and Replication supports:

Logical data type	Description
DECIMAL	An arbitrary-precision signed decimal number of the form unscaled × 10^{-scale}
DATE	A date, without reference to a time or time zone.
TIME	A time of day that has the precision of 1 millisecond or 1 microsecond, without reference to a time zone or date.
TIMESTAMP	A date and time value that has the precision of 1 millisecond or microsecond, without reference to a particular calendar or time zone.

Key concepts

Review the key concepts in this section to understand some configuration options before configuring an application ingestion and replication task.

Integrating Application Ingestion and Replication tasks with Data Integration taskflows

You can configure application ingestion and replication tasks to trigger Data Integration taskflows that process and transform the ingested data.

This feature is available for tasks that use any supported load type and have an Amazon Redshift, Oracle, SQL Server, or Snowflake target. It's also available for tasks that use the initial load type and have an Amazon S3, Azure SQL Database, Databricks, Google BigQuery, Google Cloud Storage, Kafka, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric OneLake, Oracle Cloud Object Storage, or PostgreSQL target.

When you define an application ingestion and replication task, you can select the **Execute in Taskflow** option to make the task available to add to taskflows in Data Integration. For incremental load and combined load jobs with an Amazon Redshift, Oracle, Snowflake (without Superpipe), or SQL Server target, you can optionally select the **Add Cycle ID** option to include cycle ID metadata in the target table. The Cycle ID column identifies the cycle in which the row got updated. It's passed as a parameter to the taskflow, where you can use it to filter the rows on which to execute transformation logic.

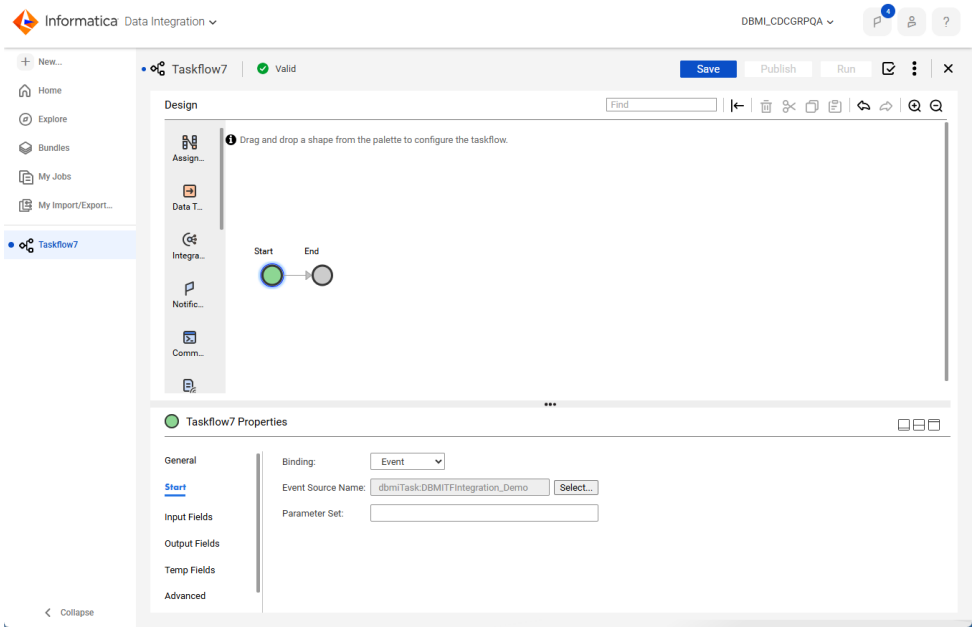
When you configure the taskflow in Data Integration, you can select the task as an event source and add any appropriate transformation type to transform the ingested data.

Configuration task flow:

1. In the Data Ingestion and Replication task configuration wizard, select the following options when defining a application ingestion and replication task:
   - (Optional) **Add Cycle ID** on the **Target** page. For incremental load jobs, this option adds the generated cycle ID for each CDC cycle to the target table. Applies to Amazon Redshift, Oracle, and SQL Server targets and to Snowflake targets that don't use the Superpipe option.
   - **Execute in Taskflow** on the **Schedule and Runtime Options** page. This option makes the task available for selection when defining a taskflow in Data Integration.
2. When done defining the task, **Save** it.
3. To define a taskflow in Data Integration, click the **Orchestrate** panel on the Home page.
4. To add the application ingestion and replication task in the taskflow, perform the following steps:
   a. Under Task Properties, click **Start**.
   b. In the **Binding** field, select **Event**.
   c. In the **Event Source Name** field, click **Select**. Then in **Select Event Source** dialog box, select the application ingestion and replication task and click **Select**.

      Note: You can filter the list of tasks by task type.

d. Check that the **Event Source Name** field and **Input Fields** display the task name. For example:



e. Save and publish the taskflow.

The taskflow is automatically triggered to start when either the initial load task successfully completes or after each CDC cycle in an incremental load operation. If a CDC cycle ends but the previous taskflow run is still running, the data is queued and waits for the previous taskflow to complete.

Handling source schema changes

You can configure Application Ingestion and Replication to automatically detect source schema changes, also called *schema drift*, and handle these changes on the target. This feature is available only for the incremental load and combined initial and incremental load tasks.

When you configure an application ingestion and replication task, on the **Let's Go** page of the application ingestion and replication task wizard, you can specify the types of source schema changes that Application Ingestion and Replication must propagate for the job associated with the task. You can also specify how the job must handle each type of source schema change. For example, you can configure the task to ignore the changes, replicate them, or stop the job when a particular type of schema change occurs on the source. For more information, see "Finalize the task definition" on page 192.

Note: In combined initial and incremental load jobs, Application Ingestion and Replication starts detecting and replicating source schema changes only after the source object reaches the Normal state.

The following table describes the types of schema changes that Application Ingestion and Replication can detect for each source type:

Source type	Supported schema changes
Marketo	Not supported
Microsoft Dynamics 365	- Add column - Modify column - Drop column

Source type	Supported schema changes
NetSuite	- Add column - Modify column
Oracle Fusion Cloud	- REST - Not supported - BICC - Not supported
Salesforce	- Add field - Modify field - Drop field - Rename field
SAP	- Using SAP ODP Extractor connector - Not supported - Using SAP Mass Ingestion connector - Not supported
ServiceNow	- Add column - Modify column - Drop column
Workday	- SOAP - Not supported - RaaS - Not supported

Application Ingestion and Replication detects a schema change in a source object only after Data Manipulation Language (DML) operations occur on the altered source object. If multiple schema changes occur without intervening DML operations, Application Ingestion and Replication detects all the schema changes together when a DML operation occurs.

Note:

- Application ingestion and replication jobs do not replicate source changes that add, remove, or modify primary key or unique key constraints. If these types of changes occur on the source, you must re-synchronize the target tables.
- For source types that allow selecting individual fields in the task configuration, field selection rules take precedence over schema drift. If your field selection rules exclude a field from replication, the job does not add, modify, drop, or rename that field in the target, and does not apply the schema change to it. Schema drift also does not modify your field selection rules. For example, if your field selection rules include only specific fields and you add a new field, the job does not replicate the new field even if you enable the Add Field schema drift option. The added field does not change your rules, so it stays excluded.
- Application ingestion and replication tasks that have Microsoft Azure Synapse Analytics targets cannot replicate rename operations on source fields. The **Replicate** option is not available.
- Application ingestion and replication tasks that have Microsoft Fabric Data Warehouse and Salesforce Data 360 targets cannot replicate drop and rename operations on source fields. The **Replicate** option is not available for these operations.
- Application ingestion and replication tasks that have Microsoft Fabric Lakehouse targets cannot replicate modify operations on source fields. The **Replicate** option is not available.
- Application ingestion and replication tasks that have Snowflake targets support modify operations on source columsn with the following limitations:
  - Snowflake targets cannot modify the scale of NUMBER columns.
  - Snowflake targets do not support changing the data type of an existing column to a different data type.

- Application ingestion and replication tasks that have Google BigQuery targets cannot replicate modify and rename operations on source fields. The schema drift options for these operations are not available.
- If you try to replicate an unsupported schema change type on the target, the application ingestion and replication jobs associated with the task will fail with an error.
- If you configured schema drift options to stop the job when Application Ingestion and Replication detects a schema change, you can use the **Resume With Options** command to resume the job with an override schema drift option.
- In jobs that load data with the Apache Iceberg Open Table format to Amazon S3 targets, the schema drift options for add, modify, drop, and rename fields are set to **Replicate** by default. To prevent the task from failing, ensure to set the schema drift options to **Ignore** so that schema changes are bypassed during processing.

Ability to apply deletes as soft deletes on the target

You can configure application ingestion and replication incremental load and combined initial and incremental load tasks to process delete operations on the source as soft deletes on the target.

Note: The **Soft Deletes** apply mode is available for tasks that have any supported source type and a Databricks, Google BigQuery, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, or Snowflake target, or an SAP Mass Ingestion source and an Amazon Redshift target.

A soft delete marks a deleted row as deleted without actually removing it from the database. The row is applied to the target with the value of "D" in the generated INFA_OPERATION_TYPE metadata column.

Important: You can use Soft Deletes only if all of the source objects have primary keys and the source does not allow the primary key values to change in any row after the row is first created. If the primary key values change, duplicate rows might be written to the target, causing the target to become corrupted.

When you apply a soft delete, the job captures the incoming DELETE event from the source table, finds the active record in the target with the matching key where the INFA_OPERATION_TYPE column is null, and updates it to "D" to mark the record as deleted. However, if the source re-inserts or undeletes a row with the same key and then deletes it again multiple times, the target might reflect each DELETE event as a separate record.

Example scenario: Your organization wants to use soft deletes in a data warehouse to mark the rows that were deleted at the source while still retaining the rows for audit purposes.

To enable soft deletes, set the **Apply Mode** field to **Soft Deletes** on **Target** page of the task wizard when you configure the ingestion and replication task.

Apply an audit history of all source table change operations to target tables

You can configure application ingestion and replication incremental load and combined initial and incremental load tasks to write an audit trail of every DML change operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image.

Note: The **Audit** apply mode is available for tasks that have Databricks, Google BigQuery, Microsoft Fabric Data Warehouse, Microsoft Fabric Lakehouse, Oracle, or Snowflake targets, and for tasks that have an SAP Mass Ingestion source and an Amazon Redshift target.

When you define the task, select **Audit** in the **Apply Mode** field on the **Target** page. The **Apply Mode** field is available for new or undeployed tasks.

To specify the audit metadata columns to add, select one or more of the check boxes under **Advanced** on the **Target** page:

- **Add Last Replicated Time**. Adds a column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp, except for Snowflake targets that use the Superpipe option where the seconds or minutes might vary slightly. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target.
- **Add Operation <*metadata_type*>**. Adds columns that contain metadata for change operations, such as the DML operation type, time, owner, transaction ID, and generated ascending sequence number. The columns are populated when data is loaded to the target tables.
- **Prefix for Metadata Columns**. Add a prefix to the names of the added audit columns to differentiate them from other table columns. Default is INFA_.

These fields are optional. Only the **Add Operation Type** check box is selected by default to add a column that shows the DML operation type: D (delete), or U (update).

The first time you run a job associated with the task, the job generates the target tables with the selected audit metadata columns. Ensure that no constraints other than indexes exist on the target tables.

For targets that support **Soft Deletes** mode, including Snowflake and Databricks targets, for each delete operation processed as a soft delete, the operation type of "D" appears in the INFA_OPERATION_TYPE column and values are written to any other metadata columns that you selected. However, for update and insert operations, the INFA_OPERATION_TYPE column and all other selected metadata columns are NULL.

Example for Audit mode

For example, assume the following DML change operations occur on a source table in the order shown:

```
Insert into tableA pkey = 1
Update tableA where pkey=1
Update tableA where pkey=1
Delete from tableA where pkey = 1
```

All of the following rows appear in the target table, providing an audit trail of all of the source DML changes:

```
opType=I, pkey=1….
opType=E, pkey=1...
opType=E, pkey=1...
opType=D, pkey=1...
```

In this example, the only audit column selected is opType.

Note that when the task's apply mode is Standard, none of these rows appears on the target table because the last DML operation is a Delete, which supersedes the prior changes.

Note: If a combined initial and incremental load job captures an incremental insert change record during the initial unload phase, the job manufactures a delete for the same row to remove any duplicate that might have been obtained from the initial unload. This manufactured activity will be reflected in audit apply mode.

Configuring an application ingestion and replication task

In Data Integration, use the application ingestion and replication task wizard to configure an application ingestion and replication task.

You can integrate initial load and incremental load tasks that have Snowflake targets with Data Integration taskflows. For more information, see "Integrating Application Ingestion and Replication tasks with Data Integration taskflows" on page 73.

To configure application ingestion and replication tasks, click the **Ingest** panel on the Home page and then complete the following configuration tasks in the task configuration wizard:

1. Choose a runtime environment, if you haven't set up a default runtime environment.
2. Select a destination connection, or configure a new connection.
3. Select a source connection, or configure a new connection.
4. Specify task details for the source and target.
5. Configure trim transformations (optional).
6. Finalize the task definition by entering a task name, task definition location, runtime environment, and some optional properties. Then save the task..

Click **Next** or **Back** to navigate from one page to another. At any point, you can click **Save** to save the information that you have entered so far under a generated task name to the Default location in Explore. When you finalize the task definition, you can enter a custom task name and location.

After you complete all wizard pages, save the task definition. You can then click **Deploy** to make the task available as an executable job instance to the Secure Agent.

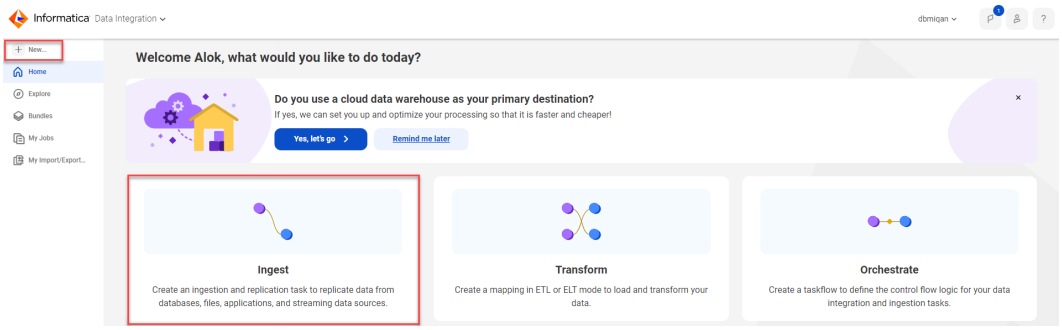
Before you begin

Before you configure an application ingestion and replication task, complete the following prerequisite tasks in Administrator:

- Verify that the Secure Agent in your runtime environment is running and you can access the Data Ingestion and Replication service.
- Define the source and target connections.

Starting the ingestion and replication task wizard

If the latest task wizard is enabled for your organization, you can start the wizard from the **Home** page.



Start the wizard in one of the following ways:

- Click the **Ingest** panel. Then in the **Ingestion and Replication Tasks** dialog box, select **Application Ingestion and Replication Task** or **Database Ingestion and Replication Task**.
- In the navigation bar on the left, click **New**. Then in the **New Asset** dialog box, click **Data Ingestion and Replication** and select either **Application Ingestion and Replication Task** or **Database Ingestion and Replication Task**.
  If you previously selected a primary cloud data warehouse and use this method to start the wizard, the wizard does not recognize the primary cloud data warehouse as the destination. You'll need to select a destination for the task.

Note: File Ingestion and Replication and Streaming Ingestion and Replication use the pre-existing wizard.

Primary cloud data warehouse setup

From the Data Integration Home page, you can configure the primary cloud data warehouse destination where you normally load data.

When you do this, the application ingestion and replication tasks and database ingestion and replication tasks that you create in the new wizard are automatically configured to load data to this destination. You can still change the destination if you need to.

The cloud data warehouse that you choose applies to the organization that you're currently logged into. If you have access to multiple organizations, you can configure a different primary cloud data warehouse for each organization and sub-organization.

The setup steps vary based on whether you've already configured a primary cloud data warehouse. If you've already configured one, you can change or deselect it.

Configuring a primary cloud data warehouse

Configure a primary cloud data warehouse from the **Home** page.

1. On the **Home** page, click **Yes, let's go** in the **Do you use a cloud data warehouse as your primary destination?** panel.
2. On the **Destination** page, select your cloud data warehouse type, for example, Snowflake Data Cloud or Databricks Delta, and click **Next**.
3. On the **Connect** page, select a connection, or click **New** and enter the connection properties.
4. Click **Connect**.

Changing or unselecting a primary cloud data warehouse

If you've already configured a primary cloud data warehouse, you can change or unselect it. Change or unselect a primary cloud data warehouse from the **Home** page.

1. On the **Home** page, click the cloud data warehouse type in the upper right corner and select **Change primary cloud data warehouse**.
2. If you want to change your primary cloud data warehouse, select **I have a primary cloud data warehouse**.
3. To change the cloud data warehouse type, complete the following steps:
   a. Click **Change** next to **Type**.
   b. On the **Destination** page, select the data warehouse type, and then click **Next**.

c. On the **Connect** page, select a connection, or click **New** and enter the connection properties.

d. Click **Connect**.

4. To change the connection, complete the following steps:

a. Click **Change** next to **Connection**.

b. On the **Connect** page, select a connection, or click **New** and enter the connection properties.

c. Click **Connect**.

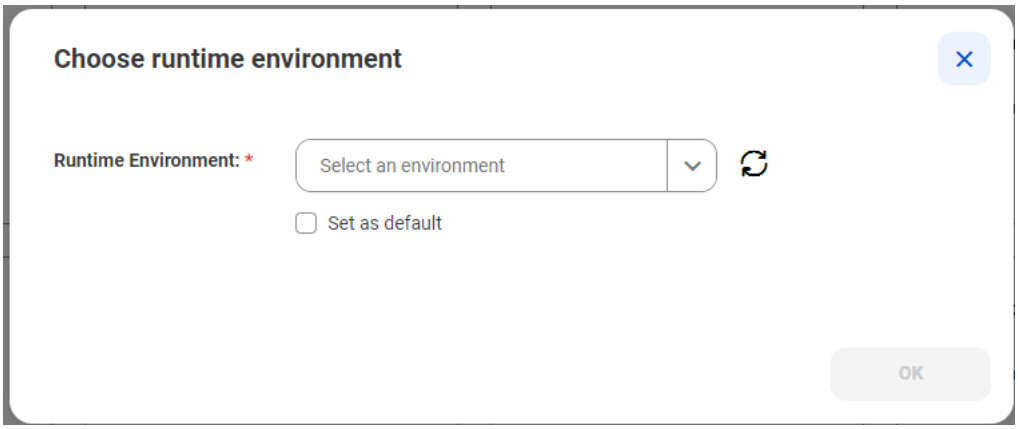
5. If you no longer wish to use a primary cloud data warehouse, select **I don't have a primary cloud data warehouse**, and click **Save**.

Choose a runtime environment

The first thing you must do after starting the task wizard is to select the runtime environment to use for retrieving the source and target metadata required to define the task. If you previously set a default runtime environment, this step is skipped.

Note: A runtime environment must have previously been configured with one or more Secure Agents in Administrator.

1. In the **Choose Runtime Environment** dialog box, select the runtime environment you want to use.



Select **Set as default** if you want to use this runtime environment as the default environment for all tasks you create. Otherwise, leave the check box cleared.

2. Click **OK**.

Note: When you finalize the task definition on the Let's Go page, you'll be prompted to enter the runtime environment for running the task. You can use this same runtime environment or select another one.

Configure the destination connection

On the **Destination** page, select an existing destination connection or add a new one.

This page displays boxes for destination connections that you previously defined from the task wizard or from Administrator.

Note: To add a new connection from the **Destination** page of the new wizard, you must have previously created at least one connection in Administrator.

Perform one of the following actions:

- To select an existing connection, select the box for the destination connection that you want to use. Then click **Next**.
- To add a new connection, complete the following steps:
  1. Click **New Connection**.
  2. On the **New Connection** page > **Destination** tab, select the destination connection type. Then click **Next**.
  3. On the **Configure** tab, enter the connection properties. To help you complete this task, the embedded Setup help on the right describes each property. To enlarge it, drag the left edge.
  4. When done, click **Test** to check that the connection definition works.
  5. Click **Add** to save it.
     The new connection appears on the **Destination** page.
  6. Select the box for the new connection and click **Next**.

After a connection has been chosent, its name is appended to the page name at the top: Destination: <connection_name>.

To manage your connections, go to Administrator.

Tip: As you proceed through the wizard, you can click **Save** to save your task entries under the generated task name at the top of the page to the Default location. On the last page of the wizard, you'll be able to enter a custom name and location for the task.

Configure the source connection

On the **Source** page, select an existing source connection or add a new one.

Note: To add a new connection from the **Source** page, you must have previously created at least one connection in Administrator.

Perform one of the following actions:

- To select an existing connection, select the box for the source connection that you want to use. Then click **Next**.
- To add a new connection, complete the following steps:
  1. Click **New Connection**.
  2. In the **New Connection** dialog box > **Source** tab, select the source connection type. Then click **Next**.
  3. On the **Configure** tab, enter the connection properties. To help you complete this task, the embedded setup help on the right describes each property. To enlarge it, drag the left edge.
  4. Click **Test** to check that the connection definition works.
  5. Click **Add** to save it.
     The new connection appears on the **Source** page.
  6. Select the box for the new connection and click **Next**.

After a connection has been chosent, its name is appended to the page name at the top: Source: <connection_name>.

Task details: Configure how to replicate data from the source

In Step 1 of Task Details, configure the data source.

Under **Source Properties**, set the required basic source properties. Under **Source Objects** or **Source Tables**, select the source objects or tables from which to replicate data. Then under **Advanced Source Properties**, set optional advanced source properties as needed. See the property descriptions for your source type:

- "Configure a Marketo source" on page 82
- "Configure a Microsoft Dynamics 365 source" on page 85
- "Configure a NetSuite source" on page 88
- "Configure an Oracle Fusion Cloud source" on page 92
- "Configure a Salesforce source" on page 95
- "Configure a Salesforce Marketing Cloud source" on page 99
- "Configure an SAP source that uses the SAP OData V2 connector" on page 101
- "Configure an SAP source that uses the SAP ODP Extractor connector" on page 103
- "Configure an SAP source that uses the SAP Mass Ingestion connector" on page 107
- "Configure a ServiceNow source" on page 110
- "Configure a Workday source" on page 113

Configure a Marketo source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.

Source Objects

Select the source objects from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Objects** tab, individually select the check box for each source object you want to include. Clear the check box for any objects you do not want to include. To select all objects, select the **Object** check box at the top.
  The **Field** count for an object shows the total number of fields in the object.

  If you select the objects you want to include, all the selected and unselected objects are displayed by default. To view the selected objects only, use the filter next to the object selection count and change the view from **All** to **Selected**.

  To find objects or fields, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns objects and fields that include CDC anywhere in their names. To narrow the search to only object names or field names, select **Object Name** or **Fields** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for objects.
  To add an object rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing object rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit an object rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the objects and fields that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the object names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
  - Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
  - If an object name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

  Note: If you define multiple object rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if an object rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC objects.

You can both manually select source objects and define selection rules. If you first manually select objects on the **Selected Objects** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any objects selected by those rules are displayed as selected on the **Selected Objects** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Objects by Rule Type	Generate and download a list of the source objects that match the object selection criteria. If you used rule-based object selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Fields** check box to include fields in the list, regardless of which object selection method you used. Click the **Download** icon to download the list.
Start Date	For initial load and combined initial and incremental load jobs, specify the date and time when the ingestion job should start replicating the source data.
Initial Start Poing for Incremental Load	For incremental load jobs, specify the point in the source data stream from which the ingestion job associated with the application ingestion and replication task starts extracting change records. **Note:** You must specify the date and time in Coordinated Universal Time (UTC).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a Microsoft Dynamics 365 source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.

Source Tables

Select the source tables from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Tables** tab, individually select the check box for each source table you want to include. Clear the check box for any tables you do not want to include. To select all tables, select the **Table** check box at the top.
  The **Column** count for a table shows the total number of columns in the table.

  If you select the tables you want to include, all the selected and unselected tables are displayed by default. To view the selected tables only, use the filter next to the table selection count and change the view from **All** to **Selected**.

  To find tables or columns, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns tables and columns that include CDC anywhere in their names. To narrow the search to only table names or column names, select **Table Name** or **Columns** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for tables.
  To add a table rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing table rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit a table rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the tables and columns that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the table names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
  - Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
  - If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

  Note: If you define multiple table rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a table rule specifies "Exclude CDC" followed by "Include C", all tables with names beginning with "C" are selected, including the CDC tables.

You can both manually select source tables and define selection rules. If you first manually select tables on the **Selected Tables** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any tables selected by those rules are displayed as selected on the **Selected Tables** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Tables by Rule Type	Generate and download a list of the source tables that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Columns** check box to include columns in the list, regardless of which table selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in Coordinated Universal Time (UTC).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a NetSuite source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.

Source Tables

Select the source tables and columns from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Tables** tab, individually select the check box for each table and column you want to include. Clear the check box for any table and column you do not want to include. To select all tables and columns, select the **Table** check box at the top.
  The **Column** count for a table shows the total number of columns in the table.

  Note: If you deselect one or more columns for a table, a minus (-) appears next to the table name in the list.

  If you select the tables you want to include, all the selected and unselected tables are displayed by default. To view the selected tables only, use the filter next to the table selection count and change the view from **All** to **Selected**.

  To find tables or columns, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns tables and columns that include CDC anywhere in their names. To narrow the search to only table names or column names, select **Table Name** or **Columns** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for tables and columns.
  - To add a table rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the wildcards, as the condition.

- To add a column rule for a table, click the plus (+) sign at the right end of the table row. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the * wildcard, as the condition.
- Validate the rule. The validation warns if your rule excludes a primary key column.

Tips:

- You can copy an existing table rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
- You can edit a table or column rule directly in its row by clicking the **Type** or **Match Condition** value.
- To view the tables and columns that match a rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

When you define a condition in a rule, use the following guidelines:

- The task wizard is case sensitive. Enter the table and column names or masks in the case with which they were defined.
- A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If a table or column name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple table rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a table rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC object.

You can both manually select tables and columns and define selection rules. If you first manually select tables and columns on the **Selected Tables** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any tables and columns selected by those rules are displayed as selected on the **Selected Tables** tab. Expand the table on the **Selection Rules** tab to view details of the rules applied to fields.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Tables by Rule Type	Generate and download a list of the source tables that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Columns** check box to include columns in the list, regardless of which table selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in Greenwich Mean Time (GMT).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. The default value is 5000.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure an Oracle Fusion Cloud source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
Oracle Fusion Replication Approach	Select one of the following replication approaches - Select **REST** to extract data from various applications of Oracle Fusion such as ERP, SCM, HCM, Sales, and Services, and transfer data to the target. - Select **BICC** (Business Intelligence Cloud Connector) to extract bulk data from the source to the target.
Oracle Fusion Application	Select the application from which you want to replicate data.

Source Objects

Select the source objects from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Objects** tab, individually select the check box for each source object you want to include. Clear the check box for any objects you do not want to include. To select all objects, select the **Object** check box at the top.
  The **Field** count for an object shows the total number of fields in the object.

  If you select the objects you want to include, all the selected and unselected objects are displayed by default. To view the selected objects only, use the filter next to the object selection count and change the view from **All** to **Selected**.

  To find objects or fields, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns objects and fields that include CDC anywhere in their names. To narrow the search to only object names or field names, select **Object Name** or **Fields** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for objects.
  To add an object rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing object rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit an object rule directly in its row by clicking the **Type** or **Match Condition** value.

- To view the objects and fields that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

When you define a condition in a rule, use the following guidelines:

- The task wizard is case sensitive. Enter the object names or masks in the case with which they were defined.
- A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If an object name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple object rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if an object rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC objects.

You can both manually select source objects and define selection rules. If you first manually select objects on the **Selected Objects** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any objects selected by those rules are displayed as selected on the **Selected Objects** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Objects by Rule Type	Generate and download a list of the source objects that match the object selection criteria. If you used rule-based object selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Fields** check box to include fields in the list, regardless of which object selection method you used. Click the **Download** icon to download the list.
Include Child Objects	Get the child object data of an object using an Oracle Fusion Cloud source. This applies only for the REST replication approach and for all load types only when the target is Google Big Query.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in the time zone configured for the Oracle Fusion Cloud instance.
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.

Property	Description
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. The default value is 50000.
Enable chunking	Select this checkbox to chunk data during the extraction process in an application ingestion and replication task. Chunking applies to initial load tasks, and in the initial portion in combined initial and incremental load tasks that use the BICC replication approach. Chunking is not applicable for CDC tasks. Options are: - **None**. Extracts data as a whole without dividing it into chunks. Default is None. - **By Primary Key**. Select this option to chunk data based on a numeric primay key. This option requires that your datastore contains a single numeric primary key column without Null values. - **By Creation Date.** Select this option to divide data into chunks at intervals defined by the number of days for date range extraction. This option requires selection of the *Is Creation Date* option in the BICC Console for a column or columns in the column list, which represent the creation date.
Number of Rows	If you select the **By Primary Key** chunking option, specify the number of rows to chunk extracts. Enter a positive integer. Make sure that the combination of total rows and chunk size results in no more than 250 chunks. For example, if you have 2500 rows with a chunk size of 10, it results in 250 chunks, which is the maximum chunks allowed.
Number of Days	If you select the **By Creation Date** chunking option, enter the number of days to set the interval for data extraction. For example, if you specify 365 days, the data is divided into segments that each cover a 365-day period, starting from the initial date. To prevent performance issues, consider specifying a larger number of days for the interval.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a Salesforce source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
Salesforce API	For initial load tasks and combined initial and incremental load tasks, select the type of Salesforce API that you want to use to retrieve the source data. Options are: - **Standard (REST) API**: Replicates source fields of Base64 data type. Informatica recommends that you use the Bulk API 2.0 unless you want to ingest fields of Base64 data type or objects that are not supported by Bulk API 2.0 during initial loading of data. All incremental load activities use only the standard REST API. - **Bulk API 2.0**: Excludes replication of source fields of Base64 data type. Bulk API 2.0 is the default API for initial load tasks and the initial load of the combined initial and incremental load tasks. - **Bulk API**: Uses Bulk API 1.0 for primary-key chunking to achieve parallel processing in Salesforce that optimizes the performance and speed of initial and combined initial and incremental load jobs. Use this option to handle large-scale data from Salesforce. **Note:** By default, incremental load tasks can capture and replicate change data from source fields of Base64 data type.

Source Objects

Select the source objects and fields from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Objects** tab, individually select the check box for each source object and field you want to include. Clear the check box for any objects and fields you do not want to include. To select all objects and fields, select the **Object** check box at the top.
  The **Field** count for an object shows the total number of fields in the object.

  Note: If you deselect one or more fields for an object, a minus (-) appears next to the object name in the list.

  If you select the objects you want to include, all the selected and unselected objects are displayed by default. To view the selected objects only, use the filter next to the object selection count and change the view from **All** to **Selected**.

  To find objects or fields, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns objects and fields that include CDC anywhere in their names. To narrow the search to only object names or field names, select **Object Name** or **Fields** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for objects and fields.
  - To add an object rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the wildcards, as the condition.
  - To add a field rule for an object, click the plus (+) sign at the right end of the object row. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the * wildcard, as the condition.
  - Validate the rule. The validation warns if your rule excludes a primary key field.

  Tips:

  - You can copy an existing object rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit an object or field rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the objects and fields that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the object and field names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
  - Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
  - If an object or field name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

  Note: If you define multiple object rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if an object rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC object.

You can both manually select source objects and fields and define selection rules. If you first manually select objects and fields on the **Selected Objects** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any objects and fields selected by those rules are displayed as selected on the **Selected Objects** tab. Expand the object on the **Selection Rules** tab to view details of the rules applied to fields.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Objects by Rule Type	Generate and download a list of the source objects that match the object selection criteria. If you used rule-based object selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Fields** check box to include fields in the list, regardless of which object selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in Greenwich Mean Time (GMT).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. The default value for initial load operations is 50000 and the default value for incremental load operations is 2000. **Note:** For combined initial and incremental load tasks, you must specify the fetch size separately for initial load operations and incremental load operations.
Include Base64 Fields	Select this check box to replicate the source fields of Base64 data type. **Note:** - You can replicate the Base64 fields only if you select **Standard (REST) API** as the Salesforce API. - Replication of Base64 data might slow down the initial load operation of the application ingestion and replication job.
Maximum Base64 Body Size	If you selected the **Include Base64 Fields** check box, specify the maximum body size in megabytes (MB) for Base64 encoded data.
Include Archived and Deleted Rows	For initial load and combined initial and incremental load jobs, select this check box to replicate the archived and soft-deleted rows from the source during the initial loading of data.

Property	Description
Enable Partitioning	For initial load and combined initial and incremental load tasks, select this check box to partition the source objects for initial loading.
Chunk Size	If you enable partitioning of source objects,enter the number of records to be processed in a single partition. Based on the chunk size, bulk jobs are created in Salesforce. The default value is 50000 and the minimum value is 100.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Salesforce Max Parallel Partition	The maximum number of partition threads that can be used to query the source for data in parallel during initial load processing or the unload phase of combined jobs. Use this property to control the number of source partition queries that can be executed against the source at the same time. For example, if a table contains data in 100 partitions, all 100 partitions are queried at the same time by default. However, you can use this property to reduce the number of concurrent queries. Default value is equal to the total number of partitions.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a Salesforce Marketing Cloud source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	**Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent.
MID	Enter the unique Member Identification code assigned to your Salesforce Marketing Cloud account.

Source Objects

Select the source objects from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Objects** tab, individually select the check box for each source object you want to include. Clear the check box for any objects you do not want to include. To select all objects, select the **Object** check box at the top.
  The **Field** count for an object shows the total number of fields in the object.

  If you select the objects you want to include, all the selected and unselected objects are displayed by default. To view the selected objects only, use the filter next to the object selection count and change the view from **All** to **Selected**.

  To find objects or fields, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns objects and fields that include CDC anywhere in their names. To narrow the search to only object names or field names, select **Object Name** or **Fields** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for objects.
  To add an object rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing object rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit an object rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the objects and fields that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the object names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
  - Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
  - If an object name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple object rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if an object rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC objects.

You can both manually select source objects and define selection rules. If you first manually select objects on the **Selected Objects** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any objects selected by those rules are displayed as selected on the **Selected Objects** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Objects by Rule Type	Generate and download a list of the source objects that match the object selection criteria. If you used rule-based object selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Fields** check box to include fields in the list, regardless of which object selection method you used. Click the **Download** icon to download the list.
Batch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. Default is 2500.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

   Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure an SAP source that uses the SAP OData V2 connector

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
OData Service Name	Select the OData service endpoint from where you want to retrieve data. The list contains a specific SAP service or a list of all available services on the SAP Gateway based on the service type you specified in the SAP OData V2 connection.

Source Tables

Select the source tables from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Tables** tab, individually select the check box for each source table you want to include. Clear the check box for any tables you do not want to include. To select all tables, select the **Table** check box at the top.
  The **Column** count for a table shows the total number of columns in the table.

  If you select the tables you want to include, all the selected and unselected tables are displayed by default. To view the selected tables only, use the filter next to the table selection count and change the view from **All** to **Selected**.

  To find tables or columns, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns tables and columns that include CDC anywhere in their names. To narrow the search to only table names or column names, select **Table Name** or **Columns** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for tables.
  To add a table rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing table rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit a table rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the tables and columns that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

When you define a condition in a rule, use the following guidelines:

- The task wizard is case sensitive. Enter the table names or masks in the case with which they were defined.
- A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple table rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a table rule specifies "Exclude CDC" followed by "Include C", all tables with names beginning with "C" are selected, including the CDC tables.

You can both manually select source tables and define selection rules. If you first manually select tables on the **Selected Tables** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any tables selected by those rules are displayed as selected on the **Selected Tables** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Tables by Rule Type	Generate and download a list of the source tables that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Columns** check box to include columns in the list, regardless of which table selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs.
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 1 day.
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure an SAP source that uses the SAP ODP Extractor connector

Define source properties for the SAP ECC or SAP S4/HANA source using the SAP ODP Extractor connector that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
Context	Select the context containing the source data sources that you want to replicate on the target.

SAP ODP Extractor Connector supports the following ODP providers or contexts for all load types:

Providers/Context	Source SAP System and ODPs
SAP Service Application Programming Interface (S-API)	SAP Data Sources/Extractors without Enterprise Search (ESH)
HANA	SAP HANA Information View
BW	SAP NetWeaver Business Warehouse
ABAP_CDS	ABAP Core Data Services
SAP SLT	SLT Queue

Source Data Sources

Select the data sources from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Data Sources** tab, individually select the check box for each data source you want to include. Clear the check box for any data sources you do not want to include. To select all data sources, select the **Data Source** check box at the top.
  The **Field** count for a data source shows the total number of fields in the data source.

  If you select the data sources you want to include, all the selected and unselected data sources are displayed by default. To view the selected data sources only, use the filter next to the data source selection count and change the view from **All** to **Selected**.

  To find data sources or fields, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns data sources and fields that include CDC anywhere in their names. To narrow the search to only object names or field names, select **Data Source Name** or **Fields** in the drop-down list adjacent to the *Find* box.
- Click **Show Fields** to display the actual columns from the SAP source. Otherwise, the columns for the selected data sources display as dummy.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for data sources.
  To add a data source rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing data source rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit a data source rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the data sources and fields that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the data sources to be included. For an Exclude rule, it shows the data sourcess to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the data source names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.

- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If an object name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple data source rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a data source rule specifies "Exclude CDC" followed by "Include C", all data sources with names beginning with "C" are selected, including the CDC data source.

You can both manually select data sources and define selection rules. If you first manually select data sources on the **Selected Data Sources** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any data sources selected by those rules are displayed as selected on the **Selected Data Sources** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Data Sources by Rule Type	Generate and download a list of the data sources that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Fields** check box to include fields in the list, regardless of which object selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** By default, the ingestion job retrieves the change records from the latest available position in the data stream.
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes. **Note:** The CDC interval must be less than the data retention period configured in the SAP system for the Operational Delta Queue (ODQ).
Fetch Size	Enter the size of data that the application ingestion and replication job associated with the task reads at a time from the source. The value must be in megabytes (MB). The default value is 8.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure an SAP source that uses the SAP Mass Ingestion connector

Define source properties for the SAP ECC or SAP S4/HANA source using the SAP Mass Ingestion connector that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
Schema	For incremental load and combined initial and incremental load jobs, enter the underlying database schema that includes the source tables. Perform the following steps to enter the schema value: - Log in to the SAP application. - Browse to **System** > **Status** - Check the **Owner** value. Enter this value in the **Schema** field.

Source Tables

Select the tables and columns from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Tables** tab, individually select the check box for each table and column you want to include. Clear the check box for any table and column you do not want to include. To select all tables and columns, select the **Table** check box at the top.
  The **Column** count for a table shows the total number of columns in the table.

  Note: If you deselect one or more columns for a table, a minus (-) appears next to the table name in the list.

  If you select the tables you want to include, all the selected and unselected tables are displayed by default. To view the selected tables only, use the filter next to the table selection count and change the view from **All** to **Selected**.

  To find tables or columns, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns tables and columns that include CDC anywhere in their names. To narrow the search to only table names or column names, select **Table Name** or **Columns** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for tables and columns.
  - To add a table rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the wildcards, as the condition.

- To add a column rule for a table, click the plus (+) sign at the right end of the table row. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without the * wildcard, as the condition.
- Validate the rule. The validation warns if your rule excludes a primary key column.

Tips:

- You can copy an existing table rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
- You can edit a table or column rule directly in its row by clicking the **Type** or **Match Condition** value.
- To view the tables and columns that match a rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

When you define a condition in a rule, use the following guidelines:

- The task wizard is case sensitive. Enter the table and column names or masks in the case with which they were defined.
- A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If a table or column name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple table rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a table rule specifies "Exclude CDC" followed by "Include C", all objects with names beginning with "C" are selected, including the CDC object.

You can both manually select tables and columns and define selection rules. If you first manually select tables and columns on the **Selected Tables** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any tables and columns selected by those rules are displayed as selected on the **Selected Tables** tab. Expand the table on the **Selection Rules** tab to view details of the rules applied to fields.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
CDC Script	For incremental loads and combined initial and incremental loads, generate a script for enabling CDC on source tables and then run or download the script. The only available option is **Enable CDC for all columns**. Click **Execute** to run the script if you have the required privileges. Or click the **Download** icon to download the script so that you can give it to your DBA to run.
List Tables by Rule Type	Generate and download a list of the source tables that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Columns** check box to include columns in the list, regardless of which table selection method you used. Click the **Download** icon to download the list.
Enable Persistent Storage	For incremental loads and combined initial and incremental loads, select this check box to enable persistent storage of transaction data in a disk buffer so that the data can be consumed continually, even when the writing of data to the target is slow or delayed. Benefits of using persistent storage are faster consumption of the source transaction logs, less reliance on log archives or backups, and the ability to still access the data persisted in disk storage after restarting an ingestion job. Persisted data is stored on the Secure Agent. It is not encrypted. The Secure Agent's files and directories are expected to be secured from unwanted access by using native file system access permissions or file system support of encryption natively.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** By default, the ingestion job retrieves the change records from the latest available position in the data stream.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a ServiceNow source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.

Source Tables

Select the source tables from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Tables** tab, individually select the check box for each source table you want to include. Clear the check box for any tables you do not want to include. To select all tables, select the **Table** check box at the top.
  The **Column** count for a table shows the total number of columns in the table.

  If you select the tables you want to include, all the selected and unselected tables are displayed by default. To view the selected tables only, use the filter next to the table selection count and change the view from **All** to **Selected**.

  To find tables or columns, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns tables and columns that include CDC anywhere in their names. To narrow the search to only table names or column names, select **Table Name** or **Columns** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for tables.
  To add a table rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing table rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit a table rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the tables and columns that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the table names or masks in the case with which they were defined.
  - A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
  - Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
  - If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

  Note: If you define multiple table rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if a table rule specifies "Exclude CDC" followed by "Include C", all tables with names beginning with "C" are selected, including the CDC tables.

You can both manually select source tables and define selection rules. If you first manually select tables on the **Selected Tables** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any tables selected by those rules are displayed as selected on the **Selected Tables** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Property	Description
List Tables by Rule Type	Generate and download a list of the source tables that match the table selection criteria. If you used rule-based table selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Select the **Include Columns** check box to include columns in the list, regardless of which table selection method you used. Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in Greenwich Mean Time (GMT).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. The default value is 10000.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Configure a Workday source

Define source properties for the source that you selected on the **Source** page.

Source Properties

Configure the basic source properties:

Property	Description
Load Type	Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task: - **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent. - **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run. - **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis.
Workday API	select the type of web service that you want to use to read source data. Options are: - **SOAP**: Uses SOAP APIs to extract Workday data. - **RaaS**: Uses Workday Report-as-a-Service (RaaS) to extract source data from custom objects and fields through custom reports. You can use Workday RaaS only in initial load jobs.

If you choose to use the SOAP API, perform the following steps:

1. From the **Product** list, select **Human Capital Management**.
2. From the **Services** list, select the Human Capital Management (HCM) services from which you want to ingest data to your target.
   You can select multiple services from the **Services** list.
3. From the **Output Type** list, select the format in which you want the data to be stored on the target.
   The ingestion jobs extract the source data in an XML structure. Based on the format that you select, the job writes the extracted data to the target as a single object in either JSON or XML format.

If you choose to use the RaaS API, perform the following steps:

1. In the **Number of Reports** field, select the number of reports you want to extract from the source.
2. If you choose to extract a single report, in the **Report Name or URL** field, enter the name or URL of the custom report you want to read from the source.
3. If you choose to extract multiple reports, in the **Report Configuration File** field, enter the path to the CSV file that you created for the list of custom reports that you want to read from the source.

Source Operations

Select the source operations from which you want to replicate data. Use one or both of the following methods:

- On the **Selected Operations** tab, individually select the check box for each source operation you want to include. Clear the check box for any ooperations you do not want to include. To select all operations, select the **Operation** check box at the top.
  The **Attribute** count for an operation shows the total number of attributes in the operation.

  If you select the operations you want to include, all the selected and unselected operations are displayed by default. To view the selected operations only, use the filter next to the operation selection count and change the view from **All** to **Selected**.

  To find operations or attributes, you can type all or part of a name in the *Find* box and click **Search**. This value is case-sensitive. If you type the beginning of a name only, a wildcard isn't required to represent the remainder. For example, CDC, CD, and CD* return the same results. However, if the search string is within the name, include the wildcard * at the beginning. For example, *CDC returns operations and attributes that include CDC anywhere in their names. To narrow the search to only operation names or field names, select **Operation Name** or **Attributes** in the drop-down list adjacent to the *Find* box.
- On the **Selection Rules** tab, you can create inclusion and exclusion rules for operations.
  To add an operation rule, click the plus (+) sign in the upper right corner. In the **Type** field, select **Include** or **Exclude** as the rule type. Then enter a string, with or without wildcards, as the condition.

  Tips:

  - You can copy an existing operation rule to use as a starting point for creating another rule. Click the copy icon at the right end of the row.
  - You can edit an operation rule directly in its row by clicking the **Type** or **Match Condition** value.
  - To view the operations and attributes that match a single rule, click **View Objects** in the row for the rule. For an Include rule, it shows the objects to be included. For an Exclude rule, it shows the objects to be excluded.

  When you define a condition in a rule, use the following guidelines:

  - The task wizard is case sensitive. Enter the source operation names or masks in the case with which they were defined.

- A mask can contain one or more wildcards. Supported wildcards are: an asterisk (*), which represents one or more characters, and a question mark (?), which represents a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- Delimiters such as quotation marks or brackets are not allowed, even if the source uses them.
- If an source operation name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

Note: If you define multiple operation rules, they're processed in the order in which they're listed (top to bottom). Be sure to define them in the correct order of processing. For example, if an operation rule specifies "Exclude CDC" followed by "Include C", all operations with names beginning with "C" are selected, including the CDC operations.

You can both manually select operations and define selection rules. If you first manually select operations on the **Selected Operations** tab, rules are generated and displayed for those selections on the **Selection Rules** tab. Similarly, if you first define rules, any operations selected by those rules are displayed as selected on the **Selected Operations** tab.

Advanced Source Properties

To configure advanced source properties, toggle on **Show Advanced Options** at the top of the page. Advanced source properties are optional or have default values.

1. Complete the following optional advanced properties as needed:

Method	Description
List Operations by Rule Type	Generate and download a list of the sourceoperation that match the operation selection criteria. If you used rule-based operation selection, you can select the type of selection rules to use. Options are: - **Include Rules Only** - **Exclude Rules Only** - **Include And Exclude Rules** Click the **Download** icon to download the list.
Initial Start Point for Incremental Load	For incremental load jobs, customize the position in the source logs from which the application ingestion and replication job starts reading change records the first time it runs. **Note:** You must specify the date and time in Coordinated Universal Time (UTC).
CDC Interval	For incremental load and combined initial and incremental load jobs, specify the time interval in which the application ingestion and replication job runs to retrieve the change records for incremental load. The default interval is 5 minutes.
Fetch Size	Enter the number of records that the application ingestion and replication job associated with the task reads at a time from the source. The default value is 100. **Note:** The **Fetch Size** field appears only for the SOAP API.
Extract Non-default Fields	Select this check box to replicate the source fields that do not contain any default value. **Note:** The **Extract Non-default Fields** check box appears only for the SOAP API.

2. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Read Event Batch Size	The number of payload events written in batch to the internal event queue during CDC processing. When the event queue is implemented as an internal ring buffer, this value is the number of payload events that the reader writes to a single internal buffer slot. **Note:** A batch size that's too small might increase contention between threads. A larger batch size can provide for more parallelism but consume more memory.
Reader Helper Thread Count	The number of reader helper threads used during CDC processing to convert change data into a canonical format that can be passed to the target. Default value is 3. You can enter a larger value to allow more threads to be available for performing conversion processing in parallel.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed to Step 2 of Task Details.

Task details: Configure how to replicate data to the target

Configure the data target in Step 2 of Task Details.

- Under **Target Properties**, set the required basic target properties. Then toggle on **Show Advanced Options** at the top of the page to set optional advanced target properties as needed. See the property descriptions for your target type:
  - "Configure an Amazon Redshift target" on page 117
  - "Configure an Amazon S3 target" on page 121
  - "Configure a Databricks target" on page 127
  - "Configure a Google BigQuery target" on page 132
  - "Configure a Google Cloud Storage target" on page 136
  - "Configure a Kafka target" on page 142
  - "Configure a Microsoft Azure Data Lake Storage Gen2 target" on page 145
  - "Configure a Microsoft Azure Synapse Analytics target" on page 151
  - "Configure a Microsoft Fabric Data Warehouse target" on page 153
  - "Configure a Microsoft Fabric Lakehouse target" on page 157
  - "Configure a Microsoft Fabric OneLake target" on page 160

- "Configure a Microsoft SQL Server target" on page 166
- "Configure an Oracle target" on page 174
- "Configure an Oracle Cloud Object Storage target" on page 168
- "Configure a PostgreSQL target" on page 178
- "Configure a Salesforce Data 360 target" on page 180
- "Configure a Snowflake Data Cloud target" on page 182

Configure an Amazon Redshift target

Define target properties for the Amazon Redshift destination.

Target Properties

Define the following required Amazon Redshift target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using the Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. **Note:** The **Audit** and **Soft Deletes** apply modes are available only for an SAP source with SAP Mass Ingestion connector. The default value is **Standard**.

Property	Description
Bucket	The name of the Amazon S3 bucket that stores, organizes, and controls access to the data objects that you load to Amazon Redshift.
Data Directory or Task Target Directory	The subdirectory where Application Ingestion and Replication stores output files for jobs associated with the task. This field is called **Data Directory** for an initial load job or **Task Target Directory** for an incremental load or combined initial and incremental load job.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected. You cannot deselect it if you are using Soft Deletes.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Add Operation Sequence	Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation. This field is available only when **Apply Mode** is set to **Audit**. By default, this check box is not selected.

Property	Description
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Enable Case Transformation	By default, target table names and column names are generated in the same case as the corresponding source names, unless cluster-level or session-level properties on the target override this case-sensitive behavior. If you want to control the case of letters in the target names, select this check box. Then select a **Case Transformation Strategy** option.
Case Transformation Strategy	If you selected **Enable Case Transformation**, select one of the following options to specify how to handle the case of letters in generated target table (or object) names and column (or field) names: - **Same as source**. Use the same case as the source table (or object) names and column (or field) names. - **UPPERCASE**. Use all uppercase. - **lowercase**. Use all lowercase. The default value is **Same as source**. **Note:** The selected strategy will override any cluster-level or session-level properties on the target for controlling case.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.

Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)

4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

   Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

   Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure an Amazon S3 target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Amazon S3 target properties:

Property	Description
Open Table Format	Controls whether to use Open Table format to replicate data to Amazon S3 storage. Select one of the following options: - **Apache Iceberg**. Replicate data to the Amazon S3 cloud storage as Apache Iceberg tables. You can access these tables directly from Amazon S3 using the AWS Glue Catalog. - **None**. Do not use Open Table format. The default value is **None**.
Namespace	If you use the Apache Iceberg Open Table format, enter the name of the database in AWS Glue Catalog where you want to store and manage your Apache Iceberg tables.
Output Format	The format of the output files. Options are: - **CSV** - **AVRO** - **PARQUET** The default value is **CSV**. **Note:** Output files in CSV format use double-quotation marks ("") as the delimiter for each field.
Warehouse Base Directory	If you use the Apache Iceberg Open Table format, enter the root directory in Amazon S3 where you want to store the target files and tables.
Add Headers to CSV File	If **CSV** is selected as the output format, select this check box to add a header with source column names to the output CSV file.
Parquet Compression Type	If the **PARQUET** output format is selected, you can select a compression type that is supported by Parquet. Options are: - **None** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **None** - **Binary** - **JSON** The default value is **Binary**.

Property	Description
Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern: *schemaname_tablename*.txt **Note:** If this directory is not specified, no Avro schema definition file is produced.
File Compression Type	Controls whether to use a file compression type for output files in CSV or AVRO output format. Options are: - **None** - **Deflate** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Encryption Type	Select the encryption type for the output files that are written to the Amazon S3 target. Options are: - **None** - **Client Side Encryption** - **Client Side Encryption with KMS** - **Server Side Encryption** - **Server Side Encryption with KMS** **Note:** If you use **Client Side Encryption** or **Client Side Encryption with KMS**, the maximum size of the file that is encrypted and decrypted is 64 GB. The default is **None**, which means no encryption is used.
Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.
Deflate Compression Level	If **Deflate** is selected in the **Avro Compression Type** field, specify a compression level from 0 to 9. The default value is 0.
Add Directory Tags	For incremental load and combined initial and incremental load tasks, select this check box to add the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. This check box is cleared by default.
Task Target Directory	For incremental load and combined initial and incremental load tasks, the root directory for the other directories that hold output data files, schema files, and CDC cycle contents and completed files. You can use it to specify a custom root directory for the task. If you enable the **Connection Directory as Parent** option, you can still optionally specify a task target directory to use with the parent directory specified in the connection properties. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.

Property	Description

Data Directory	*For initial load tasks*, define a directory structure for the directories where Application Ingestion and Replication stores output data files and optionally stores the schema.
	The default directory pattern is `{TableName)_{Timestamp}`.
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter a folder name or use variables to create a folder name.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the initial load job starts to transfer data to the target.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	Note: If you manually enter the directory expression, ensure that you enclose placeholders with curly brackets { }. Placeholder values are not case sensitive.
	For example:
	`myDir1/{SchemaName}/{TableName}`
	`myDir1/myDir2/{SchemaName}/{YYYY}/{MM}/{TableName}_{Timestamp}`
	`myDir1/{toLower(SchemaName)}/{TableName}_{Timestamp}`
	For incremental load and combined initial and incremental load tasks, define a custom path to the subdirectory that contains the cdc-data data files.
	The default directory pattern is `{TaskTargetDirectory}/data/{TableName}/data`
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the CDC cycle started.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	-
	For Amazon S3 and Microsoft Azure Data Lake Storage Gen2 targets, Application Ingestion and Replication uses the directory specified in the target connection properties as the root for the data directory path when **Connection Directory as Parent** is selected.
	For Google Cloud Storage targets, Application Ingestion and Replication uses the **Bucket** name that you specify in the target properties for the ingestion task.
	For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.
	For Amazon S3 targets with Open Table format, the data directory field is not applicable. Enabling the **Connection Directory as Parent** includes the connection directory before the warehouse base path. If disabled, files are saved directly under the warehouse base directory.

Property	Description
Connection Directory as Parent	If you use the Open Table format, select this check box to use the directory value specified in the target connection properties as the parent directory. This path appends to the file path on S3 while creating the file. This check box is selected by default. For example, if the S3 directory set in the connection is `myFolderOnS3/F1` and the **Warehouse Base Directory** is `/myFold`, files are saved to `myFolderOnS3/F1/myFold/<files>`. However, if you do not select the **Connection Directory as Parent** option, files are saved directly to `/myFold/<files>`. If you do not use the Open Table format, selecting this check box uses the directory value specified in the target connection properties as the parent directory for the custom directory paths specified in the task target properties. For initial load tasks, the parent directory is used in the **Data Directory** and **Schema Directory**. For incremental load and combined initial and incremental load tasks, the parent directory is used in the **Data Directory**, **Schema Directory**, **Cycle Completion Directory**, and **Cycle Contents Directory**. This check box is selected by default. If you clear it, for initial loads, define the full path to the output files in the **Data Directory** field. For incremental loads, optionally specify a root directory for the task in the **Task Target Directory**.
Schema Directory	Specify a custom directory for storing the schema file if you want to store it in a directory other than the default directory. For initial loads, previously used values if available are shown in a list for your convenience. This field is optional. For initial loads, the schema is stored in the data directory by default. For incremental loads and combined initial and incremental loads, the schema file is stored in the following directory by default: `{TaskTargetDirectory}/data/{TableName}/schema`. You can use the same placeholders as for the **Data Directory** field. If you manually enter placeholders, ensure that you enclose them with curly brackets { }. If you include the toUpper or toLower function, put the placeholder name in parentheses and enclose both the function and placeholder in curly brackets, for example: `{toLower(SchemaName)}` **Note:** Schema is written only to output data files in CSV format. Data files in Parquet and Avro formats contain their own embedded schema.
Cycle Completion Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle completed file. Default is `{TaskTargetDirectory}/cycle/completed`.
Cycle Contents Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle contents files. Default is `{TaskTargetDirectory}/cycle/contents`.
Use Cycle Partitioning for Data Directory	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.

Property	Description
Use Cycle Partitioning for Summary Directories	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under the summary contents and completed subdirectories.
List Individual Files in Contents	For incremental load and combined initial and incremental load tasks, lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target. For incremental loads, the job writes "I" for insert, "U" for update, or "D" for delete. For initial loads, the job always writes "I" for insert. By default, this check box is selected for incremental load and initial and incremental load jobs, and cleared for initial load jobs.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. For initial loads, the job always writes the current date and time. By default, this check box is not selected.
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

Notes:

- If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
- If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
- On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Databricks target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Databricks target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.

Property	Description
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. After enabling Soft Deletes, any update in the source table during normal or backlog mode results in the deletion of the matching record, insertion of the updated record, and marking of the INFA_OPERATION_TYPE operation as NULL in the target table. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, transaction ID, and before image. Consider using **Audit** apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. After enabling the Audit apply mode, any update in the source table during backlog or normal mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Default is **Standard**.
Data Directory or Task Target Directory	The subdirectory where Application Ingestion and Replication stores output files for jobs associated with the task. This field is called **Data Directory** for an initial load job or **Task Target Directory** for an incremental load or combined initial and incremental load job.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected and you you cannot deselect it.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Add Operation Sequence	Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation. This field is available only when **Apply Mode** is set to **Audit**. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Create Unmanaged Tables	Select this check box if you want the task to create Databricks target tables as unmanaged tables. After you deploy the task, you cannot edit this field to switch to managed tables. By default, this option is cleared and managed tables are created. If you selected **Personal Staging Location** in the **Staging Environment** field in the selected Databricks target connection, this check box is disabled. You cannot use unmanaged tables in this situation. For more information about Databricks managed and unmanaged tables, see the Databricks documentation.

Property	Description
Unmanaged Tables Parent Directory	If you choose to create Databricks unmanaged tables, you must specify a parent directory in Amazon S3 or Microsoft Azure Data Lake Storage to hold the Parquet files that are generated for each target table when captured DML records are processed. **Note:** To use Unity Catalog, you must provide an existing external directory. **Note:** For volume staging, provide the complete parent directory path.
Staging File Format	Select the format of the staging files in the staging environment specified in the Databricks connection. The files hold data before it's loaded into Databricks tables. Format options are: - CSV - Parquet Default is CSV.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Google BigQuery target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Google BigQuery target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using the Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. The default value is **Standard**.
Bucket	The name of an existing bucket container that stores, organizes, and controls access to the data objects that you load to Google Cloud Storage.
Data Directory or Task Target Directory	The subdirectory where Application Ingestion and Replication stores output files for jobs associated with the task. This field is called **Data Directory** for an initial load job or **Task Target Directory** for an incremental load or combined initial and incremental load job.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected and you you cannot deselect it.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Add Operation Sequence	Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation. This field is available only when **Apply Mode** is set to **Audit**. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. Do not include special characters in the prefix. Otherwise, task deployment will fail. The default value is INFA_.

Property	Description
Enable Case Transformation	By default, target table names and column names are generated in the same case as the corresponding source names, unless cluster-level or session-level properties on the target override this case-sensitive behavior. If you want to control the case of letters in the target names, select this check box. Then select a **Case Transformation Strategy** option.
Case Transformation Strategy	If you selected **Enable Case Transformation**, select one of the following options to specify how to handle the case of letters in generated target table (or object) names and column (or field) names: - **Same as source**. Use the same case as the source table (or object) names and column (or field) names. - **UPPERCASE**. Use all uppercase. - **lowercase**. Use all lowercase. The default value is **Same as source**. **Note:** The selected strategy will override any cluster-level or session-level properties on the target for controlling case.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)

4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Google Cloud Storage target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Google Cloud Storage target properties:

Property	Description
Output Format	The format of the output files. Options are: - **CSV** - **AVRO** - **PARQUET** The default value is **CSV**. **Note:** Output files in CSV format use double-quotation marks ("") as the delimiter for each field.
Add Headers to CSV File	If **CSV** is selected as the output format, select this check box to add a header with source column names to the output CSV file.
Parquet Compression Type	If the **PARQUET** output format is selected, you can select a compression type that is supported by Parquet. Options are: - **None** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **None** - **Binary** - **JSON** The default value is **Binary**.
Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern: *schemaname_tablename*.txt **Note:** If this directory is not specified, no Avro schema definition file is produced.
File Compression Type	Controls whether to use a file compression type for output files in CSV or AVRO output format. Options are: - **None** - **Deflate** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.

Property	Description
Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.
Deflate Compression Level	If **Deflate** is selected in the **Avro Compression Type** field, specify a compression level from 0 to 9. The default value is 0.
Add Directory Tags	For incremental load and combined initial and incremental load tasks, select this check box to add the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. This check box is cleared by default.
Bucket	The name of an existing bucket container that stores, organizes, and controls access to the data objects that you load to Google Cloud Storage.
Task Target Directory	For incremental load and combined initial and incremental load tasks, the root directory for the other directories that hold output data files, schema files, and CDC cycle contents and completed files. You can use it to specify a custom root directory for the task. If you enable the **Connection Directory as Parent** option, you can still optionally specify a task target directory to use with the parent directory specified in the connection properties. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.

Property	Description

Data Directory	*For initial load tasks*, define a directory structure for the directories where Application Ingestion and Replication stores output data files and optionally stores the schema.
	The default directory pattern is `{TableName)_{Timestamp}`.
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter a folder name or use variables to create a folder name.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the initial load job starts to transfer data to the target.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	Note: If you manually enter the directory expression, ensure that you enclose placeholders with curly brackets { }. Placeholder values are not case sensitive.
	For example:
	`myDir1/{SchemaName}/{TableName}`
	`myDir1/myDir2/{SchemaName}/{YYYY}/{MM}/{TableName}_{Timestamp}`
	`myDir1/{toLower(SchemaName)}/{TableName}_{Timestamp}`
	For incremental load and combined initial and incremental load tasks, define a custom path to the subdirectory that contains the cdc-data data files.
	The default directory pattern is `{TaskTargetDirectory}/data/{TableName}/data`
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the CDC cycle started.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	-
	For Amazon S3 and Microsoft Azure Data Lake Storage Gen2 targets, Application Ingestion and Replication uses the directory specified in the target connection properties as the root for the data directory path when **Connection Directory as Parent** is selected.
	For Google Cloud Storage targets, Application Ingestion and Replication uses the **Bucket** name that you specify in the target properties for the ingestion task.
	For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.
	For Amazon S3 targets with Open Table format, the data directory field is not applicable. Enabling the **Connection Directory as Parent** includes the connection directory before the warehouse base path. If disabled, files are saved directly under the warehouse base directory.

Schema Directory	Specify a custom directory for storing the schema file if you want to store it in a directory other than the default directory. For initial loads, previously used values if available are shown in a list for your convenience. This field is optional. For initial loads, the schema is stored in the data directory by default. For incremental loads and combined initial and incremental loads, the schema file is stored in the following directory by default: `{TaskTargetDirectory}/data/{TableName}/schema`. You can use the same placeholders as for the **Data Directory** field. If you manually enter placeholders, ensure that you enclose them with curly brackets { }. If you include the toUpper or toLower function, put the placeholder name in parentheses and enclose both the function and placeholder in curly brackets, for example: `{toLower(SchemaName)}` **Note:** Schema is written only to output data files in CSV format. Data files in Parquet and Avro formats contain their own embedded schema.

Property	Description
Cycle Completion Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle completed file. Default is `{TaskTargetDirectory}/cycle/completed`.
Cycle Contents Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle contents files. Default is `{TaskTargetDirectory}/cycle/contents`.
Use Cycle Partitioning for Data Directory	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.
Use Cycle Partitioning for Summary Directories	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under the summary contents and completed subdirectories.
List Individual Files in Contents	For incremental load and combined initial and incremental load tasks, lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target. For incremental loads, the job writes "I" for insert, "U" for update, or "D" for delete. For initial loads, the job always writes "I" for insert. By default, this check box is selected for incremental load and initial and incremental load jobs, and cleared for initial load jobs.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. For initial loads, the job always writes the current date and time. By default, this check box is not selected.
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Kafka target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Kafka target properties:

Property	Description
Use Table Name as Topic Name	Controls whether Application Ingestion and Replication writes messages that contain source data to separate topics, one for each source object, or writes all messages to a single topic. Select this check box to write messages to separate table-specific topics. The topic names match the source table names, unless you add the source schema name, a prefix, or a suffix in the **Include Schema Name**, **Table Prefix**, or **Table Suffix** properties. By default, this check box is cleared.
Include Schema Name	When **Use Table Name as Topic Name** is selected, this check box appears and is selected by default. This setting adds the source schema name in the table-specific topic names. The topic names then have the format *schemaname_tablename*. If you do *not* want to include the schema name, clear this check box.
Table Prefix	When **Use Table Name as Topic Name** is selected, this property appears so that you can optionally enter a prefix to add to the table-specific topic names. For example, if you specify myprefix_, the topic names have the format myprefix_*tablename*. If you omit the underscore (_) after the prefix, the prefix is prepended to the table name.
Table Suffix	When **Use Table Name as Topic Name** is selected, this property appears so that you can optionally enter a suffix to add to the table-specific topic names. For example, if you specify _mysuffix, the topic names have the format *tablename*_mysuffix. If you omit the underscore (_) before the suffix, the suffix is appended to the table name.
Output Format	Select the format of the output file. Options are: - **CSV** - **AVRO** - **JSON** The default value is **CSV**. **Note:** Output files in CSV format use double-quotation marks ("") as the delimiter for each field. If your Kafka target uses Confluent Schema Registry to store schemas for incremental load jobs, you must select **AVRO** as the format.
JSON Format	If **JSON** is selected as the output format, select the level of detail of the output. Options are: - **Concise**. This format records only the most relevant data in the output, such as the operation type and the column names and values. - **Verbose**. This format records detailed information, such as the table name and column types.
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.

Property	Description
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **Binary** - **JSON** - **None** The default value is **Binary**. If you have a Confluent Kafka target that uses Confluent Schema Registry to store schemas, select **None**. Otherwise, Confluent Schema Registry does not register the schema. Do not select **None** if you are not using Confluent Scheme Registry.

Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern:
	`schemaname_tablename.txt`
	Note: If this directory is not specified, no Avro schema definition file is produced.
	If a source schema change is expected to alter the target, the Avro schema definition file is regenerated with a unique name that includes a timestamp, in the following format:
	`schemaname_tablename_YYYYMMDDhhmmss.txt`
	This unique naming pattern ensures that older schema definition files are preserved for audit purposes.

Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Operation Type	Select this check box to add a metadata column that includes the source SQL operation type in the output that the job propagates to the target. The job writes "I" for insert, "U" for update, or "D" for delete. By default, this check box is selected.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. By default, this check box is not selected.

Property	Description
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.
Async Write	Controls whether to use synchronous delivery of messages to Kafka. - Clear this check box to use synchronous delivery. Kafka must acknowledge each message as received before Application Ingestion and Replication sends the next message. In this mode, Kafka is unlikely to receive duplicate messages. However, performance might be slower. - Select this check box to use asynchronous delivery. Application Ingestion and Replication sends messages as soon as possible, without regard for the order in which the changes were retrieved from the source. By default, this check box is selected.
Producer Configuration Properties	Specify a comma-separated list of *key=value* pairs to enter Kafka producer properties for Apache Kafkatargets. You can specify Kafka producer properties in either this field or in the **Additional Connection Properties** field in the Kafka connection. If you enter the producer properties in this field, the properties pertain to the application ingestion and replication jobs associated with this task only. If you enter the producer properties for the connection, the properties pertain to jobs for all tasks that use the connection definition, unless you override the connection-level properties for specific tasks by also specifying properties in the **Producer Configuration Properties** field. For information about Kafka producer properties, see the Apache Kafka documentation.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft Azure Data Lake Storage Gen2 target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Microsoft Azure Data Lake Storage Gen2 target properties:

Property	Description
Output Format	The format of the output files. Options are: - **CSV** - **AVRO** - **PARQUET** The default value is **CSV**. **Note:** Output files in CSV format use double-quotation marks ("") as the delimiter for each field.
Add Headers to CSV File	If **CSV** is selected as the output format, select this check box to add a header with source column names to the output CSV file.
Parquet Compression Type	If the **PARQUET** output format is selected, you can select a compression type that is supported by Parquet. Options are: - **None** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.

Property	Description
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **None** - **Binary** - **JSON** The default value is **Binary**.
Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern: *schemaname_tablename*.txt **Note:** If this directory is not specified, no Avro schema definition file is produced.
File Compression Type	Controls whether to use a file compression type for output files in CSV or AVRO output format. Options are: - **None** - **Deflate** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.
Deflate Compression Level	If **Deflate** is selected in the **Avro Compression Type** field, specify a compression level from 0 to 9. The default value is 0.
Add Directory Tags	For incremental load and combined initial and incremental load tasks, select this check box to add the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. This check box is cleared by default.
Task Target Directory	For incremental load and combined initial and incremental load tasks, the root directory for the other directories that hold output data files, schema files, and CDC cycle contents and completed files. You can use it to specify a custom root directory for the task. If you enable the **Connection Directory as Parent** option, you can still optionally specify a task target directory to use with the parent directory specified in the connection properties. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.

Property	Description

Data Directory	*For initial load tasks*, define a directory structure for the directories where Application Ingestion and Replication stores output data files and optionally stores the schema.
	The default directory pattern is `{TableName)_{Timestamp}`.
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter a folder name or use variables to create a folder name.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the initial load job starts to transfer data to the target.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	Note: If you manually enter the directory expression, ensure that you enclose placeholders with curly brackets { }. Placeholder values are not case sensitive.
	For example:
	`myDir1/{SchemaName}/{TableName}`
	`myDir1/myDir2/{SchemaName}/{YYYY}/{MM}/{TableName}_{Timestamp}`
	`myDir1/{toLower(SchemaName)}/{TableName}_{Timestamp}`
	For incremental load and combined initial and incremental load tasks, define a custom path to the subdirectory that contains the cdc-data data files.
	The default directory pattern is `{TaskTargetDirectory}/data/{TableName}/data`
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the CDC cycle started.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	-
	For Amazon S3 and Microsoft Azure Data Lake Storage Gen2 targets, Application Ingestion and Replication uses the directory specified in the target connection properties as the root for the data directory path when **Connection Directory as Parent** is selected.
	For Google Cloud Storage targets, Application Ingestion and Replication uses the **Bucket** name that you specify in the target properties for the ingestion task.
	For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.
	For Amazon S3 targets with Open Table format, the data directory field is not applicable. Enabling the **Connection Directory as Parent** includes the connection directory before the warehouse base path. If disabled, files are saved directly under the warehouse base directory.

Connection Directory as Parent	Select this check box to use the directory value that is specified in the target connection properties as the parent directory for the custom directory paths specified in the task target properties. For initial load tasks, the parent directory is used in the **Data Directory** and **Schema Directory**. For incremental load and combined initial and incremental load tasks, the parent directory is used in the **Data Directory**, **Schema Directory**, **Cycle Completion Directory**, and **Cycle Contents Directory**. This check box is selected by default. If you clear it, for initial loads, define the full path to the output files in the **Data Directory** field. For incremental loads, optionally specify a root directory for the task in the **Task Target Directory**.

Property	Description
Schema Directory	Specify a custom directory for storing the schema file if you want to store it in a directory other than the default directory. For initial loads, previously used values if available are shown in a list for your convenience. This field is optional. For initial loads, the schema is stored in the data directory by default. For incremental loads and combined initial and incremental loads, the schema file is stored in the following directory by default: `{TaskTargetDirectory}/data/{TableName}/schema`. You can use the same placeholders as for the **Data Directory** field. If you manually enter placeholders, ensure that you enclose them with curly brackets { }. If you include the toUpper or toLower function, put the placeholder name in parentheses and enclose both the function and placeholder in curly brackets, for example: `{toLower(SchemaName)}` **Note:** Schema is written only to output data files in CSV format. Data files in Parquet and Avro formats contain their own embedded schema.
Cycle Completion Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle completed file. Default is `{TaskTargetDirectory}/cycle/completed`.
Cycle Contents Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle contents files. Default is `{TaskTargetDirectory}/cycle/contents`.
Use Cycle Partitioning for Data Directory	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.
Use Cycle Partitioning for Summary Directories	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under the summary contents and completed subdirectories.
List Individual Files in Contents	For incremental load and combined initial and incremental load tasks, lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target. For incremental loads, the job writes "I" for insert, "U" for update, or "D" for delete. For initial loads, the job always writes "I" for insert. By default, this check box is selected for incremental load and initial and incremental load jobs, and cleared for initial load jobs.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. For initial loads, the job always writes the current date and time. By default, this check box is not selected.
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft Azure Synapse Analytics target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Microsoft Azure Synapse Analytics target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables. The schema name that is specified in the connection properties is displayed by default. This field is case sensitive. Therefore, ensure that you entered the schema name in the connection properties in the correct case.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. Do not include special characters in the prefix. Otherwise, task deployment will fail. The default value is INFA_.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).

- On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

   Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

   Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft Fabric Data Warehouse target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Microsoft Fabric Data Warehouse target properties:

Property	Description
Schema Path	Select the schema path where Application Ingestion and Replication creates the target tables.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using the Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. The default value is **Standard**.
Staging Lakehouse Name	The name of the intermediate storage location in the data lake where data is temporarily stored before it is loaded into Microsoft Fabric Data Warehouse.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Field	Description
Add Operation Type	Add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected and you you cannot deselect it.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Enable Case Transformation	Not applicable.
Case Transformation Strategy	Not applicable.
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)

4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft Fabric Lakehouse target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Microsoft Fabric Lakehouse target properties:

Property	Description
Schema Path	Select the schema path where Application Ingestion and Replication creates the target tables.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using the Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. The default value is **Standard**.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Field	Description
Add Operation Type	Add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected and you you cannot deselect it.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Enable Case Transformation	Not applicable.
Case Transformation Strategy	Not applicable.
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)

4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft Fabric OneLake target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Microsoft Fabric OneLake target properties:

Property	Description
Open Mirroring	Select this checkbox to use Microsoft Fabric open mirroring. Open mirroring creates a synchronized, near real-time copy of your source data in Microsoft Fabric OneLake in the Parquet format. For more information, see "Using open mirroring to replicate data to Microsoft Fabric OneLake" on page 52. **Note:** Make sure to select Parquet as the output format before you enable Open Mirroring and continue with the open mirroring configurations.
Landing Zone URL	If you selected **Open Mirroring**, enter the unique Microsoft Fabric OneLake path where mirrored data files are stored in Parquet format before processing in Microsoft Fabric. **Note:** The rest of the fields in this table don't apply when you enable open mirroring.
Output Format	Select the format of the output file. Options are: - **CSV** - **AVRO** - **PARQUET** The default value is **CSV**. Output files in CSV format use double-quotation marks ("") as the delimiter for each field. **Note:** Open mirroring uses the Parquet output format.

Property	Description
Add Headers to CSV File	If **CSV** is selected as the output format, select this check box to add a header with source column names to the output CSV file.
Parquet Compression Type	If the **PARQUET** output format is selected, you can select a compression type that is supported by Parquet. Options are: - **None** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **None** - **Binary** - **JSON** The default value is **Binary**.
Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern: *schemaname_tablename*.txt **Note:** If this directory is not specified, no Avro schema definition file is produced.
File Compression Type	Controls whether to use a file compression type for output files in CSV or AVRO output format. Options are: - **None** - **Deflate** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.
Deflate Compression Level	If **Deflate** is selected in the **Avro Compression Type** field, specify a compression level from 0 to 9. The default value is 0.
Add Directory Tags	For incremental load and combined initial and incremental load tasks, select this check box to add the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. This check box is cleared by default.

Property	Description
Task Target Directory	For incremental load and combined initial and incremental load tasks, the root directory for the other directories that hold output data files, schema files, and CDC cycle contents and completed files. You can use it to specify a custom root directory for the task. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.

Data Directory	*For initial load tasks*, define a directory structure for the directories where Application Ingestion and Replication stores output data files and optionally stores the schema.
	The default directory pattern is `{TableName)_{Timestamp}`.
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter a folder name or use variables to create a folder name.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the initial load job starts to transfer data to the target.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	Note: If you manually enter the directory expression, ensure that you enclose placeholders with curly brackets { }. Placeholder values are not case sensitive.
	For example:
	`myDir1/{SchemaName}/{TableName}`
	`myDir1/myDir2/{SchemaName}/{YYYY}/{MM}/{TableName}_{Timestamp}`
	`myDir1/{toLower(SchemaName)}/{TableName}_{Timestamp}`
	For incremental load and combined initial and incremental load tasks, define a custom path to the subdirectory that contains the cdc-data data files.
	The default directory pattern is `{TaskTargetDirectory}/data/{TableName}/data`
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the CDC cycle started.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	-
	For Amazon S3 and Microsoft Azure Data Lake Storage Gen2 targets, Application Ingestion and Replication uses the directory specified in the target connection properties as the root for the data directory path when **Connection Directory as Parent** is selected.
	For Google Cloud Storage targets, Application Ingestion and Replication uses the **Bucket** name that you specify in the target properties for the ingestion task.
	For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.
	For Amazon S3 targets with Open Table format, the data directory field is not applicable. Enabling the **Connection Directory as Parent** includes the connection directory before the warehouse base path. If disabled, files are saved directly under the warehouse base directory.

Property	Description
Schema Directory	Specify a custom directory for storing the schema file if you want to store it in a directory other than the default directory. For initial loads, previously used values if available are shown in a list for your convenience. This field is optional. For initial loads, the schema is stored in the data directory by default. For incremental loads and combined initial and incremental loads, the schema file is stored in the following directory by default: `{TaskTargetDirectory}/data/{TableName}/schema`. You can use the same placeholders as for the **Data Directory** field. If you manually enter placeholders, ensure that you enclose them with curly brackets { }. If you include the toUpper or toLower function, put the placeholder name in parentheses and enclose both the function and placeholder in curly brackets, for example: `{toLower(SchemaName)}` **Note:** Schema is written only to output data files in CSV format. Data files in Parquet and Avro formats contain their own embedded schema.
Cycle Completion Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle completed file. Default is `{TaskTargetDirectory}/cycle/completed`.
Cycle Contents Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle contents files. Default is `{TaskTargetDirectory}/cycle/contents`.
Use Cycle Partitioning for Data Directory	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.
Use Cycle Partitioning for Summary Directories	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under the summary contents and completed subdirectories.
List Individual Files in Contents	For incremental load and combined initial and incremental load tasks, lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target. For incremental loads, the job writes "I" for insert, "U" for update, or "D" for delete. For initial loads, the job always writes "I" for insert. By default, this check box is selected for incremental load and initial and incremental load jobs, and cleared for initial load jobs.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. For initial loads, the job always writes the current date and time. By default, this check box is not selected.
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Add Source Table Name	Adds a metadata column to each target table to store the name of the respective source table.
Add Source Schema Name	Adds a metadata column to each target table to store the name of the respective source schema.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

   Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

   Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Microsoft SQL Server target

Define Microsoft SQL Server target properties.

Target Properties

Define the following required Microsoft SQL Server target properties:

Property	Description
Target Creation	The Create Target Tables option generates the target tables based on the source tables. **Note:** After the target table is created, Application Ingestion and Replication intelligently handles the target tables on subsequent job runs. Application Ingestion and Replication might truncate or re-create the target tables depending on specific circumstances.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables. The schema name that is specified in the connection properties is displayed by default. This field is case sensitive. Therefore, ensure that you entered the schema name in the connection properties in the correct case.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Property	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. Do not include special characters in the prefix. Otherwise, task deployment will fail. The default value is INFA_.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

You can enter multiple rules.

Notes:

- If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
- If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
- On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.

Property	Description
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure an Oracle Cloud Object Storage target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Oracle Cloud Object Storage target properties:

Property	Description
Output Format	The format of the output files. Options are: - **CSV** - **AVRO** - **PARQUET** The default value is **CSV**. **Note:** Output files in CSV format use double-quotation marks ("") as the delimiter for each field.
Add Headers to CSV File	If **CSV** is selected as the output format, select this check box to add a header with source column names to the output CSV file.
Parquet Compression Type	If the **PARQUET** output format is selected, you can select a compression type that is supported by Parquet. Options are: - **None** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.

Property	Description
Avro Format	If you selected **AVRO** as the output format, select the format of the Avro schema that will be created for each source table. Options are: - **Avro-Flat**. This Avro schema format lists all Avro fields in one record. - **Avro-Generic**. This Avro schema format lists all columns from a source table in a single array of Avro fields. - **Avro-Nested**. This Avro schema format organizes each type of information in a separate record. The default value is **Avro-Flat**.
Avro Serialization Format	If **AVRO** is selected as the output format, select the serialization format of the Avro output file. Options are: - **None** - **Binary** - **JSON** The default value is **Binary**.
Avro Schema Directory	If **AVRO** is selected as the output format, specify the local directory where Application Ingestion and Replication stores Avro schema definitions for each source table. Schema definition files have the following naming pattern: *schemaname_tablename*.txt **Note:** If this directory is not specified, no Avro schema definition file is produced.
File Compression Type	Controls whether to use a file compression type for output files in CSV or AVRO output format. Options are: - **None** - **Deflate** - **Gzip** - **Snappy** The default value is **None**, which means no compression is used.
Avro Compression Type	If **AVRO** is selected as the output format, you can select an Avro compression type. Options are: - **None** - **Bzip2** - **Deflate** - **Snappy** The default value is **None**, which means no compression is used.
Deflate Compression Level	If **Deflate** is selected in the **Avro Compression Type** field, specify a compression level from 0 to 9. The default value is 0.
Add Directory Tags	For incremental load and combined initial and incremental load tasks, select this check box to add the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. This check box is cleared by default.
Task Target Directory	For incremental load and combined initial and incremental load tasks, the root directory for the other directories that hold output data files, schema files, and CDC cycle contents and completed files. You can use it to specify a custom root directory for the task. If you enable the **Connection Directory as Parent** option, you can still optionally specify a task target directory to use with the parent directory specified in the connection properties. This field is required if the {TaskTargetDirectory} placeholder is specified in patterns for any of the following directory fields.

Property	Description
Connection Directory as Parent	Select this check box to use the directory value that is specified in the target connection properties as the parent directory for the custom directory paths specified in the task target properties. For initial load tasks, the parent directory is used in the **Data Directory** and **Schema Directory**. For incremental load and combined initial and incremental load tasks, the parent directory is used in the **Data Directory**, **Schema Directory**, **Cycle Completion Directory**, and **Cycle Contents Directory**. This check box is selected by default. If you clear it, for initial loads, define the full path to the output files in the **Data Directory** field. For incremental loads, optionally specify a root directory for the task in the **Task Target Directory**.

Data Directory	*For initial load tasks*, define a directory structure for the directories where Application Ingestion and Replication stores output data files and optionally stores the schema.
	The default directory pattern is `{TableName)_{Timestamp}`.
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter a folder name or use variables to create a folder name.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the initial load job starts to transfer data to the target.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	Note: If you manually enter the directory expression, ensure that you enclose placeholders with curly brackets { }. Placeholder values are not case sensitive.
	For example:
	`myDir1/{SchemaName}/{TableName}`
	`myDir1/myDir2/{SchemaName}/{YYYY}/{MM}/{TableName}_{Timestamp}`
	`myDir1/{toLower(SchemaName)}/{TableName}_{Timestamp}`
	For incremental load and combined initial and incremental load tasks, define a custom path to the subdirectory that contains the cdc-data data files.
	The default directory pattern is `{TaskTargetDirectory}/data/{TableName}/data`
	To customize the directory pattern, click the Edit icon to select from the following listed path types and values:
	- **Folder Path**. Enter {TaskTargetDirectory} for a task-specific base directory on the target to use instead of the S3 folder path specified in the connection properties.
	- **Timestamp values**. Select data elements *Timestamp*, *yy*, *yyyy*, *mm*, or *dd*. The Timestamp values are in the format yyyymmdd_hhmissms. The generated dates and times in the directory paths indicate when the CDC cycle started.
	- **Schema Name**. Select *SchemaName*, *toUpper(SchemaName)*, or *toLower(SchemaName)*.
	- **Table Name**. Select *TableName*, *toUpper(TableName)*, and `toLower(TableName)`.
	-
	For Amazon S3 and Microsoft Azure Data Lake Storage Gen2 targets, Application Ingestion and Replication uses the directory specified in the target connection properties as the root for the data directory path when **Connection Directory as Parent** is selected.
	For Google Cloud Storage targets, Application Ingestion and Replication uses the **Bucket** name that you specify in the target properties for the ingestion task.
	For Microsoft Fabric OneLake targets, the parent directory is the path specified in the **Lakehouse Path** field in the Microsoft Fabric OneLake connection properties.
	For Amazon S3 targets with Open Table format, the data directory field is not applicable. Enabling the **Connection Directory as Parent** includes the connection directory before the warehouse base path. If disabled, files are saved directly under the warehouse base directory.

Property	Description
Schema Directory	Specify a custom directory for storing the schema file if you want to store it in a directory other than the default directory. For initial loads, previously used values if available are shown in a list for your convenience. This field is optional. For initial loads, the schema is stored in the data directory by default. For incremental loads and combined initial and incremental loads, the schema file is stored in the following directory by default: `{TaskTargetDirectory}/data/{TableName}/schema`. You can use the same placeholders as for the **Data Directory** field. If you manually enter placeholders, ensure that you enclose them with curly brackets { }. If you include the toUpper or toLower function, put the placeholder name in parentheses and enclose both the function and placeholder in curly brackets, for example: `{toLower(SchemaName)}` **Note:** Schema is written only to output data files in CSV format. Data files in Parquet and Avro formats contain their own embedded schema.
Cycle Completion Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle completed file. Default is `{TaskTargetDirectory}/cycle/completed`.
Cycle Contents Directory	For incremental load and combined initial and incremental load tasks, the path to the directory that contains the cycle contents files. Default is `{TaskTargetDirectory}/cycle/contents`.
Use Cycle Partitioning for Data Directory	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under each data directory. If this option is not selected, individual data files are written to the same directory without a timestamp, unless you define an alternative directory structure.
Use Cycle Partitioning for Summary Directories	For incremental load and combined initial and incremental load tasks, select this option to create a timestamp subdirectory for each CDC cycle, under the summary contents and completed subdirectories.
List Individual Files in Contents	For incremental load and combined initial and incremental load tasks, lists individual data files under the contents subdirectory. If **Use Cycle Partitioning for Summary Directories** is cleared, this option is selected by default. All of the individual files are listed in the contents subdirectory unless you can configure custom subdirectories by using the placeholders, such as for timestamp or date. If **Use Cycle Partitioning for Data Directory** is selected, you can still optionally select this check box to list individual files and group them by CDC cycle.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Field	Description
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target. For incremental loads, the job writes "I" for insert, "U" for update, or "D" for delete. For initial loads, the job always writes "I" for insert. By default, this check box is selected for incremental load and initial and incremental load jobs, and cleared for initial load jobs.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. For initial loads, the job always writes the current date and time. By default, this check box is not selected.
Add Orderable Sequence	Select this check box to add a metadata column that records a combined epoch value and an incremental numeric value for each change operation that the job inserts into the target tables. The sequence value is always ascending, but not guaranteed to be sequential and gaps may exist. The sequence value is used to identify the order of activity in the target records. By default, this check box is not selected.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure an Oracle target

Define target properties for the Oracle destination.

Target Properties

Define the following required Oracle target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using the Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. **Note:** The **Audit** apply mode applies for an SAP source with SAP Mass Ingestion connector. The default value is **Standard**.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Field	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp in UTC format at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Select this check box to add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the target table. This field is available only when the **Apply Mode** option is set to **Audit**. In Audit mode, the job writes "I" for insert, "U" for update, or "D" for delete. By default, this check box is selected.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit**. By default, this check box is not selected.
Add Operation Sequence	Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation. This field is available only when **Apply Mode** is set to **Audit**. By default, this check box is not selected.
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.

Field	Description
Enable Case Transformation	By default, target table names and column names are generated in the same case as the corresponding source names. If you want to control the case of letters in the target names, select this check box. Then select a **Case Transformation Strategy** option.
Case Transformation Strategy	If you selected **Enable Case Transformation**, select one of the following options to specify how to handle the case of letters in generated target table (or object) names and column (or field) names: - **Same as source**. Use the same case as the source table (or object) names and column (or field) names. - **UPPERCASE**. Use all uppercase. - **lowercase**. Use all lowercase. The default value is **Same as source**.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)

4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a PostgreSQL target

Define target properties for the PostgreSQL destination.

Target Properties

Define the following required PostgreSQL target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
2. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
3. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Salesforce Data 360 target

Define target properties for the destination that you selected on the **Destination** page.

Target Properties

Define the following required Salesforce Data 360 target properties:

Property	Description
Schema Path	The schema path where the Application Ingestion and Replication creates the target tables.
Data Space	Name of the data space in Salesforce Data 360 where you want to load data. If you do not enter a data space, the default data space is used.
Category	The data category assigned to data from source streams when it is loaded into Salesforce Data 360. Options are: - Profile. Applies to customer attributes such as name, email, or address. - Engagement. Applies to behavioral and interaction data such as website visits, email opens, clicks, and other activity logs. - Other. Applies to data that doesn't fit into Profile or Engagement, such as transactional records, order histories, CRM system logs, and other system or contextual data.
Event Time Field	Name of the event date and time field for engagement records if you selected the Engagement category. It defines exactly when an interaction occurred, such as a purchase, click, or login, and helps identify the most recent or relevant event.
Recorded Modified Field	The name of the recorded modified field that captures the date and time when a record was last updated if you selected the Profile or Other category.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. You can use only the Standard apply mode. This mode accumulates the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define the following optional advanced target property that you want to use:

Property	Description
Ingestion Mode	The ingestion mode to load data into Salesforce 360. You can select from the following options based on the data volume, processing speed, and near real-time requirements: - Bulk API. Loads large datasets. - Streaming API. Processes near-real-time micro-batch datasets.
Enable Case Transformation	Not applicable.
Case Transformation Strategy	Not applicable.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.
   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.
   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).
   - On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.
3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
The following table describes the properties that are available:

Property	Description
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Configure a Snowflake Data Cloud target

Define target properties for the Snowflake destination.

Target Properties

Define the following required Snowflake target properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source objects.
Schema	Select the target schema in which Application Ingestion and Replication creates the target tables.

Property	Description
Stage	The name of internal staging area that holds the data read from the source before the data is written to the target tables. This name must not include spaces. If the staging area does not exist, it will be automatically created. **Note:** This field is not available if you selected the **Superpipe** option in the **Advanced Target Properties**.
Apply Mode	For incremental load and combined initial and incremental load jobs, this property indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. After enabling Soft Deletes, any update in the source table during normal or backlog mode results in the deletion of the matching record, insertion of the updated record, and marking of the INFA_OPERATION_TYPE operation as NULL in the target table. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, transaction ID, and before image. Consider using **Audit** apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. After enabling the Audit apply mode, any update in the source table during backlog or normal mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. **Note:** The **Audit** apply mode applies for SAP source with SAP Mass Ingestion connector. Default is **Standard**.

Advanced Target Properties

To view advanced properties, toggle on **Show Advanced Options** at the top of the page.

1. Under **Advanced Options**, define any of the following optional advanced target properties that you want to use:

Field	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp, except for Snowflake targets that use the Superpipe option where minutes and seconds might vary slightly. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Add Operation Type	Add a metadata column that records the source SQL operation type in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column. In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts, updates, and upserts. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values. By default, this check box is selected and you you cannot deselect it.
Add Operation Time	Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target database or inserts into the audit table on the target system. This field is available only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**. By default, this check box is not selected.
Add Operation Sequence	Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation. This field is available only when **Apply Mode** is set to **Audit**. By default, this check box is not selected.
Add Cycle ID	Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.

Field	Description
Superpipe	Select this check box to use the Snowpipe Streaming API to quickly stream rows of data directly to Snowflake Data Cloud target tables with low latency instead of first writing the data to stage files. This option is available for all load types. When you configure the target connection, select KeyPair authentication. By default, this check box is selected. Deselect it if you want to write data to intermediate stage files. **Note:** If you enable the **Superpipe** option for a task that uses the **Soft Deletes** apply mode, make sure the source tables contain a primary key.
Merge Frequency	When **Superpipe** is selected, you can optionally set the frequency, in seconds, at which change data rows are merged and applied to the Snowflake target tables. The merge frequency affects how often the stream change data is merged to the Snowflake base table. A Snowflake view joins the stream change data with the base table. Set this value to balance the costs of merging data to the base table with the performance of view join processing. This field applies to incremental load and combined initial and incremental load tasks. Valid values are 60 through 604800 seconds. Default is 3600 seconds.
Enable Case Transformation	By default, target table names and column names are generated in the same case as the corresponding source names, unless cluster-level or session-level properties on the target override this case-sensitive behavior. If you want to control the case of letters in the target names, select this check box. Then select a **Case Transformation Strategy** option.
Case Transformation Strategy	If you selected **Enable Case Transformation**, select one of the following options to specify how to handle the case of letters in generated target table (or object) names and column (or field) names: - **Same as source**. Use the same case as the source table (or object) names and column (or field) names. - **UPPERCASE**. Use all uppercase. - **lowercase**. Use all lowercase. The default value is **Same as source**. **Note:** The selected strategy will override any cluster-level or session-level properties on the target for controlling case.

2. Under **Table Renaming Rules**, if you want to rename the target objects that are associated with the selected source tables, define renaming rules.

   a. Click the + (Add new row) icon and enter a source table name or name mask and enter a corresponding target table name or name mask.

   b. To define a mask, include one or more the asterisk (*) wildcards. Then, press Enter.

   For example, to add the prefix "PROD_" to the names of target tables that correspond to all selected source tables, enter the * wildcard for the source table and enter PROD_* for the target table.

   You can enter multiple rules.

   Notes:

   - If you enter the wildcard for a source table mask, you must also enter the wildcard for a target table mask.
   - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).

- On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Data Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables do not exceed 232 characters.

3. Under **Data Type Rules**, if you want to override the default mappings of source data types to target data types, define data type rules. Click the + (Add new row) icon and enter a source data type and corresponding target data type. Then press Enter.
   Also, in the Source Data Type value, you can include the percent (%) wildcard to represent the data type precision, scale, or size, for example, NUMBER(%,4), NUMBER(8,%), or NUMBER(%). Use the wildcard to cover all source columns that have the same data type but use different precision, scale, or size values, instead of specifying each one individually. For example, enter FLOAT(%) to cover FLOAT(16), FLOAT(32), and FLOAT(84). You cannot enter the % wildcard in the target data type. A source data type that uses the % wildcard must map to a target data type that uses specific precision, scale, or size value. For example, you could map the source data type FLOAT(%) to a target data type specification such as NUMBER(38,10)
4. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.
   The following table describes the properties that are available, depending on the load type:

Property	Description
Writer Distributor Count	The number of distributors that can run on separate threads in parallel to process data during an initial load job or the unload phase of a combined load job when the **Writer Unload Multiple Distributors** custom property is set to true. Using parallel distributor threads can improve job performance, particularly for high-volume data transfers. Default value is 3. If your system has ample resources, Informatica recommends that you set this parameter to 8. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal distributor count. You can override the value from auto-tuning by setting this custom property.
Writer Helper Thread Count	The number of writer helper threads that are used to convert incoming change data rows or initial unload rows to the output format configured for the target, such as Avro, CSV, or Parquet. Default value is 2. If two threads can't keep up with the incoming volume of data, you can increase the number of threads. Consider increasing the number of threads in the following situations: 1) the incoming volume is high, 2) multiple writer distributors are in use, or 3) some rows are very large or wide, which increases conversion time. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal thread count. You can override the value from auto-tuning by setting this custom property.

Property	Description
Writer Unload Multiple Distributors	Indicates whether multiple distributor threads can be used to process data in parallel during initial load jobs and the unload phase of combined load jobs. The distributors perform work such as uploading data files to staging areas and flushing data to the target. Set this property to true to use multiple distributor threads. Default value is false. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine whether to use multiple distributors. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

Click **Next** to proceed, or click **Save**.

Transform the data

You can apply trim transformations to selected tables and columns to remove spaces to the left or right of character column values. You can also define row-level filter rules to filter out data rows for source tables based on column conditions you define before the data is applied to the target.

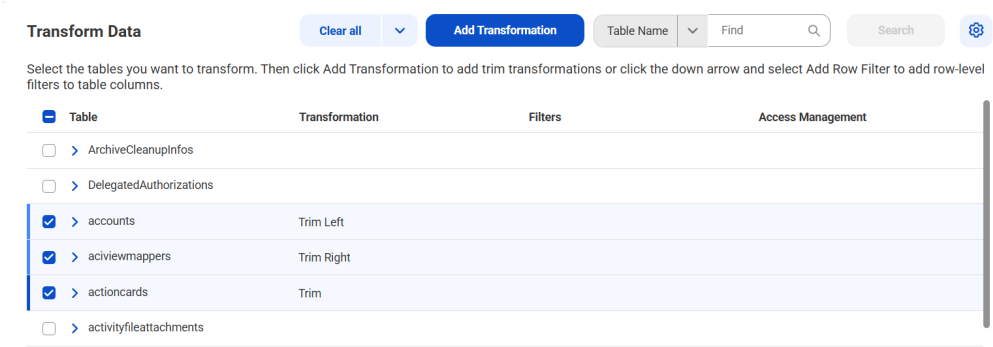
Note: If you edit row-level filters in the task for a deployed job, you must Redeploy the job afterwards for the updated filters to take effect.

1. On the **Transform Data** page, select the tables and columns to which you want to assign a transformation.

   Note: You can apply trim transformations and row-level filters to the same tables and columns.
2. To add a trim transformation, click **Add Transformation**.

   The **How do you want to transform your data?** dialog box appears.
3. Click the + (Add a new row) icon to add a row. Then, in the **Transformation Type** list, select one of the following options:
   - **Trim Left**. Trim spaces to the left of character column values.
   - **Trim Right**. Trim spaces to the right of character column values.
   - **Trim**. Trim spaces to the left of and to the right of character column values.

   Click the Save icon to add the entry.
4. Click **Next** to go to the **Summary** tab where you can review your transformation settings.
5. If the settings are correct on the **Summary** tab, click **Save** to save them and return to the initial **Transform Data** page.

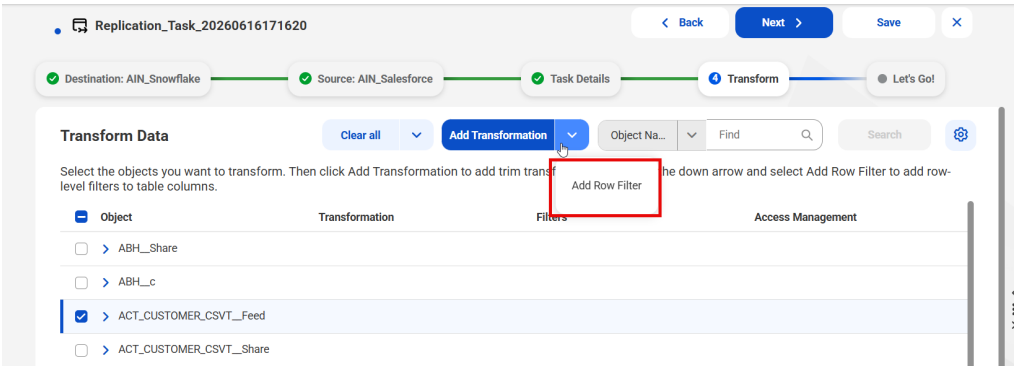


6. To add another transformation type for a different table or set of tables, repeat steps 1 through 5.

   Tip: You can remove a transformation assignment on the **Transform Data** page. Select the table with the unwanted transformation and click **Clear All**.

7. To add row-level filters to the selected tables and columns, click the down arrow next to **Add Transformation** and select **Add Row Filter**.

   The **Add Row Filter** option is available only only for application ingestion and replication tasks that have an SAP source (with an Oracle or HANA database) and use the SAP Mass Ingestion connector or that have a Salesforce source and use the Salesforce Mass Ingestion connector. The tasks can use any load type.



   The **How do you want to filter your data?** dialog box appears.

8. Select the table and filter type to apply the filter conditions.
   a. From the set of tables you previously selected, select the table that you want to assign a filter to.
   b. Select one of the following filter types:
      - Basic
      - Advanced

      The default option is **Basic**.

9. To add a **Basic** filter, complete the following substeps:
   a. Click the + (Add a new row) icon to add a row.
   b. Under **Column Name**, select a column.

      Columns with unsupported data types for row filtering are marked as "Not supported."
   c. Under **Operator**, select an operator type to use with the value.

d. Under **Value**, select or enter a value, depending on the column type. Then click the Save icon on the right end of the row to save the condition.

The following table describes the values that are valid for each column data type supported for filtering:

Column data type	Description
INTEGER	Enter a numeric value. You can use "+" and "-" only once before the number. The value must be between -2147483648 and 2147483647.
LONG	Enter a numeric value. You can use "+" and "-" only once before the number. The value must be between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807.
BIGINT	Enter a numeric value. You can use "+" and "-" only once before the number. Maximum length is 50 digits.
BIGDEC	Enter a numeric value. You can use "+" and "-" only once before the number. A decimal is allowed. Maximum length is 50 digits.
STRING	Enter text.
DATE	Use the date picker to select the date.
TIME	Enter the value in the format HH:MM:SS.MS, with milliseconds being optional and up to a maximum length is 9 digits. For example, 13:14:15.123456789
DATETIME	Use the date picker to select the date and time.
OFFSET_DATETIME	Use the date picker to select the date, time, and time zone.

Note: Application Ingestion and Replication does not support BOOLEAN, BINARY, BLOB, CLOB, and graphic column data types.

e. Click **Validate** to test the syntax of the specified condition.

f. To add another Basic condition, repeat steps a through e.

The AND operator is used to combine the multiple conditions.

g. Click **Save** to validate and save the changes.

h. When done defining Basic filter conditions, click **OK** to return to the **Transform Data** page.

10. To define an **Advanced** filter that consists of multiple conditions combined with the AND or OR operator, manually enter the conditions in the box.

Note: If you entered a Basic filter conditions for a column and then switched to the Advanced filter, the Basic condition is displayed so that you can add to it to make a more complex filter.

a. Under **Column Name**, select a column and click the > arrow.

The column name appears in the **Filter Condition** box.

Note: For combined load tasks, do not include columns that you expect will be updated during CDC processing. If the column is updated, it might become ineligible for replication and cause unpredictable results. In this case, you'd need to Resync the job.

b. In the **Filter Condition** box, type one or more conditions for the selected column. Manually enter conditions using the supported syntax and the appropriate operators, which can vary based on the column data type. You can also nest conditions using parentheses. See "Syntax for row-level filtering" on page 191. When done, click the Save icon on the right end of the row to save the advanced filter.

The following table describes the values that are valid for each column data type supported for filtering:

Column data type	Description
INTEGER	Enter a numeric value. You can use "+" and "-" only once before the number. The value must be between -2147483648 and 2147483647.
LONG	Enter a numeric value. You can use "+" and "-" only once before the number. The value must be between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807.
BIGINT	Enter a number value. You can use "+" and "-" only once before the number. Maximum length is 50 digits.
BIGDEC	Enter a numeric value. You can use "+" and "-" only once before the number. A decimal is allowed. Maximum length is 50 digits.
STRING	Enter an input attribute in single quotes (').
DATE	Enter the value in the format YYYY-MM-DD. Enter the input attribute in single quotes (').
TIME	Enter the value in the format HH:MM:SS.MS, with milliseconds (MS) being optional and up to a maximum length is 9 digits. Enter the input attribute in single quotes ('). For example, 13:14:15.123456789
DATETIME	Enter the date and time in the following format: YYYY-MM-DDTHH:MM:SS:MS For example, 2024-12-31T03:04:05.123456789 Enter the input attribute in single quotes (').
OFFSET_DATETIME	Enter the date, time, and time zone in the following format: YYYY-MM-DDTHH:MM:SS.MS+05:00 For example, 2024-03-15T10:03:04.123456789+05:00 Enter the input attribute in single quotes (').

Notes:

- Application Ingestion and Replication does not support BOOLEAN, BINARY, BLOB, CLOB, and graphic column data types.
- All date, time, and datetime values are matched against the source date and time. Time changes between Daylight Saving Time and standard time are not accommodated.

c. Click **Validate** to test the syntax of the specified conditions.

Note: Switching from **Advanced** to **Basic** filter type after creating or editing an Advanced filter condition, deletes all changes to the filter condition, even if you saved it.

d. Click **Save** to validate and save the changes and then click **OK** to return to the **Transform Data** page.

Note: Do not modify any column included in the filter after the task has been deployed. If you do so, the row-level filtering might not work properly.

The **Filters** column on the **Transform Data** page shows the applied filters as hyperlinks. Clicking the link opens the selected filter in edit mode. Tables with an advanced filter display **Advanced** next to their filter conditions in the **Filters** column.
On the **Transform Data** page, clicking the **Clear All** button at the top right hand corner removes all filters, including trim transformations and row-level filters from the selected tables.

11. When done, click **Next**.

Syntax for row-level filtering

If you create Advanced row-level filters when you define an application ingestion and replication task, ensure that you enter the filter conditions using the correct syntax. Otherwise, filter validation is likely to fail.

Operators

In an Advanced filter, you can use the following operators within a condition, depending on the column data type:

Operator	Description
=	Equals
!=	Does not equal
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
IS NULL	Contains a null
IS NOT NULL	Cannot contain a null
BETWEEN x AND y	Greater than or equal to *x* and less than or equal to *y*
NOT BETWEEN %s AND %s	Not greater than or equal to *x* and less than or equal to *y*
LIKE	A comparative operative for string columns only. Example: `LIKE '%06%7__' `. This condition matches against the following values: 06789, A06X789, AB06XY789", "06X789, and A06789. However, it does not match against these values: A06789Z, A0678, A6789, "".
NOT LIKE	A comparative operative for string columns only.
IN	True if the operand is equal to one of a list of expressions
NOT IN	True if the operand is NOT equal to one of a list of expressions
`+ - / *`	Numeric computation operators for addition, subtraction, division, and multiplication

Syntax rules

In Advanced filters, use with the following syntax rules:

- Enclose string, date, time, datetime, offset_datetime values in single quotation marks. For example: 'SomeName'
- In datetime and offset_datetime values, enter the character "T" between date and time. For example: '2024-11-15T00:00:00.000000001'

Finalize the task definition

Almost done! On the **Let's Go!** page, complete a few more properties. Then you can Save and Deploy the task.

1. Under **General Properties**, set the following properties:

Property	Description
Task Name	Enter a name that you want to use to identify theapplication ingestion and replication task, if you do not want to use the generated name. Using a descriptive name will make finding the task easier later. Task names can contain Latin alphanumeric characters, spaces, periods (.), commas (,), underscores (_), plus signs (+), and hyphens (-). Task names cannot include other special characters. Task names are not case sensitive. Maximum length is 50 characters. **Note:** If you include spaces in the task name, after you deploy the task, the spaces do not appear in the corresponding job name.
Location	The project or project\folder in Explore that will contain the task definition. If you do not specify a project, the "Default" project is used.
Runtime Environment	Select the runtime environment that you want to use to run the task. By default, the runtime environment that you initially entered when you began defining the task is displayed. You can use this runtime environment or select another one. **Tip:** To refresh the list of runtime environments, click **Refresh**. The runtime environment can be a Secure Agent group that consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs tasks and enables secure communication. Alternatively, for application ingestion and replication that have selected source types, you can use a serverless runtime environment hosted on Microsoft Azure. **Note:** You cannot choose a serverless runtime environment if a local runtime environment was previously selected. The Cloud Hosted Agent is not supported. Select **Set as default** to use the specified runtime environment as your default environment for all tasks you create. Otherwise, leave this check box cleared.
Description	Optionally, enter a description you want to use for the task. Maximum length is 4,000 characters.

Property	Description
Schedule	If you want to run an initial load task based on a schedule instead of manually starting it, select **Run this task based on a schedule**. Then select a schedule that was previously defined in Administrator. The default option is **Do not run this task based on a schedule**. **Note:** This field is not available for incremental load and combined initial and incremental load tasks. To view and edit the schedule options, go to Administrator. If you edit the schedule, the changes will apply to all jobs that use the schedule. If you edit the schedule after deploying the task, you do not need to redeploy the task. If the schedule criteria for running the job is met but the previous job run is still active, Application Ingestion and Replication skips the new job run.
Execute in Taskflow	Select this check box to make the task available in Data Integration to add to a taskflow as an event source.You can then include transformations in the taskflow to transform the ingested data. Available for initial load and incremental load tasks with Snowflake targets that don't use the **Superpipe** option.

2. To display advanced properties, toggle on **Show Advanced Options**.
3. For initial load jobs that have Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage, or Microsoft Fabric OneLake targets, optionally clear the **File Extension Based on File Type** check box if you want the output data files to have the .dat extension. This check box is selected by default, which causes the output files to have file-name extensions based on their file types.

   Note: For incremental load jobs with these target types, this option is not available. Application Ingestion and Replication always uses output file-name extensions based on file type.
4. Optionally, configure an apply cycle. An apply cycle is a cycle of applying change data that starts with fetching the intermediate data from the source and ends with the commit of the data to the target. For continuous replication, the source processes the data in multiple low-latency apply cycles.

For application ingestion and replication incremental load tasks that have Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, or Microsoft Fabric OneLake targets, you can configure the following apply cycle options:

Option	Description
Apply Cycle Interval	Specifies the amount of time that must elapse before an application ingestion and replication job ends an apply cycle. You can specify days, hours, minutes, and seconds or specify values for a subset of these time fields leaving the other fields blank. The default value is 15 minutes. **Note:** If you're using an Amazon S3 target with the **Apache Iceberg** option for the **Open Table Format** target property, this field is ignored.
Apply Cycle Change Limit	Specifies the number of records that must be processed before an application ingestion and replication job ends an apply cycle. When this record limit is reached, the ingestion job ends the apply cycle and writes the change data to the target. The default value is 10000 records. During startup, jobs might reach this limit more frequently than the apply cycle interval if they need to catch up on processing a backlog of older data. **Note:** If you're using an Amazon S3 target with the **Apache Iceberg** option for the **Open Table Format** target property, this field is ignored.
Low Activity Flush Interval	Specifies the amount of time, in hours, minutes, or both, that must elapse during a period of no change activity on the source before an application ingestion and replication job ends an apply cycle. When this time limit is reached, the ingestion job ends the apply cycle and writes the change data to the target. If you do not specify a value for this option, an application ingestion and replication job ends apply cycles only after either the **Apply Cycle Change Limit** or **Apply Cycle Interval** limit is reached. No default value is provided.

Note:

- Either the **Apply Cycle Interval** or **Apply Cycle Change Limit** field must have a non-zero value or use the default value.
- An apply cycle ends when the job reaches any of the three limits, whichever limit is met first.

5. For incremental load jobs that have an Apache Kafka target, configure the following Checkpoint Options:

Option	Description
Checkpoint All Rows	Indicates whether an application ingestion and replication job performs checkpoint processing for every message that is sent to the Kafka target. **Note:** If this check box is selected, the **Checkpoint Every Commit**, **Checkpoint Row Count**, and **Checkpoint Frequency (secs)** options are ignored.
Checkpoint Every Commit	Indicates whether an application ingestion and replication job performs checkpoint processing for every commit that occurs on the source.

Option	Description
Checkpoint Row Count	Specifies the maximum number of messages that an application ingestion and replication job sends to the target before adding a checkpoint. If you set this option to 0, the job does not perform checkpoint processing based on the number of messages. If you set this option to 1, the job adds a checkpoint for each message.
Checkpoint Frequency (secs)	Specifies the maximum number of seconds that must elapse before an application ingestion and replication job adds a checkpoint. If you set this option to 0, an application ingestion and replication does not perform checkpoint processing based on elapsed time.

6. Under **Schema Drift Options**, if the detection of schema drift is supported for your source and target combination, specify the schema drift option to use for each of the supported types of DDL operations.

   Note: The **Schema Drift Options** section appears only for incremental load and combined initial and incremental load tasks. Additionally, this section appears only for the sources that support automatic detection of schema changes.

   The following table describes the schema drift options that you can set for a DDL operation type:

Option	Description
Ignore	Do not replicate DDL changes that occur on the source database to the target.
Replicate	Allow the application ingestion and replication job to replicate the DDL changes to the target. The types of supported DDL operations are: - Add Field - Modify Field - Drop Field - Rename Field **Note:** - Add Field operations that add a primary-key field are not supported and might cause unpredictable results. - Modify Field operations that change the NULL or NOT NULL constraint of a field are not replicated to the target. - Tasks that have Microsoft Azure Synapse Analytics targets cannot replicate rename operations on source fields. - Tasks that have Microsoft Fabric Data Warehouse and Salesforce Data 360 targets cannot replicate drop and rename operations on source fields. - Tasks that have Microsoft Fabric Lakehouse targets cannot replicate modify operations on source fields. - Tasks that have Google BigQuery targets cannot replicate modify and rename operations on source fields. The schema drift options for these operations are not available. - Application ingestion and replication jobs don't support adding fields for Oracle targets.
Stop Job	Stop the application ingestion and replication job.
Stop Table	Stop processing the source object on which the DDL change occurred. **Note:** When one or more objects are excluded from replication because of the Stop Object schema drift option, the status of the job changes to **Running with Warning**. The application ingestion and replication job cannot retrieve the data changes that occurred on the source object after the job stops processing the changes. This action leads to data loss on the target. To avoid data loss, you must re-synchronize the source and target objects that the job stopped processing before you resume the application ingestion and replication job.

7. Under **Custom Properties**, you can enter one or more custom properties that Informatica provides to improve performance or to meet your special requirements. To add a property, click the + icon to add a

row. In the **Property Name** field, select a property and then enter a property value, or select the **Custom** option and manually enter both the property name and value.

The following table describes the properties that are available:

Property	Description
Number of Rows in Output File	Specify the maximum number of rows that the task writes to an output data file. **Note:** This field is not displayed for jobs that have an Apache Kafka target or if you use the Superpipe option for a Snowflake target. For incremental load operations and combined initial and incremental load operations, change data is flushed to the target either when this number of rows is reached or when the flush latency period expires and the job is not in the middle of processing a transaction. The flush latency period is the time that the job waits for more change data before flushing data to the target. The latency period is internally set to 10 seconds and cannot be changed. Valid values are 1 through 100000000. Default is 100000 rows. **Note:** For Microsoft Azure Synapse Analytics targets, the data is first sent to a Microsoft Azure Data Lake Storage staging file before being written to the target tables. After data is written to the target, the entire contents of the table-specific directory that includes the staging files are emptied. For Snowflake targets, the data is first stored in an internal stage area before being written to the target tables. **Note:** If you select the **Auto-Tune** option on the **Let's Go!** page of the task wizard, auto-tuning can determine the optimal number of rows to flush. You can override the value from auto-tuning by setting this custom property.
Custom	Select this option to manually enter the name of a property and its value. Use this option to enter properties that Informatica Global Customer Support or a technical staff member has provided to you for a special case.

Custom properties are intended to address performance or special processing needs. A property name can contain only alphanumeric characters and the following special characters: periods (.), hyphens (-), and underscores (_).

Tip: To delete a custom property after you've entered it, click the Delete icon at the right end of the property row.

8. Click **Save** to save the task.
9. Click **Deploy** to deploy a job instance for the task, or click the Actions menu button and select **View Details** to view or edit the task.

   You can run a job with a status of Deployed from the actions menu or from the **My Jobs** page.

Deploying an application ingestion and replication task

After you define an application ingestion and replication task, deploy the task to create an executable job instance on the on-premises system that contains the Secure Agent and the Database Ingestion agent service. You can run an application ingestion and replication job only after you deploy the associated task. When you deploy the task, Application Ingestion and Replication also validates the task definition.

If you undeploy a job and then want to run the job again, you must deploy the task again to create a new job instance. The new job instance name ends with an incremented number in the format

taskname-job_instance_number. The job instance number is incremented each time you deploy the ingestion task by adding 1 to the maximum instance number across all ingestion jobs.

Before you deploy a task that is configured for a Snowflake target, you must drop the existing target tables that do not match the structure of source objects. The existing target tables might not match the structure of source objects because of newly added source fields, dropped source or target fields, or modified field null constraints or data types. When you deploy the task after dropping the existing target tables, a new set of target tables are generated based on the source object selection rules and target table renaming rules.

▶ To deploy an application ingestion and replication task, in the task wizard, save the completed task definition and then click **Deploy**.

Deployment considerations:

- If you included spaces in the application ingestion and replication task name, the spaces are omitted from the corresponding job name for the generated job instance.
- If the deployment process fails, the job status switches to Failed. You can then Undeploy the job. To diagnose the error, you can download the error log from one of the monitoring interfaces: 1) the **My Jobs** page accessed from the unified Home page, 2) the **All Jobs** page in Monitor, or 3) the **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights. To download the error log, on the **Actions** menu for the job, click **Error Log**. After you resolve the issue, deploy the task again from the application ingestion and replication task wizard or from the **Actions** menu for the job.
- If the Secure Agent is restarted while the task is deploying, the job status switches to Failed. Avoid restarting the Secure Agent while tasks are being deployed.
- If a task appears to be hung in the Deploying state, restart the Secure Agent. The associated job instance acquires the status of Failed. You can undeploy it and then deploy it again.

Running a deployed application ingestion and replication job

You can run an application ingestion and replication job that has been previously deployed and is in a state other than Undeployed.

After you deploy a task successfully in the task wizard, the associated job instance is in the Deployed state. You can run it directly from the task wizard using the **Run** option in the Actions menu.

You can also run the job from any of the following monitoring interfaces: the **My Jobs** page that's accessed from the navigation bar of the Home page, the **All Jobs** page in Monitor, or the **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights .

For initial load jobs, you can specify a schedule for running the job when you configure the corresponding application ingestion and replication task.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to run.
2. In the Actions menu for the row, click **Run**.

   A subtask is started for each source object.

Notes:

- If the initial load job fails to load data to a target table, the application ingestion and replication job retries the subtask for the object up to three times. The minimum interval between the retries is 60 seconds. If all the initial load retries fail, Application Ingestion and Replication excludes the object from replication.
- If an initial load job detects inconsistencies between field definitions in the source and target objects, the job drops the target table and then re-creates it to be consistent with the source object before loading the source data to the target.
- Initial load jobs may take a long time to complete the ingestion if the source objects contain many records.

Managing application ingestion and replication jobs

After you configure and run application ingestion and replication jobs, you might occasionally need to perform some job management tasks such stopping, resuming, undeploying, or redeploying jobs.

Stopping an application ingestion and replication job

You can stop an application ingestion and replication job of any load type that is in the **Up and Running**, **Running with Warning**, or **On Hold** status.

When you stop an incremental load job, Application Ingestion and Replication records an identifier for the position in the change stream where it has stopped the incremental processing. The identifier is stored in a recovery table named INFORMATICA_CDC_RECOVERY on the target. If you restart the job, Application Ingestion and Replication uses this identifier to identify the last change record that was loaded to the target and starts loading the changes that were made after that point in the change stream.

Note: Before stopping an application ingestion and replication incremental load or combined load job that has a Snowflake target and uses the **Superpipe** option, ensure that all ingested data in the change stream has been merged into the target. Otherwise, if the job is not restarted for an extended period or before the stream expires, any data that remains in the stream is lost and can't be recovered.

For initial load jobs, the job stops only after its subtasks that are already running complete their operation. The subtasks that are not running remain in their current states.

You can stop the job from any of the following monitoring interfaces:

- **My Jobs** page that's accessed from the navigation bar of the Home page
- **All Jobs** page in Monitor
- **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

You can also stop the job from the Actions menu in the upper-right corner of the application ingestion and replication task wizard.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to stop.
2. In the **Actions** menu for the row, select **Stop**, or click the Stop icon next to the menu.

   The job state switches to **Stopping** and then to **Stopped**.

   Tip: If the Stop operation is taking too long, you can abort the job.

Resuming an application ingestion and replication job

You can resume an application ingestion and replication job that is in the Stopped, Aborted, or Failed status.

When you resume an initial load job that has multiple subtasks, Application Ingestion and Replication starts only the subtasks that are in the Failed, Stopped, Aborted, or Queued status.

When you resume an incremental load job or a combined initial and incremental load job, Application Ingestion and Replication resumes replicating source data change from the last position recorded in the checkpoint file or target recovery table. A checkpoint will not be available unless a change record was processed for at least one of the tables during the first job run after deployment. If a checkpoint is not available, the job resumes processing from the configured restart point, which is the latest available position in the change stream by default.

When you resume an incremental load job, Application Ingestion and Replication resumes propagating source data from where it last left off.

You can resume the job from any of the following monitoring interfaces:

- **My Jobs** page that's accessed from the navigation bar of the Home page
- **All Jobs** page in Monitor
- **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

You can also resume the job from the Actions menu in the upper-right corner of the application ingestion and replication task wizard.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to resume.
2. In the **Actions** menu for the row, click **Resume**, or click the Resume icon next to the menu.

   Note: The **Resume** command is not available if the job is in the **Failed** state because the task deployment failed.

   A subtask is started for each source table.

   If an error occurs, an error message is displayed at the top of the page.

Restart and recovery for incremental load jobs

Application Ingestion and Replication can restart the incremental load jobs that stopped because of an error and the jobs that were stopped or aborted by users without any loss of change data.

After the first job run, Application Ingestion and Replication continually records an identifier for the processing position in the change stream as changes are applied to the target. The identifier is stored in a recovery table named INFORMATICA_CDC_RECOVERY on the target.

When you resume an incremental load job, the job uses the last position recorded in the recovery table to identify the change records that it must load to the target. This process ensures that all changes are ingested to the target.

Overriding schema drift options when resuming an application ingestion and replication job

You can override the schema drift options when you resume an application ingestion and replication job that is in the Stopped, Aborted, or Failed state. The overrides affect only those objects that are currently in the Error state because of the **Stop Object**, **Stop Table**, **Stop Report**, or **Stop Job** schema drift option. Use the overrides to correct or resolve these errors.

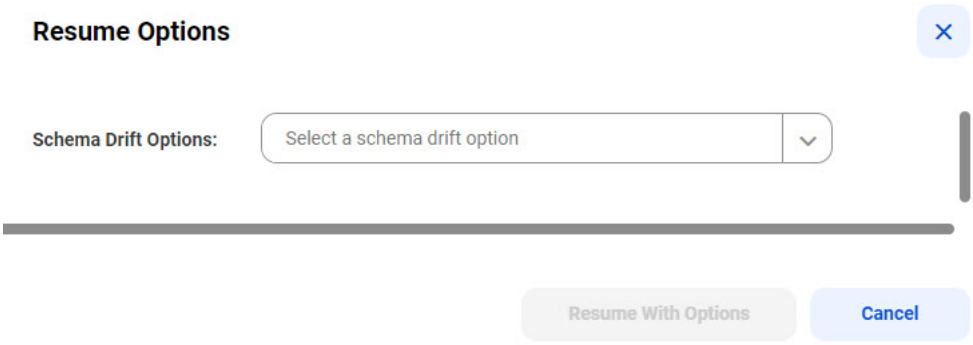
You can override schema drift options and resume an incremental load job or a combined initial and incremental load job from the **My Jobs** page that's accessed from the navigation bar of the Home page, the

All Jobs page in Monitor, or the **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights .

1. Navigate to the row for the job that you want to resume with an override.
2. In the Actions menu for the row, click **Resume With Options**.

   Note: The **Resume With Options** command is not available if the job is in the **Failed** state because the task deployment failed.

   The **Resume Options** dialog box appears.



3. In the **Schema Drift Options** list, select the schema drift option that will be used to process the DDL operation on the source that caused the application ingestion and replication job to stop.

   The following table describes the schema drift options:

Option	Description
Ignore	Do not replicate DDL changes that occur on the source to the target.
Stop Table	Stop processing the source object on which the DDL change occurred. **Important:** The application ingestion and replication job cannot retrieve the data changes that occurred on the source object after the job stopped processing it. Consequently, data loss might occur on the target. To avoid data loss, you will need to resynchronize the source and target objects that the job stopped processing. Use the **Resume With Options** > **Resync** option.
Resync	Resynchronize the target table with the source object. Use this option for objects that the job stopped processing because of the **Stop Object**, **Stop Table**, or **Stop Report** setting for a **Schema Drift** option. **Important:** This option is available only for combined initial and incremental load jobs.
Replicate	Allow the application ingestion and replication job to replicate the DDL change to the target. **Important:** If you specify the **Replicate** option for Rename Column operations on Microsoft Azure Synapse Analytics targets, the job will end with an error.

4. Click **Resume With Options**.

   The resumed job will use the schema drift option that you specified in step 3 to process the schema change that caused the job to stop. Thereafter, the schema drift options that you specified when creating the task take effect again.

   Important: Application Ingestion and Replication processes a schema change to a source object only after a DML operation occurs on the object. Therefore, after you resume a job, the object subtask state remains unchanged until the first DML operation occurs on the object.

Redeploying an application ingestion and replication job

Some fields in application ingestion and replication tasks are editable even when you do not undeploy the associated application ingestion and replication jobs. If you edit any available field in an application ingestion and replication task without undeploying the associated job, you must redeploy the job so that the changes can take effect.

If the fields that you want to edit are uneditable, you must undeploy the associated job and then edit them. For more information about undeploying a job, see "Undeploying an application ingestion and replication job" on page 201.

The redeploy operation stops each job subtask for a source table, deploys the updated ingestion task, and then starts the subtasks. The subtasks that are started includes the subtasks that were previously stopped and the subtasks that are newly created due to the configuration changes in the task.

Note: For incremental load jobs and combined initial and incremental load jobs, the redeployment does not change the source objects that were selected for ingestion during the previous deployment. To update the list of objects, you must edit the object selection rules in the associated task, and then redeploy the job.

You can redeploy the job from any of the following monitoring interfaces:

- **My Jobs** page that's accessed from the navigation bar of the Home page
- **All Jobs** page in Monitor
- **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

You can also redeploy the job from the Actions menu in the upper-right corner of the application ingestion and replication task wizard.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to redeploy.
2. In the **Actions** menu for the row, select **Redeploy**.

   The job instance automatically starts running.
   If the job was running when you selected **Redeploy**, Application Ingestion and Replication stops the job and then redeploys the ingestion task and restarts the job.

Undeploying an application ingestion and replication job

You can undeploy an application ingestion and replication job that has a status of Aborted, Completed, Deployed, Failed, or Stopped. You might want to undeploy a job if you no longer need to run the job or you need to change a connection or property in the associated task that cannot be edited without first undeploying the job.

Before you attempt to undeploy a job, ensure that it is not running.

After you undeploy a job, you cannot run it again or redeploy it. If you want to run a job that is undeployed, you must deploy the associated task again from the application ingestion and replication task wizard to create a new job instance. For example, if you want to change the target connection for a job, you must undeploy the job, edit the ingestion task to change the connection, deploy the task again, and then run the new job instance.

You can undeploy the job from any of the following monitoring interfaces:

- **My Jobs** page that's accessed from the navigation bar of the Home page
- **All Jobs** page in Monitor
- **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

You can also undeploy the job from the Actions menu in the upper-right corner of the application ingestion and replication task wizard.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to undeploy.
2. In the **Actions** menu for the row, click **Undeploy**.

   A confirmation dialog prompts you to confirm that you want to undeploy the job.
3. Click **OK** to undeploy the job, or click **Cancel**.

   If the undeploy operation fails, the job status switches to Failed, even if it was in the Aborted state, or remains as Failed.

   Note: After undeploying jobs, do not immediately shut down the Secure Agent. Application Ingestion and Replication requires some time to clean up files for tasks in the /root/infaagent/apps/Database_Ingestion/data/tasks directory.

Aborting an application ingestion and replication job

You can abort an application ingestion and replication job of any load type that is in the Up and Running, Running with Warning, On Hold, or Stopping status.

When you abort an incremental load job, Application Ingestion and Replication records an identifier for the position in the change stream where it has stopped the incremental processing. The identifier is stored in a recovery table named INFORMATICA_CDC_RECOVERY on the target. If you restart the job, Application Ingestion and Replication uses this identifier to identify the last change record that was loaded to the target and starts loading the changes that were made after that point in the change stream.

For initial load jobs, the subtasks that are already running stop immediately, and then the job stops. The subtasks that are not running remain in their current states.

You can abort the job from any of the following monitoring interfaces:

- **My Jobs** page that's accessed from the navigation bar of the Home page
- **All Jobs** page in Monitor
- **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

You can also abort the job from the Actions menu in the upper-right corner of the application ingestion and replication task wizard.

1. In any of the monitoring interfaces, navigate to the row for the job that you want to abort.
2. In the **Actions** menu for the row, select **Abort**.

   The status of the job changes to Aborting and then changes to Aborted.
   For initial load jobs, the status of the subtasks that were running change to Aborted. For incremental load jobs, the status of the subtasks change to Stopped.

Resynchronizing source and target objects

You can resynchronize source and target objects for a subtask that is part of a running application ingestion and replication job of combined initial and incremental load type. The subtask must be in a state other than Queued or Starting.

Resynchronization loads the latest changes from the source object to the target to make sure the source and target are consistent. Usually, the target object contents is truncated before the current source data is applied. However, for data lake targets, a T (truncate) operation is replicated instead of actually truncating the target contents.

For example, you might want to resynchronize the target with the source if initial load or incremental load processing failed or if you want to start the job over again from a specific restart point.

Important: To resynchronize objects that stopped and are currently in the **Error** state because of the **Schema Drift** setting of **Stop Object**, **Stop Table**, or **Stop Report**, you must use the **Resume With Options** > **Resync** option in the Actions menu. For more information, see "Overriding schema drift options when resuming an application ingestion and replication job" on page 199.

1. Drill down on the application ingestion and replication job that you want to resynchronize from one of the following monitoring interfaces:
   - **My Jobs** page that's accessed from the navigation bar of the Home page
   - **All Jobs** page in Monitor
   - **All Jobs** tab on the **Data Ingestion and Replication** page in Operational Insights

   The job must be in the **Up and Running** state and be for a combined initial and incremental load operation.
2. Click the **Object Detail** tab.
3. In the subtask row for the source and target objects that you want to resynchronize, click the Actions menu and select **Resync**. The resync operation resynchronizes the target table with the latest source table definition, including any DDL changes.

   Note: For the Actions menu and **Resync** option to be available, the subtask must be in a state other than Queued or Starting.

   If you resynchronize a subtask for an application ingestion and replication combined load job that has a NetSuite, Salesforce, or SAP Mass Ingestion source, use one of the following resync options instead of the **Resync** option:
   - **Resync (refresh)**. Use this option to resynchronize the target table with the latest source object definition, including any DDL change that schema drift ignored. After the target table is refreshed, the target table structure matches the current source object structure. This option mimics the behavior of the **Resync** option.
   - **Resync (retain)**. Use this option to resynchronize the same fields that have been processed for CDC, retaining the current structure of the source and target tables. No checks for changes in the source or target table definitions are performed. If source DDL changes affected the source object structure, those changes are not processed.

Notes:

- If the source table contains many rows, the resynchronization might take a long time to perform.
- If the source table schema does not match the target table schema, the ingestion subtask drops the target table and creates a new table that matches the source schema. Regardless of whether the target tables are re-created, the subtask truncates the target tables and then reloads source data to the tables.
- When you resync an application ingestion and replication subtask with a Snowflake target and use Audit apply mode, you can retain the audit information. Data Ingestion and Replication re-creates the target table and renames the existing table that contains the audit information using an appended timestamp in the format <target_table_name>_<current_UTC_timestamp>. If you want the audit information in the actual target table, you need to load it, for example, with a join operation. If adding the timestamp to the existing table name causes the name to exceed the maximum number of characters, the subtask fails with an error. If you enable schema drift and a schema drift change, such as Add Column, occurs, the new column will be in the re-created target table but not in the renamed table. To enable this behavior, set the backupTargetTableBeforeResync custom property to **true** on the **Target** page of the task wizard.

  Consider the following limitations when you resync a combined load job that has an existing audit table:

  - Storing the existing audit table on the target consumes extra database storage.

- To obtain a unified view of audit information, you need to join the multiple versions of the audit tables.

Default data-type mappings

This reference provides default data-type mappings for source applications and Amazon Redshift, Databricks, Google BigQuery, Microsoft Azure Synapse Analytics, Oracle, and Snowflake targets. When Application Ingestion and Replication generates the target data, it uses these mappings.

When you configure a target, you can optionally define data-type mapping rules to customize the default mappings of source data types to the target tables. For more information, see the target-specific topics accessed from "Task details: Configure how to replicate data to the target" on page 116.

If a source data type is not listed, Application Ingestion and Replication either cannot extract data from the source columns with this data type or cannot apply the extracted data to any appropriate target data type.

Marketo Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Marketo source and an Amazon Redshift target:

Marketo Source Data Type	Amazon Redshift Target Data Type
boolean	boolean
currency(38,18)	numeric(38,18)
date	date
datetime	timestamp without time zone
email	character varying(65535)
float	double precision
integer	bigint
long	bigint
percent(38,18)	numeric(38,18)
phone	character varying(65535)
reference	character varying(65535)
score	bigint
string	character varying(65535)
text	character varying(65535)
url	character varying(65535)

Marketo Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Marketo source and a Databricks target:

Marketo Source Data Type	Databricks Target Data Type
boolean	boolean
currency(38,18)	decimal(38,18)
date	string
datetime	timestamp
email	string
float	double
integer	integer
long	long
percent(38,18)	decimal(38,18)
phone	string
reference	string
score	integer
string	string
text	string
url	string

Marketo Source and Google BigQuery Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Marketo source and a Google BigQuery target:

Marketo Source Data Type	Google BigQuery Target Data Type
boolean	bool
currency(38,18)	bignumeric
date	date
datetime	datetime
email	string

Marketo Source Data Type	Google BigQuery Target Data Type
float	float64
integer	int64
long	int64
percent(38,18)	bignumeric
phone	string
reference	string
score	int64
string	string
text	string
url	string

Marketo Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Marketo source and a Microsoft Azure Synapse Analytics target:

Marketo Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
boolean	bit
currency(38,18)	decimal(38,18)
date	date
datetime	datetime
email	nvarchar(max)
float	float
integer	bigint
long	bigint
percent(38,18)	decimal(38,18)
phone	nvarchar(max)
reference	nvarchar(max)
score	bigint
string	nvarchar(max)

Marketo Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
text	nvarchar(max)
url	nvarchar(max)

Marketo Source and Oracle Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Marketo source and an Oracle target:

Marketo Source Data Type	Oracle Target Data Type
boolean	char(1 char)
currency(38,18)	number(38,18)
date	date
datetime	timestamp(3)
email	clob
float	binary_double
integer	number(10)
long	number(19)
percent(38,18)	number(38,18)
phone	clob
reference	clob
score	number(10)
string	clob
text	clob
url	clob

Marketo Source and Snowflake Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Marketo source and a Snowflake target:

Marketo Source Data Type	Snowflake Target Data Type
boolean	boolean
currency(38,18)	number(38,18)
date	date
datetime	datetime(3)
email	char(16777216)
float	float
integer	integer
long	integer
percent(38,18)	number(38,18)
phone	char(16777216)
reference	char(16777216)
score	integer
string	char(16777216)
text	char(16777216)
url	char(16777216)

Microsoft Dynamics 365 Source and Amazon Redshift Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Amazon Redshift target:

Microsoft Dynamics 365 Source Data Type	Amazon Redshift Target Data Type
biginttype	bigint
datetimetype	timestamp without time zone
decimaltype(38,38)	character varying(41)
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 37	numeric(p,s), 1 <= p <= 38, 0 <= s <= 37
doubletype	double precision

Microsoft Dynamics 365 Source Data Type	Amazon Redshift Target Data Type
filetype(*precision*), 1 <= p <= 134217728	binary varying(*size*), 1 <= s <= 1024000
imagetype(*precision*), 1 <= p <= 31457280	binary varying(*size*), 1 <= s <= 1024000
integertype	integer
managedpropertytype(*precision*), 1 <= p <= 10485760	character varying(*size*), 4 <= s <= 65535
memotype(*precision*), 1 <= p <= 1048576	character varying(*size*), 4 <= s <= 65535
moneytype(38,38)	character varying(41)
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 37	numeric(p,s), 1 <= p <= 38, 0 <= s <= 37
picklisttype	integer
statetype	integer
statustype	integer
stringtype(*precision*), 1 <= p <= 3901	character varying(*size*), 4 <= s <= 15604
uniqueidentifier(16)	character(36)

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Amazon Redshift target” on page 117.

Microsoft Dynamics 365 Source and Databricks Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Databricks target:

Microsoft Dynamics 365 Source Data Type	Databricks Target Data Type
biginttype	long
datetimetype	timestamp
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
doubletype	double
filetype(*precision*), 1 <= p <= 134217728	binary
imagetype(*precision*), 1 <= p <= 31457280	binary
integertype	integer
managedpropertytype(*precision*), 1 <= p <= 10485760	string

Microsoft Dynamics 365 Source Data Type	Databricks Target Data Type
memotype(*precision*), 1 <= p <= 1048576	string
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
picklisttype	integer
statetype	integer
statustype	integer
stringtype(*precision*), 1 <= p <= 3901	string
uniqueidentifier(16)	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

Microsoft Dynamics 365 Source and Google BigQuery Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Google BigQuery target:

MSD365 Source Data Type	Google BigQuery Target Data Type
biginttype	int64
datetimetype	datetime
decimaltype(10,10)	bignumeric
decimaltype(11,s), 10 <= s <= 11	bignumeric
decimaltype(12,s), 10 <= s <= 12	bignumeric
decimaltype(13,s), 10 <= s <= 13	bignumeric
decimaltype(14,s), 10 <= s <= 14	bignumeric
decimaltype(15,s), 10 <= s <= 15	bignumeric
decimaltype(16,s), 10 <= s <= 16	bignumeric
decimaltype(17,s), 10 <= s <= 17	bignumeric
decimaltype(18,s), 10 <= s <= 18	bignumeric
decimaltype(19,s), 10 <= s <= 19	bignumeric
decimaltype(20,s), 10 <= s <= 20	bignumeric
decimaltype(21,s), 10 <= s <= 21	bignumeric

MSD365 Source Data Type	Google BigQuery Target Data Type
decimaltype(22,s), 10 <= s <= 22	bignumeric
decimaltype(23,s), 10 <= s <= 23	bignumeric
decimaltype(24,s), 10 <= s <= 24	bignumeric
decimaltype(25,s), 10 <= s <= 25	bignumeric
decimaltype(26,s), 10 <= s <= 26	bignumeric
decimaltype(27,s), 10 <= s <= 27	bignumeric
decimaltype(28,s), 10 <= s <= 28	bignumeric
decimaltype(38,s), 10 <= s <= 38	bignumeric
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 9	numeric
decimaltype(p,s), 29 <= p <= 30, 0 <= s <= 29	bignumeric
decimaltype(p,s), 30 <= p <= 31, 0 <= s <= 30	bignumeric
decimaltype(p,s), 31 <= p <= 32, 0 <= s <= 31	bignumeric
decimaltype(p,s), 32 <= p <= 33, 0 <= s <= 32	bignumeric
decimaltype(p,s), 33 <= p <= 34, 0 <= s <= 33	bignumeric
decimaltype(p,s), 34 <= p <= 35, 0 <= s <= 34	bignumeric
decimaltype(p,s), 35 <= p <= 36, 0 <= s <= 35	bignumeric
decimaltype(p,s), 36 <= p <= 37, 0 <= s <= 36	bignumeric
decimaltype(p,s), 37 <= p <= 38, 0 <= s <= 37	bignumeric
doubletype	float64
filetype(*precision*), 1 <= p <= 134217728	bytes
imagetype(*precision*), 1 <= p <= 31457280	bytes
integertype	int64
managedpropertytype(*precision*), 1 <= p <= 10485760	string
memotype(*precision*), 1 <= p <= 1048576	string
moneytype(10,10)	bignumeric
moneytype(11,s), 10 <= s <= 11	bignumeric
moneytype(12,s), 10 <= s <= 12	bignumeric

MSD365 Source Data Type	Google BigQuery Target Data Type
moneytype(13,s), 10 <= s <= 13	bignumeric
moneytype(14,s), 10 <= s <= 14	bignumeric
moneytype(15,s), 10 <= s <= 15	bignumeric
moneytype(16,s), 10 <= s <= 16	bignumeric
moneytype(17,s), 10 <= s <= 17	bignumeric
moneytype(18,s), 10 <= s <= 18	bignumeric
moneytype(19,s), 10 <= s <= 19	bignumeric
moneytype(20,s), 10 <= s <= 20	bignumeric
moneytype(21,s), 10 <= s <= 21	bignumeric
moneytype(22,s), 10 <= s <= 22	bignumeric
moneytype(23,s), 10 <= s <= 23	bignumeric
moneytype(24,s), 10 <= s <= 24	bignumeric
moneytype(25,s), 10 <= s <= 25	bignumeric
moneytype(26,s), 10 <= s <= 26	bignumeric
moneytype(27,s), 10 <= s <= 27	bignumeric
moneytype(28,s), 10 <= s <= 28	bignumeric
moneytype(38,s), 10 <= s <= 38	bignumeric
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 9	numeric
moneytype(p,s), 29 <= p <= 30, 0 <= s <= 29	bignumeric
moneytype(p,s), 30 <= p <= 31, 0 <= s <= 30	bignumeric
moneytype(p,s), 31 <= p <= 32, 0 <= s <= 31	bignumeric
moneytype(p,s), 32 <= p <= 33, 0 <= s <= 32	bignumeric
moneytype(p,s), 33 <= p <= 34, 0 <= s <= 33	bignumeric
moneytype(p,s), 34 <= p <= 35, 0 <= s <= 34	bignumeric
moneytype(p,s), 35 <= p <= 36, 0 <= s <= 35	bignumeric
moneytype(p,s), 36 <= p <= 37, 0 <= s <= 36	bignumeric
moneytype(p,s), 37 <= p <= 38, 0 <= s <= 37	bignumeric

MSD365 Source Data Type	Google BigQuery Target Data Type
picklisttype	int64
statetype	int64
statustype	int64
stringtype(*precision*), 1 <= p <= 3901	string
uniqueidentifier(16)	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Google BigQuery target" on page 132.

Microsoft Dynamics 365 Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Microsoft Azure Synapse Analytics target:

MSD365 Source Data Type	Microsoft Azure SQL Data Warehouse or Synapse Analytics Target Data Type
biginttype	bigint
datetimetype	datetime
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
doubletype	float
filetype(1)	binary
filetype(*precision*), 100001 <= p <= 134217728	varbinary(max)
imagetype(1)	binary
imagetype(*precision*), 1000001 <= p <= 31457280	varbinary(max)
integertype	int
managedpropertytype(*precision*), 1 <= p <= 10485760	varchar(*size*), 4 <= s <= max
memotype(*precision*), 1 <= p <= 1048576	varchar(*size*), 4 <= s <= max
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
picklisttype	int

MSD365 Source Data Type	Microsoft Azure SQL Data Warehouse or Synapse Analytics Target Data Type
statetype	int
statustype	int
stringtype(*precision*), 1 <= p <= 3901	varchar(*size*), 4 <= s <= max
uniqueidentifier(16)	uniqueidentifier

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

Microsoft Dynamics 365 Source and Microsoft Azure SQL Database Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Microsoft Azure SQL Database target:

Microsoft Dynamics 365 Source Data Type	Microsoft Azure SQL Database Target Data Type
biginttype	bigint
booleantype	bit
datetimetype	datetime2(0)
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
doubletype	float
filetype(*precision*), 1 <= p <= 134217728	varbinary(max)
imagetype(*precision*), 1 <= p <= 31457280	varbinary(max)
integertype	int
managedpropertytype(*precision*), 1 <= p <= 10485760	varchar(*size*), 4 <= s <= max
memotype(*precision*), 1 <= p <= 1048576	varchar(*size*), 4 <= s <= max
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 38	decimal(p,s), 1 <= p <= 38, 0 <= s <= 38
picklisttype	int
statetype	int
statustype	int

Microsoft Dynamics 365 Source Data Type	Microsoft Azure SQL Database Target Data Type
stringtype(*precision*), 1 <= p <= 3901	varchar(*size*), 4 <= s <= max
uniqueidentifier(16)	uniqueidentifier

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Microsoft SQL Server target” on page 166.

Microsoft Dynamics 365 Source and Oracle Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and an Oracle target:

MSD365 Source Data Type	Oracle Target Data Type
biginttype	number(19)
datetimetype	date
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 38	number(p,s), 1 <= p <= 38, 0 <= s <= 38
doubletype	binary_double
filetype(*precision*), 1 <= p <= 134217728	blob
imagetype(*precision*), 1 <= p <= 31457280	blob
integertype	number(10)
managedpropertytype(*precision*), 1 <= p <= 1001	char(s char), 4 <= s <= 2000
managedpropertytype(*precision*), 2001 <= p <= 10485760	clob
memotype(*precision*), 1 <= p <= 1001	char(s char), 4 <= s <= 2000
memotype(*precision*), 2001 <= p <= 1048576	clob
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 38	number(p,s), 1 <= p <= 38, 0 <= s <= 38
picklisttype	number(10)
statetype	number(10)
statustype	number(10)
stringtype(*precision*), 1 <= p <= 1901	char(s char), 4 <= s <= 2000
stringtype(*precision*), 2001 <= p <= 3901	clob
uniqueidentifier(16)	char(36 char)

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Oracle target" on page 174.

Microsoft Dynamics 365 Source and Snowflake Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Microsoft Dynamics 365 source and a Snowflake target:

MSD365 Source Data Type	Snowflake Target Data Type
biginttype	number(19)
datetimetype	datetime(0)
decimaltype(38,38)	char(41)
decimaltype(p,s), 1 <= p <= 38, 0 <= s <= 37	number(p,s), 1 <= p <= 38, 0 <= s <= 37
doubletype	float
filetype(*precision*), 1 <= p <= 134217728	binary(*size*), 1 <= s <= 8300001
imagetype(*precision*), 1 <= p <= 31457280	binary(*size*), 1 <= s <= 8000001
integertype	number(10)
managedpropertytype(*precision*), 1 <= p <= 10485760	variant
memotype(*precision*), 1 <= p <= 1048576	char(*size*), 4 <= s <= 4194304
moneytype(38,38)	char(41)
moneytype(p,s), 1 <= p <= 38, 0 <= s <= 37	number(p,s), 1 <= p <= 38, 0 <= s <= 37
picklisttype	number(10)
statetype	number(10)
statustype	number(10)
stringtype(*precision*), 1 <= p <= 3901	char(*size*), 4 <= s <= 15604
uniqueidentifier(16)	char(36)

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

NetSuite Source and Amazon Redshift Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and an Amazon Redshift target:

NetSuite Source Data Type	Amazon Redshift Target Data Type
bigint	bigint
char	character varying(4)
clob	binary varying(*size*), 262144 <= size <= 400000
clob	character varying(*size*), 4 <= size <= 65535
double(p,s), 1 <= p <= 38, 0 <= s <= 18	numeric(p,s), 1 <= p <= 38, 0 <= s <= 18
number	character varying(*size*), 40 <= size <= 41
number	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
timestamp	timestamp without time zone
varchar2	character varying(*size*), 4 <= size <= 65535

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Amazon S3 target" on page 121.

NetSuite Source and Databricks Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and a Databricks target:

NetSuite Source Data Type	Databricks Target Data Type
bigint	byte
bigint	integer
bigint	long
char	string
clob	string
double(p,s), 1 <= p <= 38, 0 <= s <= 18	decimal(p,s), 1 <= p <= 38, 1 <= s <= 18
number	decimal(p,s), 1 <= p <= 38, 0 <= s <= 16
number	string

NetSuite Source Data Type	Databricks Target Data Type
timestamp	timestamp
varchar2	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

NetSuite Source and Google BigQuery Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and a Google BigQuery target:

NetSuite Source Data Type	Google BigQuery Target Data Type
bigint	int64
char	string
clob	string
double(p,s), 1 <= p <= 38, 0 <= s <= 18	bignumeric
number	bignumeric
number	bignumeric
number	numeric
timestamp	datetime
varchar2	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Google BigQuery target" on page 132.

NetSuite Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and a Microsoft Azure Synapse Analytics target:

NetSuite Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
bigint	bigint
char	nchar
clob	char(8000)

NetSuite Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
clob	nchar(*size*), 1 <= size <= 4000
clob	nvarchar(max)
clob	nvarchar(max)
double(p,s), 1 <= p <= 38, 0 <= s <= 18	decimal(p,s), 1 <= p <= 38, 1 <= s <= 18
number	decimal(p,s), 1 <= p <= 38, 0 <= s <= 16
number	varchar(*size*), 40 <= size <= 41
timestamp	datetime
varchar2	varchar(*size*), 4 <= size <= max

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

NetSuite Source and Oracle Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and an Oracle target:

NetSuite Source Data Type	Oracle Target Data Type
bigint	char(s char), 2 <= s <= 13
char	char(4 char)
clob	char(s char), 4 <= s <= 2000
clob	clob
double(p,s), 1 <= p <= 38, 0 <= s <= 18	number(p,s), 1 <= p <= 38, 1 <= s <= 18
number	char(s char), 40 <= s <= 41
number	number(p,s), 1 <= p <= 38, 0 <= s <= 16
timestamp	timestamp(1)
varchar2	char(s char), 4 <= s <= 2000
varchar2	clob

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Oracle target" on page 174.

NetSuite Source and Snowflake Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a NetSuite source and a Snowflake target:

NetSuite Source Data Type	Snowflake Target Data Type
bigint	char(*size*), 2 <= size <= 13
char	char(4)
clob	char(*size*), 4 <= size <= 400000
double(p,0), 1 <= p <= 38	integer
double(p,s), 1 <= p <= 38, 1 <= s <= 18	number(p,s), 1 <= p <= 38, 1 <= s <= 18
number	number(p,s), 1 <= p <= 38, 0 <= s <= 16
number	varchar(*size*), 40 <= size <= 41
timestamp	datetime(1)
varchar2	varchar(*size*), 4 <= size <= 262136

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

Oracle Fusion Cloud source and Amazon Redshift target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and Amazon Redshift target:

Oracle Fusion Cloud Source Data Type	Amazon Redshift Target Data Type
boolean	boolean
char(p), 1 <= p <= 2000	character varying(p), 4 <= p <= 8000
date	date
datetime	timestamp without time zone
datetime	timestamp without time zone
double	double precision
float	real
integer(p), 1 <= p <= 64	bigint
integer(p), 128 <= p <= 255	character varying(p), 40 <= p <= 78

Oracle Fusion Cloud Source Data Type	Amazon Redshift Target Data Type
integer(p), 65 <= p <= 127	numeric(p,0), 20 <= p <= 38
json	super
longvarchar(p), 1 <= p <= 2147483647	character varying(p), 4 <= p <= 65535
number(p,s), 2 <= p <= 38, 1 <= s <= 10	numeric(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	numeric(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2147483647	character varying(p), 4 <= p <= 65535
varchar(p), 1 <= p <= 2147483647	character varying(p), 4 <= p <= 65535

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Amazon Redshift target" on page 117.

Oracle Fusion Cloud source and Databricks target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and a Databricks target:

Oracle Fusion Cloud Source Data Type	Databricks Target Data Type
boolean	boolean
char(p), 1 <= p <= 2000	string
date	string
datetime	timestamp
datetime	timestamp
double	double
float	float
integer(p), 1 <= p <= 9	byte
integer(p), 10 <= p <= 32	integer
integer(p), 128 <= p <= 255	string
integer(p), 33 <= p <= 64	long
integer(p), 65 <= p <= 127	decimal(p), 20 <= p <= 38
longvarchar(p), 1 <= p <= 2147483647	string

Oracle Fusion Cloud Source Data Type	Databricks Target Data Type
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2147483647	string
varchar(p), 1 <= p <= 2147483647	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

Oracle Fusion Cloud source and Google BigQuery target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and Google BigQuery target:

Oracle Fusion Cloud Source Data Type	Google BigQuery Target Data Type
boolean	bool
char(p), 1 <= p <= 2000	string
date	date
datetime	datetime
datetime	datetime
double	float64
float	float64
integer(p), 1 <= p <= 64	int64
integer(p), 128 <= p <= 255	string
integer(p), 65 <= p <= 127	bignumeric
json	json
longvarchar(p), 1 <= p <= 2147483647	string
number(p,s), 2 <= p <= 38, 1 <= s <= 10	bignumeric
number(p,s), 2 <= p <= 38, 1 <= s <= 10	bignumeric
string(p), 1 <= p <= 2147483647	string
varchar(p), 1 <= p <= 2147483647	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Google BigQuery target" on page 132.

Oracle Fusion Cloud source and Oracle target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and Oracle target:

Oracle Fusion Cloud Source Data Type	Oracle Target Data Type
boolean	char(1 char)
char(p), 1 <= p <= 2000	char(s char), 4 <= s <= 2000
date	date
datetime	timestamp(3)
datetime	timestamp(3)
double	binary_double
float	binary_double
integer(1)	char(1 char)
integer(p), 128 <= p <= 255	char(s char), 40 <= s <= 78
integer(p), 2 <= p <= 127	number(p), 1 <= p <= 38
longvarchar(p), 1 <= p <= 2000	char(s char), 4 <= s <= 2000
longvarchar(p), 2001 <= p <= 2147483647	clob
number(p,s), 2 <= p <= 38, 1 <= s <= 10	number(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	number(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2000	char(s char), 4 <= s <= 2000
string(p), 2001 <= p <= 2147483647	clob
varchar(p), 1 <= p <= 2000	char(s char), 4 <= s <= 2000
varchar(p), 2001 <= p <= 2147483647	clob

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Oracle target" on page 174.

Oracle Fusion Cloud source and Microsoft Azure Synapse Analytics target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and a Microsoft Azure Synapse Analytics target:

Oracle Fusion Cloud Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
boolean	bit
char(p), 1 <= p <= 2000	varchar(p), 4 <= p <= 8000
date	date
datetime	datetime
datetime	datetime
double	float
float	real
integer(p), 1 <= p <= 64	bigint
integer(p), 128 <= p <= 255	char(p), 40 <= p <= 78
integer(p), 65 <= p <= 127	decimal(p), 20 <= p <= 38
longvarchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max
varchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

Oracle Fusion Cloud source and Microsoft Azure SQL Database target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and a Microsoft Azure SQL Database target:

Oracle Fusion Cloud Source Data Type	Microsoft Azure SQL Database Target Data Type
boolean	bit
char(p), 1 <= p <= 2000	varchar(p), 4 <= p <= 8000

Oracle Fusion Cloud Source Data Type	Microsoft Azure SQL Database Target Data Type
date	date
datetime	datetime
datetime	datetime
double	float
float	real
integer(p), 1 <= p <= 64	bigint
integer(p), 128 <= p <= 255	char(p), 40 <= p <= 78
integer(p), 65 <= p <= 127	decimal(p), 20 <= p <= 38
longvarchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	decimal(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max
varchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= max

Oracle Fusion Cloud source and Microsoft SQL Server target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and Microsoft SQL Server target:

Oracle Fusion Cloud Source Data Type	Microsoft SQL Server Target Data Type
boolean	bit
char(*size*), 1 <= s <= 2000	char(*size*), 4 <= s <= 8000
date	date
datetime	datetime2(3)
datetime	datetime2(3)
double	float
float	real
integer(*precision*), 1 <= p <= 64	bigint
integer(*precision*), 128 <= p <= 255	char(*size*), 40 <= s <= 78
integer(*precision*), 65 <= p <= 127	decimal(p,0), 20 <= p <= 38

Oracle Fusion Cloud Source Data Type	Microsoft SQL Server Target Data Type
longvarchar(2147483647)	ntext
longvarchar(*size*), 1 <= s <= 8000	char(*size*), 4 <= s <= 8000
longvarchar(*size*), 8001 <= s <= 32000	nvarchar(max)
number(2,s), 1 <= s <= 2	decimal(2,s), 1 <= s <= 2
number(2,s), 1 <= s <= 2	decimal(2,s), 1 <= s <= 2
number(3,s), 1 <= s <= 3	decimal(3,s), 1 <= s <= 3
number(3,s), 1 <= s <= 3	decimal(3,s), 1 <= s <= 3
number(4,s), 1 <= s <= 4	decimal(4,s), 1 <= s <= 4
number(4,s), 1 <= s <= 4	decimal(4,s), 1 <= s <= 4
number(5,s), 1 <= s <= 5	decimal(5,s), 1 <= s <= 5
number(5,s), 1 <= s <= 5	decimal(5,s), 1 <= s <= 5
number(6,s), 1 <= s <= 6	decimal(6,s), 1 <= s <= 6
number(6,s), 1 <= s <= 6	decimal(6,s), 1 <= s <= 6
number(7,s), 1 <= s <= 7	decimal(7,s), 1 <= s <= 7
number(7,s), 1 <= s <= 7	decimal(7,s), 1 <= s <= 7
number(8,s), 1 <= s <= 8	decimal(8,s), 1 <= s <= 8
number(8,s), 1 <= s <= 8	decimal(8,s), 1 <= s <= 8
number(9,10)	char(13)
number(9,10)	char(13)
number(9,s), 1 <= s <= 9	decimal(9,s), 1 <= s <= 9
number(9,s), 1 <= s <= 9	decimal(9,s), 1 <= s <= 9
number(p,s), 10 <= p <= 38, 1 <= s <= 10	decimal(p,s), 10 <= p <= 38, 1 <= s <= 10
number(p,s), 10 <= p <= 38, 1 <= s <= 10	decimal(p,s), 10 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 3, 3 <= s <= 4	money
number(p,s), 2 <= p <= 3, 3 <= s <= 4	money
number(p,s), 2 <= p <= 9, 5 <= s <= 10	char(*size*), 8 <= s <= 13
number(p,s), 2 <= p <= 9, 5 <= s <= 10	char(*size*), 8 <= s <= 13

Oracle Fusion Cloud Source Data Type	Microsoft SQL Server Target Data Type
number(p,s), 5 <= p <= 9, 6 <= s <= 10	char(*size*), 9 <= s <= 13
number(p,s), 5 <= p <= 9, 6 <= s <= 10	char(*size*), 9 <= s <= 13
number(p,s), 6 <= p <= 9, 7 <= s <= 10	char(*size*), 10 <= s <= 13
number(p,s), 6 <= p <= 9, 7 <= s <= 10	char(*size*), 10 <= s <= 13
number(p,s), 7 <= p <= 9, 8 <= s <= 10	char(*size*), 11 <= s <= 13
number(p,s), 7 <= p <= 9, 8 <= s <= 10	char(*size*), 11 <= s <= 13
number(p,s), 8 <= p <= 9, 9 <= s <= 10	char(*size*), 12 <= s <= 13
number(p,s), 8 <= p <= 9, 9 <= s <= 10	char(*size*), 12 <= s <= 13
string(2147483647)	ntext
string(*precision*), 1 <= p <= 8000	char(*size*), 4 <= s <= 8000
string(*precision*), 8001 <= p <= 32000	nvarchar(max)
varchar(2147483647)	ntext
varchar(*size*), 1 <= s <= 8000	char(*size*), 4 <= s <= 8000
varchar(*size*), 8001 <= s <= 32000	nvarchar(max)

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft SQL Server target" on page 166.

Oracle Fusion Cloud source and Snowflake target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an Oracle Fusion Cloud source and Snowflake target:

Oracle Fusion Cloud Source Data Type	Snowflake Target Data Type
boolean	boolean
char(p), 1 <= p <= 2000	varchar(p), 4 <= p <= 8000
date	date
datetime	datetime(3)
datetime	datetime(3)
double	float

Oracle Fusion Cloud Source Data Type	Snowflake Target Data Type
float	float
integer(1)	char
integer(p), 128 <= p <= 255	char(p), 40 <= p <= 78
integer(p), 2 <= p <= 127	integer
json	variant
longvarchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= 127996
number(p,s), 2 <= p <= 38, 1 <= s <= 10	number(p,s), 2 <= p <= 38, 1 <= s <= 10
number(p,s), 2 <= p <= 38, 1 <= s <= 10	number(p,s), 2 <= p <= 38, 1 <= s <= 10
string(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= 127996
varchar(p), 1 <= p <= 2147483647	varchar(p), 4 <= p <= 127996

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

Salesforce Source and Amazon Redshift Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and an Amazon Redshift target:

Salesforce Source Data Type	Amazon Redshift Target Data Type
anytype(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
base64 **Note:** The precision depends on the base64 body size you select.	character varying(*size*), 1 <= size <= 65535
boolean	boolean
combobox(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 0 <= s <= 17
date	date
datetime	timestamp without time zone
double(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 0 <= s <= 17
email(*precision*), 1 <= p <= 80	character varying(*size*), 1 <= size <= 80
encryptedstring(*precision*), 1 <= p <= 175	character varying(*size*), 1 <= size <= 175

Salesforce Source Data Type	Amazon Redshift Target Data Type
id(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
int	integer
long	bigint
multipicklist(*precision*), 1 <= p <= 4099	character varying(*size*), 1 <= size<= 4099
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 0 <= s <= 17
phone(*precision*), 1 <= p <= 40	character varying(*size*), 1 <= size <= 40
picklist(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
reference(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
string(*precision*), 1 <= p <= 100	character varying(*size*), 1 <= size <= 100
textarea(*precision*), 1 <= p <= 131072	character varying(*size*), 1 <= size <= 65535
time	time without time zone
url(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Amazon Redshift target" on page 117.

Salesforce Source and Databricks Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and a Databricks target:

Salesforce Source Data Type	Databricks Target Data Type
anytype(*precision*), 1 <= p <= 255	string
base64 **Note:** The precision depends on the base64 body size you select.	string
boolean	boolean
combobox(*precision*), 1 <= p <= 255	string
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
date	string
datetime	timestamp
double(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17

Salesforce Source Data Type	Databricks Target Data Type
email(*precision*), 1 <= p <= 80	string
encryptedstring(*precision*), 1 <= p <= 175	string
id(*precision*), 1 <= p <= 18	string
int	integer
long	long
multipicklist(*precision*), 1 <= p <= 4099	string
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	string
picklist(*precision*), 1 <= p <= 255	string
reference(*precision*), 1 <= p <= 18	string
string(*precision*), 1 <= p <= 100	string
textarea(*precision*), 1 <= p <=131072	string
time	string
url(*precision*), 1 <= p <= 255	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

Salesforce Source and Google BigQuery Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and a Google BigQuery target:

Salesforce Source Data Type	Google BigQuery Target Data Type
anytype(*precision*), 1 <= p <= 255	string
base64 **Note:** The precision depends on the base64 body size you select.	string
boolean	bool
combobox(*precision*), 1 <= p <= 255	string
currency(p,0), 1 <= p <= 18	numeric
currency(p,s), 1 <= p <= 18, 1 <= s <= 9	numeric
currency(p,s), 1 <= p <= 18, 10 <= s <= 17	bignumeric

Salesforce Source Data Type	Google BigQuery Target Data Type
date	date
datetime	datetime
double(p,0), 1 <= p <= 18	numeric
double(p,s), 1 <= p <= 18, 1 <= s <= 9	numeric
double(p,s), 1 <= p <= 18, 10 <= s <= 17	bignumeric
email(*precision*), 1 <= p <= 80	string
encryptedstring(*precision*), 1 <= p <= 175	string
id(*precision*), 1 <= p <= 18	string
int	int64
long	int64
multipicklist(*precision*), 1 <= p <= 4099	string
percent(p,0), 1 <= p <= 18	numeric
percent(p,s), 1 <= p <= 18, 1 <= s <= 9	numeric
percent(p,s), 1 <= p <= 18, 10 <= s <= 17	bignumeric
phone(*precision*), 1 <= p <= 40	string
picklist(*precision*), 1 <= p <= 255	string
reference(*precision*), 1 <= p <= 18	string
string(*precision*), 1 <= p <= 100	string
textarea(*precision*), 1 <= p <=131072	string
time	time
url(*precision*), 1 <= p <= 255	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Google BigQuery target” on page 132.

Salesforce Source and Microsoft Azure Synapse Analytics Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and a Microsoft Azure Synapse Analytics target:

Salesforce Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
anytype(*precision*), 1 <= p <= 255	varchar(*size*), 1 <= size <= 255
base64 **Note:** The precision depends on the base64 body size you select.	nchar(*size*), 1 <= size <= 255
boolean	bit
combobox(*precision*), 1 <= p <= 255	varchar(*size*), 1 <= size <= 255
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
date	date
datetime	datetime2(3)
double(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
email(*precision*), 1 <= p <= 80	varchar(*size*), 1 <= size <= 80
encryptedstring(*precision*), 1 <= p <= 175	varchar(*size*), 1 <= size <= 175
id(*precision*), 1 <= p <= 18	varchar(*size*), 1 <= size <= 18
int	int
long	bigint
multipicklist(*precision*), 1 <= p <= 4099	varchar(*size*), 1 <= size <= 4099
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	varchar(*size*), 1 <= size <= 40
picklist(*precision*), 1 <= p <= 255	varchar(*size*), 1 <= size <= 255
reference(*precision*), 1 <= p <= 18	varchar(*size*), 1 <= size <= 18
string(*precision*), 1 <= p <= 100	varchar(*size*), 1 <= size <= 100
textarea(*precision*), 1 <= p <=131072	varchar(*size*), 1 <= size <= max
time	time(3)
url(*precision*), 1 <= p <= 255	varchar(*size*), 1 <= size <= 255

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Microsoft Azure Synapse Analytics target” on page 151.

Salesforce Source and Microsoft Azure SQL Database Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and Microsoft Azure SQL Database target:

Salesforce Source Data Type	Microsoft Azure SQL Database Target Data Type
anytype(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
base64 **Note:** The precision depends on the base64 body size you select.	character varying(*size*), 1 <= size <= max
boolean	bit
combobox(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
date	date
datetime	datetime2 (3)
double(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
email(*precision*), 1 <= p <= 80	character varying(*size*), 1 <= size <= 80
encryptedstring(*precision*), 1 <= p <= 175	character varying(*size*), 1 <= size <= 175
id(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
int	int
long	long
multipicklist(*precision*), 1 <= p <= 4099	character varying(*size*), 1 <= size <= 4099
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	decimal(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	character varying(*size*), 1 <= size <= 40
picklist(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
reference(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
string(*precision*), 1 <= p <= 100	character varying(*size*), 1 <= size <= 100
textarea(*precision*), 1 <= p <= 131072	character varying(*size*), 1 <= size <= 131072
time	time(*precision*) without time zone, 3 <= p <= null
url(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft SQL Server target" on page 166.

Salesforce Source and Oracle Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and an Oracle target:

Salesforce Source Data Type	Oracle Target Data Type
anytype(*precision*), 1 <= p <= 255	char(s char), 1 <= s <= 255
base64 **Note:** The precision depends on the base64 body size you select.	clob
boolean	char(1 char)
combobox(*precision*), 1 <= p <= 255	char(s char), 1 <= s <= 255
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
date	date
datetime	timestamp(3)
double(p,s), 1 <= p <= 18, 0 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
email(*precision*), 1 <= p <= 80	char(s char), 1<= s <= 80
encryptedstring(*precision*), 1 <= p <= 175	char(s char), 1 <= s <= 175
id(*precision*), 1 <= p <= 18	char(s char), 1 <= s <= 18
int	number(10)
long	number(19)
multipicklist(*precision*),1 <= p <= 4099	clob
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	char(s char), 1 <= s <= 40
picklist(*precision*), 1 <= p <= 255	char(s char), 1 <= s <= 255
reference(*precision*), 1 <= p <= 18	char(s char), 1 <= s <= 18
string(*precision*), 1 <= p <= 100	char(s char), 1 <= s <= 100
textarea(*precision*), 1 <= p <= 4000	char(s char), 1 <= s <= 4000
textarea(*precision*), 4001 <= p <= 131072	clob
time	timestamp(3)
url(*precision*), 1 <= p <= 255	char(s char), 1<= s <= 255

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Oracle target” on page 174.

Salesforce Source and PostgreSQL Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and a PostgreSQL target:

Salesforce Source Data Type	PostgreSQL Target Data Type
anytype(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
base64 **Note:** The precision depends on the base64 body size you select.	character varying(*size*), 1 <= size <= max
boolean	boolean
combobox(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
currency(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 1 <= s <= 17
date	date
datetime	timestamp(*precision*) without time zone, 3 <= p <= null
double(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 1 <= s <= 17
email(*precision*), 1 <= p <= 80	character varying(*size*), 1 <= size <= 80
encryptedstring(*precision*), 1 <= p <= 175	character varying(*size*), 1 <= size <= 175
id(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
int	integer
long	bigint
multipicklist(*precision*), 1 <= p <= 4099	character varying(*size*), 1 <= size <= 4099
percent(p,s), 1 <= p <= 18, 0 <= s <= 17	numeric(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	character varying(*size*), 1 <= size <= 40
picklist(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255
reference(*precision*), 1 <= p <= 18	character varying(*size*), 1 <= size <= 18
string(*precision*), 1 <= p <= 100	character varying(*size*), 1 <= size <= 100
textarea(*precision*), 1 <= p <= 131072	character varying(*size*), 1 <= size <= 131072
time	time(*precision*) without time zone, 3 <= p <= null
url(*precision*), 1 <= p <= 255	character varying(*size*), 1 <= size <= 255

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a PostgreSQL target” on page 178.

Salesforce Source and Snowflake Target

The following table identifies the default data-type mappings for Application Ingestion and Replication configurations with a Salesforce source and a Snowflake target:

Salesforce Source Data Type	Snowflake Target Data Type
anytype(*precision*), 1 <= p <= 255	char(*size*), 4 <= size <= 1020
base64 **Note:** The precision depends on the base64 body size you select.	char(*size*), 4 <= size <= 1020
boolean	boolean
combobox(*precision*), 1 <= p <= 255	char(*size*), 4 <= size <= 1020
currency(p,0), 1 <= p <= 18	integer
currency(p,s), 1 <= p <= 18, 1 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
date	date
datetime	datetime(3)
double(p,0), 1 <= p <= 18	integer
double(p,s), 1 <= p <= 18, 1 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
email(*precision*), 1 <= p <= 80	char(*size*), 4 <= size <= 320
encryptedstring(*precision*), 1 <= p <= 175	char(*size*), 4 <= size <= 700
id(*precision*), 1 <= p <= 18	char(*size*), 4 <= size <= 72
int	number(10)
long	number(19)
multipicklist(*precision*), 1 <= p <= 4099	char(*size*), 4 <= size <= 16396
percent(p,0), 1 <= p <= 18	integer
percent(p,s), 1 <= p <= 18, 1 <= s <= 17	number(p,s), 1 <= p <= 18, 1 <= s <= 17
phone(*precision*), 1 <= p <= 40	char(*size*), 4 <= size <= 160
picklist(*precision*), 1 <= p <= 255	char(*size*), 4 <= size <= 1020
reference(*precision*), 1 <= p <= 18	char(*size*), 4 <= size <= 72
string(*precision*), 1 <= p <= 100	char(*size*), 4 <= size <= 400
textarea(*precision*), 1 <= p <= 131072	char(*size*), 4 <= size <= 131072

Salesforce Source Data Type	Snowflake Target Data Type
time	time(3)
url(*precision*), 1 <= p <= 255	char(*size*), 4 <= size <= 1020

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

Salesforce Marketing Cloud Source and Amazon Redshift Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and an Amazon Redshift target:

Salesforce Marketing Cloud Source Data Type	Amazon Redshift Target Data Type
boolean	boolean
date(0)	timestamp without time zone
decimal(p,s), 1 <= p <= 29, 0 <= s <= 8	numeric(p,s), 1 <= p <= 29, 0 <= s <= 8
emailaddress(*precision*), 1 <= p <= 254	character varying(*size*), 4 <= size <= 1016
locale(*precision*), 1 <= p <= 5	character varying(*size*), 4 <= size <= 20
number	integer
phone(*precision*), 1 <= p <= 50	character varying(*size*), 4 <= size <= 200
text(*precision*), 1 <= p <= 4000	character varying(*size*), 4 <= size <= 16000

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Amazon Redshift target" on page 117.

Salesforce Marketing Cloud Source and Databricks Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and a Databricks target:

Salesforce Marketing Cloud Source Data Type	Databricks Target Data Type
boolean	boolean
date(0)	timestamp
decimal(p,s), 1 <= p <= 29, 0 <= s <= 8	decimal(p,s), 1 <= p <= 29, 1 <= s <= 8
emailaddress(*precision*), 1 <= p <= 254	string

Salesforce Marketing Cloud Source Data Type	Databricks Target Data Type
locale(*precision*), 1 <= p <= 5	string
number	integer
phone(*precision*), 1 <= p <= 50	string
text(*precision*), 1 <= p <= 4000	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

Salesforce Marketing Cloud Source and Google BigQuery Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and a Google BigQuery target:

Salesforce Marketing Cloud Source Data Type	Google BigQuery Target Data Type
boolean	bool
date(0)	datetime
decimal(p,s), 1 <= p <= 29, 0 <= s <= 8	bignumeric
emailaddress(*precision*), 1 <= p <= 254	string
locale(*precision*), 1 <= p <= 5	string
number	int64
phone(*precision*), 1 <= p <= 50	string
text(*precision*), 1 <= p <= 4000	string

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Google BigQuery target" on page 132.

Salesforce Marketing Cloud Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and a Microsoft Azure Synapse Analytics target:

Salesforce Marketing Cloud Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
boolean	bit
date(0)	datetime
decimal(p,s), 1 <= p <= 29, 0 <= s <= 8	decimal(p,s), 1 <= p <= 29, 1 <= s <= 8
emailaddress(*precision*), 1 <= p <= 254	nchar(*size*), 1 <= size <= 254
locale(*precision*), 1 <= p <= 5	nchar(*size*), 1 <= size <= 5
number	int
phone(*precision*), 1 <= p <= 50	nchar(*size*), 1 <= size <= 50
text(*precision*), 1 <= p <= 4000	nchar(*size*), 1 <= size <= 4000

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

Salesforce Marketing Cloud Source and Oracle Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and an Oracle target:

Salesforce Marketing Cloud Source Data Type	Oracle Target Data Type
boolean	char(1 char)
date(0)	date
decimal(p,s), 1 <= p <= 29, 0 <= s <= 8	number(p,s), 1 <= p <= 29, 1 <= s <= 8
emailaddress(*precision*), 1 <= p <= 254	char(s char), 4 <= s <= 1016
locale(*precision*), 1 <= p <= 5	char(s char), 4 <= s <= 20
number	number(10)
phone(*precision*), 1 <= p <= 50	char(s char), 4 <= s <= 200
text(*precision*), 1 <= p <= 2000	char(s char), 4 <= s <= 2000
text(*precision*), 2001 <= p <= 4000	clob

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Oracle target" on page 174.

Salesforce Marketing Cloud Source and Snowflake Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a Salesforce Marketing Cloud source and a Snowflake target:

Salesforce Marketing Cloud Source Data Type	Snowflake Target Data Type
boolean	boolean
date(0)	datetime(0)
decimal(p,0), 1 <= p <= 29	integer
decimal(p,s), 1 <= p <= 29, 1 <= s <= 8	number(p,s), 1 <= p <= 29, 1 <= s <= 8
emailaddress(*precision*), 1 <= p <= 254	char(*size*), 4 <= size <= 1016
locale(*precision*), 1 <= p <= 5	char(*size*), 4 <= size <= 20
number	number(10)
phone(*precision*), 1 <= p <= 50	char(*size*), 4 <= size <= 200
text(*precision*), 1 <= p <= 4000	char(*size*), 4 <= size <= 16000

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

SAP Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP source and an Amazon Redshift target:

SAP Source Data Type	Amazon Redshift Target Data Type
d16r	double precision
d34r	double precision
fltp	double precision
int1	smallint
int2	smallint
int4	integer
int8	bigint

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure an Amazon Redshift target" on page 117.

SAP Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP source and a Databricks target:

SAP Source Data Type	Databricks Target Data Type
d16r	double
d34r	double
fltp	double
int1	integer
int2	integer
int4	integer
int8	long

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Databricks target" on page 127.

SAP Source and Google BigQuery Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP source and a Google BigQuery target:

SAP Source Data Type	Google BigQuery Target Data Type
d16r	float64
d34r	float64
fltp	float64
int1	int64
int2	int64
int4	int64
int8	int64

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Google BigQuery target” on page 132.

SAP Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP source and a Microsoft Azure Synapse Analytics target:

SAP Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
d16r	float
d34r	float
fltp	float
int1	tinyint
int2	smallint

SAP Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
int4	int
int8	bigint

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

SAP Source and Oracle Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with an SAP source and an Oracle target:

SAP Source Data Type	Oracle Target Data Type
CHAR	VARCHAR2(20)
ACCP	VARCHAR2(24)
CURR	NUMBER(31,8)
CUKY	VARCHAR2(20)
DF16_DEC	BINARY_DOUBLE
DF16_RAW	BINARY_DOUBLE
DF34_DEC	BINARY_DOUBLE
DF34_RAW	BINARY_DOUBLE
DEC	NUMBER(31,9)
DATS	DATE
CLNT	VARCHAR2(12)
INT2	NUMBER(5,0)
LANG	VARCHAR2(4)

SAP Source Data Type	Oracle Target Data Type
NUMC	VARCHAR2(40)
RAW	BLOB
SSTRING	CLOB
STRING	CLOB
TIMS	TIMESTAMP
QUAN	NUMBER(31,6)
UNIT	VARCHAR2(12)
INT1	NUMBER(3,0)
INT4	NUMBER(10,0)
LCHR	VARCHAR2(2000)

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Oracle target” on page 174.

SAP Source and Snowflake Target

The following table identifies the recommended data-type mappings for Application Ingestion and Replication configurations with a SAP source and a Snowflake target:

SAP Source Data Type	Snowflake Target Data Type
d16r	float
d34r	float
fltp	float
int1	number(3)
int2	number(5)

SAP Source Data Type	Snowflake Target Data Type
int4	number(10)
int8	number(19)

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- D16S
- D34S
- PREC
- VARC

Additionally, application ingestion and replication jobs do not support the RSTR data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

SAP Mass Ingestion connection Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and an Amazon Redshift target:

SAP Mass Ingestion connection Source Data Type	Amazon Redshift Target Data Type
accp(6)	character varying(24)
char(p), 1 <= p <= 255	character varying(p), 4 <= p <= 1020
clnt(p), 1 <= p <= 3	character varying(p), 4 <= p <= 12
cuky(p), 1 <= p <= 5	character varying(p), 4 <= p <= 20
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d16r	double precision
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d34r	double precision
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
fltp	double precision
int1	smallint

SAP Mass Ingestion connection Source Data Type	Amazon Redshift Target Data Type
int2	smallint
int4	integer
int8	bigint
lang(1)	character varying(4)
lchr *	character varying (65535)
lraw *	binary varying(1024000)
numc(p), 1 <= p <= 255	character varying(p), 4 <= p <= 1020
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
raw(p), 1 <= p <= 255	binary varying(p), 1 <= p <= 255
sstr(p), 1 <= p <= 1333	character varying(p), 4 <= p <= 5332
strg *	character varying (65535)
tims	time without time zone
unit(p), 1 <= p <= 5	character varying(p), 4 <= p <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Amazon Redshift target” on page 117.

SAP Mass Ingestion connection Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and a Microsoft Azure Synapse Analytics target:

SAP Mass Ingestion connection Source Data Type	Databricks Target Data Type
accp(6)	string
char(p), 1 <= p <= 255	string
clnt(p), 1 <= p <= 3	string
cuky(p), 1 <= p <= 5	string
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	double
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	double
dats	string
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	double
int1	integer
int2	integer
int4	integer
int8	long
lang(1)	string
lchr *	string
lraw *	binary
numc(p), 1 <= p <= 255	string
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	binary
sstr(p), 1 <= p <= 1333	string
strg *	string

SAP Mass Ingestion connection Source Data Type	Databricks Target Data Type
tims	string
unit(p), 1 <= p <= 5	string
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Databricks target” on page 127.

SAP Mass Ingestion connection Source and Google BigQuery Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and a Google BigQuery target:

SAP Mass Ingestion connection Source Data Type	Google BigQuery Target Data Type
accp(6)	string
char(p), 1 <= p <= 255	string
clnt(p), 1 <= p <= 3	string
cuky(p), 1 <= p <= 5	string
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
d16r	float64
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
fltp	float64
int1	int64
int2	int64

SAP Mass Ingestion connection Source Data Type	Google BigQuery Target Data Type
int4	int64
int8	int64
lang(1)	string
lchr *	string
lraw *	bytes
numc(p), 1 <= p <= 255	string
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
raw(p), 1 <= p <= 255	bytes
sstr(p), 1 <= p <= 1333	string
strg *	string
tims	time
unit(p), 1 <= p <= 5	string
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Google BigQuery target” on page 132.

SAP Mass Ingestion connection source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and an Microsoft Azure Synapse Analytics target:

SAP Mass Ingestion connection Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
accp(6)	varchar(24)
char(1)	nchar null
char(p), 2 <= p <= 255	nchar(p), 2 <= p <= 255
clnt(1)	nchar null
clnt(p), 2 <= p <= 3	nchar(p), 2 <= p <= 3
cuky(1)	nchar null
cuky(p), 2 <= p <= 5	nchar(p), 2 <= p <= 5
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	float
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	float
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	float
int1	bigint
int2	bigint
int4	bigint
int8	bigint
lang(1)	nchar null
lchr *	varchar(max)
lraw *	varbinary(max)
numc(1)	nchar null
numc(p), 2 <= p <= 255	nchar(p), 2 <= p <= 255

SAP Mass Ingestion connection Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(1)	binary null
raw(p), 2 <= p <= 255	binary(p), 2 <= p <= 255
sstr(1)	nchar null
sstr(p), 2 <= p <= 1333	nchar(p), 2 <= p <= 1333
strg *	char null
tims	time(0)
unit(1)	nchar null
unit(p), 2 <= p <= 5	nchar(p), 2 <= p <= 5
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Microsoft Azure Synapse Analytics target" on page 151.

SAP Mass Ingestion connection Source and Oracle Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion source and an Oracle target:

SAP Mass Ingestion Source Data Type	Oracle Target Data Type
accp(6)	varchar2(24 char)
char(p), 1 <= p <= 255	varchar2(s char), 4 <= s <= 1020
clnt(p), 1 <= p <= 3	varchar2(s char), 4 <= s <= 12
cuky(p), 1 <= p <= 5	varchar2(s char), 4 <= s <= 20
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16

SAP Mass Ingestion Source Data Type	Oracle Target Data Type
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	binary_double
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	binary_double
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	binary_double
int1	number(3)
int2	number(5)
int4	number(10)
int8	number(19)
lang(1)	varchar2(4 char)
lchr *	varchar2(1 char)
lraw *	blob null
numc(p), 1 <= p <= 255	varchar2(s char), 4 <= s <= 1020
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	blob null
sstr(p), 1 <= p <= 1333	varchar2(s char), 4 <= s <= 4000
strg *	varchar2(1 char)
tims	timestamp(0)
unit(p), 1 <= p <= 5	varchar2(s char), 4 <= s <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Oracle target” on page 174.

SAP Mass Ingestion connection Source and Snowflake Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and a Snowflake target:

SAP Mass Ingestion connection Source Data Type	Snowflake Target Data Type
accp(6)	varchar(24)
char(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
clnt(p), 1 <= p <= 3	varchar(p), 4 <= p <= 12
cuky(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	float
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	float
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	float
int1	number(3)
int2	number(5)
int4	number(10)
int8	number(19)
lang(1)	varchar(4)
lchr *	varchar(4)
lraw *	binary(1)
numc(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	binary(p), 1 <= p <= 255
sstr(p), 1 <= p <= 1333	varchar(p), 4 <= p <= 5332

SAP Mass Ingestion connection Source Data Type	Snowflake Target Data Type
strg *	varchar(4)
tims	time(0)
unit(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

SAP ODP Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP ODP source and an Amazon Redshift target:

SAP ODP Source Data Type	Amazon Redshift Target Data Type
accp(6)	character varying(24)
char(p), 1 <= p <= 255	character varying(p), 4 <= p <= 1020
clnt(p), 1 <= p <= 3	character varying(p), 4 <= p <= 12
cuky(p), 1 <= p <= 5	character varying(p), 4 <= p <= 20
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d16r	double precision
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
d34r	double precision
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16

SAP ODP Source Data Type	Amazon Redshift Target Data Type
fltp	double precision
int1	smallint
int2	smallint
int4	integer
int8	bigint
lang(1)	character varying(4)
lchr *	character varying (65535)
lraw *	binary varying (1024000)
numc(p), 1 <= p <= 255	character varying(p), 4 <= p <= 1020
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	numeric(p,s), 1 <= p <= 38, 0 <= s <= 16
raw(p), 1 <= p <= 255	binary varying(p), 1 <= p <= 255
sstr(p), 1 <= p <= 1333	character varying(p), 4 <= p <= 5332
strg *	character varying (65535)
tims	time without time zone
unit(p), 1 <= p <= 5	character varying(p), 4 <= p <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Amazon Redshift target” on page 117.

SAP ODP Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP ODP source and a Databricks target:

SAP ODP Source Data Type	Databricks Target Data Type
accp(6)	string
char(p), 1 <= p <= 255	string
clnt(p), 1 <= p <= 3	string
cuky(p), 1 <= p <= 5	string
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	double
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	double
dats	string
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	double
int1	integer
int2	integer
int4	integer
int8	long
lang(1)	string
lchr *	string
lraw *	binary
numc(p), 1 <= p <= 255	string
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	binary
sstr(p), 1 <= p <= 1333	string
strg *	string
tims	string

SAP ODP Source Data Type	Databricks Target Data Type
unit(p), 1 <= p <= 5	string
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Databricks target” on page 127.

SAP ODP Source and Google BigQuery Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP Mass Ingestion connection source and a Google BigQuery target:

SAP Mass Ingestion connection Source Data Type	Google BigQuery Target Data Type
accp(6)	string
char(p), 1 <= p <= 255	string
clnt(p), 1 <= p <= 3	string
cuky(p), 1 <= p <= 5	string
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
d16r	float64
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
fltp	float64
int1	int64
int2	int64
int4	int64
int8	int64

SAP Mass Ingestion connection Source Data Type	Google BigQuery Target Data Type
lang(1)	string
lchr *	string
lraw *	string
numc(p), 1 <= p <= 255	string
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	bignumeric
raw(p), 1 <= p <= 255	bytes
sstr(p), 1 <= p <= 1333	string
strg *	string
tims	time
unit(p), 1 <= p <= 5	string
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Google BigQuery target" on page 132.

SAP ODP Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP ODP source and a Microsoft Azure Synapse Analytics target:

SAP ODP Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
accp(6)	varchar(24)
char(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
clnt(p), 1 <= p <= 3	varchar(p), 4 <= p <= 12
cuky(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20

SAP ODP Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	float
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	float
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	float
int1	tinyint
int2	smallint
int4	int
int8	bigint
lang(1)	varchar(4)
lchr *	varchar(max)
lraw *	varbinary(max)
numc(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	decimal(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(1)	binary null
raw(p), 2 <= p <= 255	binary(p), 2 <= p <= 255
sstr(p), 1 <= p <= 1333	varchar(p), 4 <= p <= 5332
strg *	varchar null
tims	time(0)
unit(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s

- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure a Microsoft Azure Synapse Analytics target” on page 151.

SAP ODP Source and Oracle Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP ODP source and an Oracle target:

SAP ODP Source Data Type	Oracle Target Data Type
accp(6)	varchar2(24 char)
char(p), 1 <= p <= 255	varchar2(s char), 4 <= s <= 1020
clnt(p), 1 <= p <= 3	varchar2(s char), 4 <= s <= 12
cuky(p), 1 <= p <= 5	varchar2(s char), 4 <= s <= 20
curr(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	binary_double
d34d(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	binary_double
dats	date
dec(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	binary_double
int1	number(3)
int2	number(5)
int4	number(10)
int8	number(19)
lang(1)	varchar2(4 char)
lchr *	varchar2(1 char)
lraw *	blob null
numc(p), 1 <= p <= 255	varchar2(s char), 4 <= s <= 1020

SAP ODP Source Data Type	Oracle Target Data Type
quan(p,s), 1 <= p <= 38, 0 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	blob null
sstr(p), 1 <= p <= 1333	varchar2(s char), 4 <= s <= 4000
strg *	varchar2(1 char)
tims	timestamp(0)
unit(p), 1 <= p <= 5	varchar2(s char), 4 <= s <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types
Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec
- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in “Configure an Oracle target” on page 174.

SAP ODP Source and Snowflake Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with an SAP ODP source and a Snowflake target:

SAP ODP Source Data Type	Snowflake Target Data Type
accp(6)	varchar(24)
char(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
clnt(p), 1 <= p <= 3	varchar(p), 4 <= p <= 12
cuky(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20
curr(p,0), 1 <= p <= 38	integer
curr(p,s), 1 <= p <= 38, 1 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16d(p,0), 1 <= p <= 38	integer
d16d(p,s), 1 <= p <= 38, 1 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d16r	float

SAP ODP Source Data Type	Snowflake Target Data Type
d34d(p,0), 1 <= p <= 38	integer
d34d(p,s), 1 <= p <= 38, 1 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
d34r	float
dats	date
dec(p,0), 1 <= p <= 38	integer
dec(p,s), 1 <= p <= 38, 1 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
fltp	float
int1	number(3)
int2	number(5)
int4	number(10)
int8	number(19)
lang(1)	varchar(4)
lchr *	varchar(4)
lraw *	binary(1)
numc(p), 1 <= p <= 255	varchar(p), 4 <= p <= 1020
quan(p,0), 1 <= p <= 38	integer
quan(p,s), 1 <= p <= 38, 1 <= s <= 16	number(p,s), 1 <= p <= 38, 1 <= s <= 16
raw(p), 1 <= p <= 255	binary(p), 1 <= p <= 255
sstr(p), 1 <= p <= 1333	varchar(p), 4 <= p <= 5332
strg *	varchar(4)
tims	time(0)
unit(p), 1 <= p <= 5	varchar(p), 4 <= p <= 20
* SAP doesn't specify the maximum length for these data types.	

Unsupported source data types

Application Ingestion and Replication does not support the following obsolete SAP data types:

- d16s
- d34s
- prec

- varc

Additionally, application ingestion and replication jobs do not support the rstr data type.

If necessary, you can create custom data-type mappings when you define an application ingestion and replication task. For more information, see "Data Type Rules" in "Configure a Snowflake Data Cloud target" on page 182.

ServiceNow Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and an Amazon Redshift target:

ServiceNow Source Data Type	Amazon Redshift Target Data Type
audio	character varying(p), 4 <= p <= 16000
boolean	boolean
choice	character varying(p), 4 <= p <= 16000
color	character varying(p), 4 <= p <= 16000
compressed	character varying(p), 4 <= p <= 16000
condition_string	character varying(p), 4 <= p <= 16000
conditions	character varying(p), 4 <= p <= 16000
currency(p,s), 1 <= p <= 18, 0 <= s <= 2	numeric(p,s), 1 <= p <= 18, 0 <= s <= 2
currency2	character varying(p), 4 <= p <= 16000
decimal(p,s), 1 <= p <= 20, 0 <= s <= 2	numeric(p,s), 1 <= p <= 20, 0 <= s <= 2
document_id	character varying(p), 4 <= p <= 16000
domain_id	character varying(p), 4 <= p <= 16000
domain_path	character varying(p), 4 <= p <= 16000
due_date	timestamp without time zone
field_name	character varying(p), 4 <= p <= 16000
file_attachment	character varying(p), 4 <= p <= 16000
float(p,s), 1 <= p <= 19, 0 <= s <= 7	numeric(p,s), 1 <= p <= 20, 0 <= s <= 7
glide_date	date
glide_date_time	timestamp without time zone
glide_duration	timestamp without time zone
glide_encrypted	character varying(p), 4 <= p <= 16000

ServiceNow Source Data Type	Amazon Redshift Target Data Type
glide_list	character varying(p), 4 <= p <= 16000
glide_time	timestamp without time zone
guid	character varying(p), 4 <= p <= 16000
html	character varying(p), 4 <= p <= 16000
icon	character varying(p), 4 <= p <= 16000
image	character varying(p), 4 <= p <= 16000
integer	integer
ip_addr	character varying(p), 4 <= p <= 16000
journal	character varying(p), 4 <= p <= 16000
journal_input	character varying(p), 4 <= p <= 16000
journal_list	character varying(p), 4 <= p <= 16000
longint	bigint
password	character varying(p), 4 <= p <= 16000
password2	character varying(p), 4 <= p <= 16000
percent_complete(p,s), 1 <= p <= 16, 0 <= s <= 2	numeric(p,s), 1 <= p <= 16, 0 <= s <= 2
phone_number_e164	character varying(p), 4 <= p <= 16000
price(p,s), 1 <= p <= 18, 0 <= s <= 2	numeric(p,s), 1 <= p <= 18, 0 <= s <= 2
reference	character varying(p), 4 <= p <= 16000
script	character varying(p), 4 <= p <= 16000
script_plain	character varying(p), 4 <= p <= 16000
simple_name_values	character varying(p), 4 <= p <= 16000
string	character varying(p), 4 <= p <= 16000
string_full_utf8	character varying(p), 4 <= p <= 16000
sys_class_name	character varying(p), 4 <= p <= 16000
sys_class_path	character varying(p), 4 <= p <= 16000
table_name	character varying(p), 4 <= p <= 16000
translated_field	character varying(p), 4 <= p <= 16000

ServiceNow Source Data Type	Amazon Redshift Target Data Type
translated_html	character varying(p), 4 <= p <= 16000
translated_text	character varying(p), 4 <= p <= 16000
url	character varying(p), 4 <= p <= 16000
user_image	character varying(p), 4 <= p <= 16000
user_roles	character varying(p), 4 <= p <= 16000
video	character varying(p), 4 <= p <= 16000
wiki_text	character varying(p), 4 <= p <= 16000
workflow	character varying(p), 4 <= p <= 16000

ServiceNow Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and a Databricks target:

ServiceNow Source Data Type	Databricks Target Data Type
audio	string
boolean	boolean
choice	string
color	string
compressed	string
condition_string	string
conditions	string
currency(p,s), 1 <= p <= 18, 0 <= s <= 2	decimal(p,s), 1 <= p <= 18, 1 <= s <= 2
currency2	string
decimal(p,s), 1 <= p <= 20, 0 <= s <= 2	decimal(p,s), 1 <= p <= 20, 1 <= s <= 2
document_id	string
domain_id	string
domain_path	string
due_date	timestamp
field_name	string

ServiceNow Source Data Type	Databricks Target Data Type
file_attachment	string
float(p,s), 1 <= p <= 19, 0 <= s <= 7	decimal(p,s), 1 <= p <= 19, 1 <= s <= 7
glide_date	string
glide_date_time	timestamp
glide_duration	timestamp
glide_encrypted	string
glide_list	string
glide_time	timestamp
guid	string
html	string
icon	string
image	string
integer	integer
ip_addr	string
journal	string
journal_input	string
journal_list	string
longint	long
password	string
password2	string
percent_complete(p,s), 1 <= p <= 16, 0 <= s <= 2	decimal(p,s), 1 <= p <= 16, 1 <= s <= 2
phone_number_e164	string
price(p,s), 1 <= p <= 18, 0 <= s <= 2	decimal(p,s), 1 <= p <= 18, 1 <= s <= 2
reference	string
script	string
script_plain	string
simple_name_values	string

ServiceNow Source Data Type	Databricks Target Data Type
string	string
string_full_utf8	string
sys_class_name	string
sys_class_path	string
table_name	string
translated_field	string
translated_html	string
translated_text	string
url	string
user_image	string
user_roles	string
video	string
wiki_text	string
workflow	string

ServiceNow Source and Google BigQuery Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and a Google BigQuery target:

ServiceNow Source Data Type	Google BigQuery Target Data Type
audio	string
boolean	bool
choice	string
color	string
compressed	string
condition_string	string
conditions	string
currency(p,s), 1 <= p <= 18, 0 <= s <= 2	bignumeric
currency2	string

ServiceNow Source Data Type	Google BigQuery Target Data Type
decimal(p,s), 1 <= p <= 20, 0 <= s <= 2	bignumeric
document_id	string
domain_id	string
domain_path	string
due_date	datetime
field_name	string
file_attachment	string
float(p,s), 1 <= p <= 19, 0 <= s <= 7	bignumeric
glide_date	date
glide_date_time	datetime
glide_duration	datetime
glide_encrypted	string
glide_list	string
glide_time	datetime
guid	string
html	string
icon	string
image	string
integer	int64
ip_addr	string
journal	string
journal_input	string
journal_list	string
longint	int64
password	string
password2	string
percent_complete(p,s), 1 <= p <= 16, 0 <= s <= 2	bignumeric

ServiceNow Source Data Type	Google BigQuery Target Data Type
phone_number_e164	string
price(p,s), 1 <= p <= 18, 0 <= s <= 2	bignumeric
reference	string
script	string
script_plain	string
simple_name_values	string
string	string
string_full_utf8	string
sys_class_name	string
sys_class_path	string
table_name	string
translated_field	string
translated_html	string
translated_text	string
url	string
user_image	string
user_roles	string
video	string
wiki_text	string
workflow	string

ServiceNow Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and a Microsoft Azure Synapse Analytics target:

ServiceNow Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
audio	varchar(p), 4 <= p <= max
boolean	bit
choice	varchar(p), 4 <= p <= max

ServiceNow Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
color	varchar(p), 4 <= p <= max
compressed	varchar(p), 4 <= p <= max
condition_string	varchar(p), 4 <= p <= max
conditions	varchar(p), 4 <= p <= max
currency(p,s), 1 <= p <= 18, 0 <= s <= 2	decimal(p,s), 1 <= p <= 18, 1 <= s <= 2
currency2	varchar(p), 4 <= p <= max
decimal(p,s), 1 <= p <= 20, 0 <= s <= 2	decimal(p,s), 1 <= p <= 20, 1 <= s <= 2
document_id	varchar(p), 4 <= p <= max
domain_id	varchar(p), 4 <= p <= max
domain_path	varchar(p), 4 <= p <= max
due_date	datetime
field_name	varchar(p), 4 <= p <= max
file_attachment	varchar(p), 4 <= p <= max
float(p,s), 1 <= p <= 19, 0 <= s <= 7	decimal(p,s), 1 <= p <= 19, 1 <= s <= 7
glide_date	date
glide_date_time	datetime
glide_duration	datetime
glide_encrypted	varchar(p), 4 <= p <= max
glide_list	varchar(p), 4 <= p <= max
glide_time	datetime
guid	varchar(p), 4 <= p <= max
html	varchar(p), 4 <= p <= max
icon	varchar(p), 4 <= p <= max
image	varchar(p), 4 <= p <= max
integer	int
ip_addr	varchar(p), 4 <= p <= max

ServiceNow Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
journal	varchar(p), 4 <= p <= max
journal_input	varchar(p), 4 <= p <= max
journal_list	varchar(p), 4 <= p <= max
longint	bigint
password	varchar(p), 4 <= p <= max
password2	varchar(p), 4 <= p <= max
percent_complete(p,s), 1 <= p <= 16, 0 <= s <= 2	decimal(p,s), 1 <= p <= 16, 1 <= s <= 2
phone_number_e164	varchar(p), 4 <= p <= max
price(p,s), 1 <= p <= 18, 0 <= s <= 2	decimal(p,s), 1 <= p <= 18, 1 <= s <= 2
reference	varchar(p), 4 <= p <= max
script	varchar(p), 4 <= p <= max
script_plain	varchar(p), 4 <= p <= max
simple_name_values	varchar(p), 4 <= p <= max
string	varchar(p), 4 <= p <= max
string_full_utf8	varchar(p), 4 <= p <= max
sys_class_name	varchar(p), 4 <= p <= max
sys_class_path	varchar(p), 4 <= p <= max
table_name	varchar(p), 4 <= p <= max
translated_field	varchar(p), 4 <= p <= max
translated_html	varchar(p), 4 <= p <= max
translated_text	varchar(p), 4 <= p <= max
url	varchar(p), 4 <= p <= max
user_image	varchar(p), 4 <= p <= max
user_roles	varchar(p), 4 <= p <= max
video	varchar(p), 4 <= p <= max
wiki_text	varchar(p), 4 <= p <= max
workflow	varchar(p), 4 <= p <= max

ServiceNow Source and Oracle Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and an Oracle target:

ServiceNow Source Data Type	Oracle Target Data Type
udio	char(s char), 4 <= s <= 2000
audio	clob
boolean	char(1 char)
choice	char(s char), 4 <= s <= 2000
choice	clob
color	char(s char), 4 <= s <= 2000
color	clob
compressed	char(s char), 4 <= s <= 2000
compressed	clob
condition_string	char(s char), 4 <= s <= 2000
condition_string	clob
conditions	char(s char), 4 <= s <= 2000
conditions	clob
currency(p,s), 1 <= p <= 18, 0 <= s <= 2	number(p,s), 1 <= p <= 18, 1 <= s <= 2
currency2	char(s char), 4 <= s <= 2000
currency2	clob
decimal(p,s), 1 <= p <= 20, 0 <= s <= 2	number(p,s), 1 <= p <= 20, 1 <= s <= 2
document_id	char(s char), 4 <= s <= 2000
document_id	clob
domain_id	char(s char), 4 <= s <= 2000
domain_id	clob
domain_path	char(s char), 4 <= s <= 2000
domain_path	clob
due_date	date
field_name	char(s char), 4 <= s <= 2000

ServiceNow Source Data Type	Oracle Target Data Type
field_name	clob
file_attachment	char(s char), 4 <= s <= 2000
file_attachment	clob
float(p,s), 1 <= p <= 19, 0 <= s <= 7	number(p,s), 1 <= p <= 19, 1 <= s <= 7
glide_date	date
glide_date_time	date
glide_duration	date
glide_encrypted	char(s char), 4 <= s <= 2000
glide_encrypted	clob
glide_list	char(s char), 4 <= s <= 2000
glide_list	clob
glide_time	date
guid	char(s char), 4 <= s <= 2000
guid	clob
html	char(s char), 4 <= s <= 2000
html	clob
icon	char(s char), 4 <= s <= 2000
icon	clob
image	char(s char), 4 <= s <= 2000
image	clob
integer	number(10)
ip_addr	char(s char), 4 <= s <= 2000
ip_addr	clob
journal	char(s char), 4 <= s <= 2000
journal	clob
journal_input	char(s char), 4 <= s <= 2000
journal_input	clob

ServiceNow Source Data Type	Oracle Target Data Type
journal_list	char(s char), 4 <= s <= 2000
journal_list	clob
longint	number(19)
password	char(s char), 4 <= s <= 2000
password	clob
password2	char(s char), 4 <= s <= 2000
password2	clob
percent_complete(p,s), 1 <= p <= 16, 0 <= s <= 2	number(p,s), 1 <= p <= 16, 1 <= s <= 2
phone_number_e164	char(s char), 4 <= s <= 2000
phone_number_e164	clob
price(p,s), 1 <= p <= 18, 0 <= s <= 2	number(p,s), 1 <= p <= 18, 1 <= s <= 2
reference	char(s char), 4 <= s <= 2000
reference	clob
script	char(s char), 4 <= s <= 2000
script	clob
script_plain	char(s char), 4 <= s <= 2000
script_plain	clob
simple_name_values	char(s char), 4 <= s <= 2000
simple_name_values	clob
string	char(s char), 4 <= s <= 2000
string	clob
string_full_utf8	char(s char), 4 <= s <= 2000
string_full_utf8	clob
sys_class_name	char(s char), 4 <= s <= 2000
sys_class_name	clob
sys_class_path	char(s char), 4 <= s <= 2000
sys_class_path	clob

ServiceNow Source Data Type	Oracle Target Data Type
table_name	char(s char), 4 <= s <= 2000
table_name	clob
translated_field	char(s char), 4 <= s <= 2000
translated_field	clob
translated_html	char(s char), 4 <= s <= 2000
translated_html	clob
translated_text	char(s char), 4 <= s <= 2000
translated_text	clob
url	char(s char), 4 <= s <= 2000
url	clob
user_image	char(s char), 4 <= s <= 2000
user_image	clob
user_roles	char(s char), 4 <= s <= 2000
user_roles	clob
video	char(s char), 4 <= s <= 2000
video	clob
wiki_text	char(s char), 4 <= s <= 2000
wiki_text	clob
workflow	char(s char), 4 <= s <= 2000
workflow	clob

ServiceNow Source and Snowflake Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a ServiceNow source and a Snowflake target:

ServiceNow Source Data Type	Snowflake Target Data Type
audio	char(p), 4 <= p <= 16000
boolean	boolean
choice	char(p), 4 <= p <= 16000

ServiceNow Source Data Type	Snowflake Target Data Type
color	char(p), 4 <= p <= 16000
compressed	char(p), 4 <= p <= 16000
condition_string	char(p), 4 <= p <= 16000
conditions	char(p), 4 <= p <= 16000
currency(p,0), 1 <= p <= 18	integer
currency(p,s), 1 <= p <= 18, 1 <= s <= 2	number(p,s), 1 <= p <= 18, 1 <= s <= 2
currency2	char(p), 4 <= p <= 16000
decimal(p,0), 1 <= p <= 20	integer
decimal(p,s), 1 <= p <= 20, 1 <= s <= 2	number(p,s), 1 <= p <= 20, 1 <= s <= 2
document_id	char(p), 4 <= p <= 16000
domain_id	char(p), 4 <= p <= 16000
domain_path	char(p), 4 <= p <= 16000
due_date	datetime(0)
field_name	char(p), 4 <= p <= 16000
file_attachment	char(p), 4 <= p <= 16000
float(p,0), 1 <= p <= 19	integer
float(p,s), 1 <= p <= 19, 1 <= s <= 7	number(p,s), 1 <= p <= 19, 1 <= s <= 7
glide_date	date
glide_date_time	datetime(0)
glide_duration	datetime(0)
glide_encrypted	char(p), 4 <= p <= 16000
glide_list	char(p), 4 <= p <= 16000
glide_time	datetime(0)
guid	char(p), 4 <= p <= 16000
html	char(p), 4 <= p <= 16000
icon	char(p), 4 <= p <= 16000
image	char(p), 4 <= p <= 16000

ServiceNow Source Data Type	Snowflake Target Data Type
integer	number(10)
ip_addr	char(p), 4 <= p <= 16000
journal	char(p), 4 <= p <= 16000
journal_input	char(p), 4 <= p <= 16000
journal_list	char(p), 4 <= p <= 16000
longint	number(19)
password	char(p), 4 <= p <= 16000
password2	char(p), 4 <= p <= 16000
percent_complete(p,0), 1 <= p <= 16	integer
percent_complete(p,s), 1 <= p <= 16, 1 <= s <= 2	number(p,s), 1 <= p <= 16, 1 <= s <= 2
phone_number_e164	char(p), 4 <= p <= 16000
price(p,0), 1 <= p <= 18	integer
price(p,s), 1 <= p <= 18, 1 <= s <= 2	number(p,s), 1 <= p <= 18, 1 <= s <= 2
reference	char(p), 4 <= p <= 16000
script	char(p), 4 <= p <= 16000
script_plain	char(p), 4 <= p <= 16000
simple_name_values	char(p), 4 <= p <= 16000
string	char(p), 4 <= p <= 16000
string_full_utf8	char(p), 4 <= p <= 16000
sys_class_name	char(p), 4 <= p <= 16000
sys_class_path	char(p), 4 <= p <= 16000
table_name	char(p), 4 <= p <= 16000
translated_field	char(p), 4 <= p <= 16000
translated_html	char(p), 4 <= p <= 16000
translated_text	char(p), 4 <= p <= 16000
url	char(p), 4 <= p <= 16000
user_image	char(p), 4 <= p <= 16000

ServiceNow Source Data Type	Snowflake Target Data Type
user_roles	char(p), 4 <= p <= 16000
video	char(p), 4 <= p <= 16000
wiki_text	char(p), 4 <= p <= 16000
workflow	char(p), 4 <= p <= 16000

Workday Source and Amazon Redshift Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and an Amazon Redshift target:

Workday Source Data Type	Amazon Redshift Target Data Type
json	super
string	character varying(p), 4 <= p <= 4000
xml	character varying(65535)
xsd	numeric(p,s), 1 <= p <= 38, 0 <= s <= 37
xsd:boolean	boolean
xsd:date	date
xsd:datetime	timestamp with time zone
xsd:string	character varying(p), 4 <= p <= 4000

Workday Source and Databricks Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and a Databricks target:

Workday Source Data Type	Databricks Target Data Type
json	string
string	string
xml	string
xsd	decimal(p,s), 1 <= p <= 38, 0 <= s <= 37
xsd:boolean	boolean
xsd:date	string

Workday Source Data Type	Databricks Target Data Type
xsd:datetime	timestamp
xsd:string	string

Workday Source and Google BigQuery Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and a Google BigQuery target:

Workday Source Data Type	Google BigQuery Target Data Type
json	string
string	string
xml	string
xsd	bignumeric
xsd:boolean	bool
xsd:date	date
xsd:datetime	timestamp
xsd:string	string

Workday Source and Microsoft Azure Synapse Analytics Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and a Microsoft Azure Synapse Analytics target:

Workday Source Data Type	Microsoft Azure Synapse Analytics Target Data Type
json	nvarchar(max)
string	nchar(p), 1 <= p <= 1000
xml	nvarchar(max)
xsd	decimal(p,s), 1 <= p <= 38, 0 <= s <= 37
xsd:boolean	bit
xsd:date	date
xsd:datetime	datetimeoffset(3)
xsd:string	nchar(p), 1 <= p <= 1000

Workday Source and Oracle Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and an Oracle target:

Workday Source Data Type	Oracle Target Data Type
json	clob
string	char(s char), 4 <= s <= 2000
xml	clob
xsd	number(p,s), 1 <= p <= 38, 0 <= s <= 37
xsd:boolean	char(1 char)
xsd:date	date
xsd:datetime	timestamp(p) with time zone, 3 <= p <= null
xsd:string	char(s char), 4 <= s <= 2000

Workday Source and Snowflake Target

The following table identifies the recommended data type mappings for Application Ingestion and Replication configurations with a Workday source and a Snowflake target:

Workday Source Data Type	Snowflake Target Data Type
json	variant
string	char(p), 4 <= p <= 4000
xml	variant
xsd	integer
xsd	number(38,s), 1 <= s <= 37
xsd	number(p,s), 1 <= p <= 37, 1 <= s <= 37
xsd:boolean	boolean
xsd:date	date
xsd:datetime	timestamp_tz(3)
xsd:string	char(p), 4 <= p <= 4000

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